author2hash=81a1ee7fe5360e509724fc7036704d31family=Anderson, familyi=A., given=Matthew, giveni=M.hash=afd0a
familyi=D., given=Anuj, giveni=A. author4hash=bde39ffefcd0d3b6044b0a997e8a3e76family=Bhargav,
familyi=B., given=C.S., giveni=C.hash=0418dabebba8265e5207aedd22178949family=Chen, familyi=C., given=Shiteng,
giveni=S.hash=8c5e29ecca41273d433a754d8117a5cffamily=Curticapean, familyi=C., given=Radu, giveni=R.hash=080c3a2
familyi=D., given=Prateek, giveni=P. author3hash=e4be97472702d5f94b6d5fd932ccedd9family=Blass,
familyi=B., given=Andreas, giveni=A.hash=d36b6995ab536537094bfe3e5596722bfamily=Gurevich, familyi=G.,
given=Yuri, giveni=Y.hash=7d5fc2482a23e1353705eea5a284444efamily=Shelah, familyi=S., given=Saharon,
giveni=S. author1hash=85d4ff4ae6234ff61c5dfd4020e20bbbfamily=Bodlaender, familyi=B.,
given=Hans L., giveni=H. L. author2hash=707b89f782f1f6754e5e05aa91f73489family=Bulian,
familyi=B., given=Jannis, giveni=J.hash=afd0a68f50fd8efe5c2163bedba4a264family=Dawar, familyi=D., given=Anuj,
giveni=A. author3hash=432ecd624d7b0bb1926744a2a227b7e8family=Cai, familyi=C.,
given=Jin-Yi, giveni=J.-Y.hash=874b8a33b4334e74c94598fefeae5cb0family=Fürer, familyi=F., given=Martin,
giveni=M.hash=3c332bcb565e4c67d3d4bb506c8ea7a9family=Immerman, familyi=I., given=Neil, giveni=N.
author1hash=8c5e29ecca41273d433a754d8117a5cffamily=Curticapean, familyi=C., given=Radu, giveni=R.
author1hash=8c5e29ecca41273d433a754d8117a5cffamily=Curticapean, familyi=C., given=Radu, giveni=R.
author3hash=8c5e29ecca41273d433a754d8117a5cffamily=Curticapean, familyi=C., given=Radu, giveni=R.hash=436902e4
familyi=D., given=Holger, giveni=H.hash=e31719b087ab896a11d4b30ad542eab8family=Marx, familyi=M.,
given=Dániel, giveni=D. author3hash=8c5e29ecca41273d433a754d8117a5cffamily=Curticapean,
familyi=C., given=Radu, giveni=R.hash=7c6e0542f0cdc6ea6a2350e30ed4f422family=Limaye, familyi=L., given=Nutan,
giveni=N.hash=7f16294a215160a4add6dc00f88a3800family=Srinivasan, familyi=S., given=Srikanth, giveni=S.
author1hash=afd0a68f50fd8efe5c2163bedba4a264family=Dawar, familyi=D., given=Anuj, giveni=A. au-
thor1hash=afd0a68f50fd8efe5c2163bedba4a264family=Dawar, familyi=D., given=Anuj, giveni=A. au-
thor1hash=afd0a68f50fd8efe5c2163bedba4a264family=Dawar, familyi=D., given=Anuj, giveni=A. au-
thor1hash=afd0a68f50fd8efe5c2163bedba4a264family=Dawar, familyi=D., given=Anuj, giveni=A. au-
thor4hash=afd0a68f50fd8efe5c2163bedba4a264family=Dawar, familyi=D., given=Anuj, giveni=A.hash=f87eb334e06f67872
familyi=G., given=Erich, giveni=E.hash=ec9f59bf3f003881bec28550746b305efamily=Kullmann, familyi=K.,
given=Leon, giveni=L.hash=c470b33ea5873c4d7480486821574a5dfamily=Pago, familyi=P., given=Benedikt,
giveni=B. author3hash=afd0a68f50fd8efe5c2163bedba4a264family=Dawar, familyi=D.,
given=Anuj, giveni=A.hash=dbdf10a8064053fb4fd43156f970d9b8family=Jakl, familyi=J., given=Tomáš, giveni=T.hash=a
familyi=R., given=Luca, giveni=L. author3hash=afd0a68f50fd8efe5c2163bedba4a264family=Dawar,
familyi=D., given=Anuj, giveni=A.hash=c470b33ea5873c4d7480486821574a5dfamily=Pago, familyi=P., given=Benedikt,
giveni=B.hash=9e8819a6b30f14d244201901dd2c98f3family=Seppelt, familyi=S., given=Tim, giveni=T. au-
thor2hash=afd0a68f50fd8efe5c2163bedba4a264family=Dawar, familyi=D., given=Anuj, giveni=A.hash=2957b2a78df1c0e18
familyi=W., given=Pengming, giveni=P. author2hash=afd0a68f50fd8efe5c2163bedba4a264family=Dawar,
familyi=D., given=Anuj, giveni=A.hash=ec25e618384f5e386b9b14107e1be0e2family=Wilsenach, familyi=W.,
given=Gregory, giveni=G. author2hash=afd0a68f50fd8efe5c2163bedba4a264family=Dawar, fam-
ilyi=D., given=Anuj, giveni=A.hash=ec25e618384f5e386b9b14107e1be0e2family=Wilsenach, familyi=W., given=Gregory,
giveni=G. author2hash=afd0a68f50fd8efe5c2163bedba4a264family=Dawar, familyi=D.,
given=Anuj, giveni=A.hash=ec25e618384f5e386b9b14107e1be0e2family=Wilsenach, familyi=W., given=Gregory,
giveni=G. author2hash=afd0a68f50fd8efe5c2163bedba4a264family=Dawar, familyi=D.,
given=Anuj, giveni=A.hash=ec25e618384f5e386b9b14107e1be0e2family=Wilsenach, familyi=W., given=Gregory,
giveni=G. author3hash=436902e40aa3939fe77770b323c53564family=Dell, familyi=D.,
given=Holger, giveni=H.hash=09c9a63ce22980640e97ed9ed81ea815family=Grohe, familyi=G., given=Martin,
giveni=M.hash=5946bff9b80e45dd4a37477605ec45e1family=Rattan, familyi=R., given=Gaurav, giveni=G.
author2hash=fca84357df9d9260f5ec6e2f768b4adffamily=Derksen, familyi=D., given=Harm, giveni=H.hash=4d4bdeb1dac2
familyi=K., given=Gregor, giveni=G. author5hash=d308252f21654f81bd0346be7e852dcffamily=Durand,
familyi=D., given=Arnaud, giveni=A.hash=e7fe313ac0c465db915055d225d643d5family=Mahajan, familyi=M.,
given=Meena, giveni=M.hash=85bb050681604a45da31680f528aebd8family=Malod, familyi=M., given=Guillaume,
giveni=G.hash=b46d649e3d59915df18cfe023890271ffamily=De Rugy-Altherre, familyi=D. R.-A., given=Nicholas,
giveni=N.hash=e226bf9701c989ee65e670df1d890f97family=Saurabh, familyi=S., given=Nitin, giveni=N. au-
thor1hash=67971cf70262cbe218a2596692e7a59ffamily=Dvořák, familyi=D., given=Zdeněk, giveni=Z. au-
thor3hash=b48eed7b77c494eefc2a72fc80fb13d7family=Fluck, familyi=F., given=Eva, giveni=E.hash=9e8819a6b30f14d2442
familyi=S., given=Tim, giveni=T.hash=20b5e60c519694047f91454652d57af0family=Spitzer, familyi=S., given=Gian
Luca, giveni=G. L. author1hash=cbf2b5aaa03a9056694348250084316 6family=Forbes, fam-
ilyi=F., given=Michael A, giveni=M. A. author2hash=73506996fa0f0bd09d7bac3c932080f5family=Grochow,
familyi=G., given=Joshua A., giveni=J. A.hash=1813007d96ae80284b22fd39a7996df9family=Pitassi, fam-

ilyi=P., given=Toniann, giveni=T. author1hash=09c9a63ce22980640e97ed9ed81ea815family=Grohe, familyi=G., given=Martin, giveni=M. author3hash=09c9a63ce22980640e97ed9ed81ea815family=Grohe, familyi=G., given=Martin, giveni=M.hash=5946bff9b80e45dd4a37477605ec45e1family=Rattan, familyi=R., given=Gaurav, giveni=G.hash=9e8819a6b30f14d244201901dd2c98f3family=Seppelt, familyi=S., given=Tim, giveni=T. author1hash=b5efaf7b0e1834612a9e61e0daf14e1bfamily=Hartmann, familyi=H., given=Werner, giveni=W. author2hash=d4c2e5fde8c027e675d30976f8fe3b41family=He, familyi=H., given=William, giveni=W.hash=c545d7824479095ebb97455d2aba396dfamily=Rossman, familyi=R., given=Benjamin, giveni=B. author1hash=6f68eead71c19b021ce89bf7a72045d5family=Hella, familyi=H., given=Lauri, giveni=L. author2hash=ebc1c91740e5dcf530d41f9715a6f586family=Jerrum, familyi=J., given=Mark, giveni=M.hash=136ed2f6da407027ebf1447180c5fba6family=Snir, familyi=S., given=Marc, giveni=M. author2hash=e06725e7b98e9180538a01c054a5160afamily=Kawalek, familyi=K., given=Piotr, giveni=P.hash=397e07922b5ee1a890324499ca317475family=Weiß, familyi=W., given=Armin, giveni=A. author2hash=132a91296a099d008b0c152b456ec45afamily=Kayal, familyi=K., given=Neeraj, giveni=N.hash=4b1b2fca2e7303dfae4e5c7108a1c1cafamily=Saptharishi, familyi=S., given=Ramprasad, giveni=R. author1hash=8417d2752ad9cb67e1b75180b803243bfamily=Kiefer, familyi=K., given=Sandra, giveni=S. author3hash=0319e6b76ab66eb0bad6ad971f92dac2family=Komarath, familyi=K., given=Balagopal, giveni=B.hash=d2bd59aeec5cb7d1e2f29fd3f0e9fafdfamily=Pandey, familyi=P., given=Anurag, giveni=A.hash=97a55d4be5f51855930df59a9c65babefamily=Rahul, familyi=R., given=Chengot Sankaramenon, giveni=C. S. author2hash=d610eca99444930c101ac630a0e86db0family=Lanzinger, familyi=L., given=Matthias, giveni=M.hash=2aa697fa2f1df98ecb7b6bac366c56c7family=Barcelo, familyi=B., given=Pablo, giveni=P. author1hash=0d1784a2f9a807b1da8d15d5b1029c08family=Lovász, familyi=L., given=László, giveni=L. author1hash=0d1784a2f9a807b1da8d15d5b1029c08family=Lovász, familyi=L., given=László, giveni=L. author2hash=da3ba4d89c6630093e2951dedb53b5c2family=Mančinska, familyi=M., given=Laura, giveni=L.hash=269a09e114433113515edd303665e2d8family=Roberson, familyi=R., given=David E., giveni=D. E. author2hash=092281e384ed2467d1d5bb96b2eba907family=Montacute, familyi=M., given=Yoàv, giveni=Y.hash=a25867e2cf7761d82252b0a43a8a2694family=Shah, familyi=S., given=Nihil, giveni=N. author1hash=ac29d53c37dea1f61c7810cfea3b5ee6family=Neuen, familyi=N., given=Daniel, giveni=D. author1hash=3ade8687e2f53bbb90d3c0b056258d04family=Pultr, familyi=P., given=Ales, giveni=A. author1hash=2be204ff183c79ca47c4fb3d72aed23bfamily=Raz, familyi=R., given=Ran, giveni=R. author2hash=2be204ff183c79ca47c4fb3d72aed23bfamily=Raz, familyi=R., given=Ran, giveni=R.hash=8fea19bbf1f84f9f59e959335022dd6ffamily=Yehudayoff, familyi=Y., given=Amir, giveni=A. author1hash=269a09e114433113515edd303665e2d8family=Roberson, familyi=R., given=David E., giveni=D. E. author2hash=30004d929a781bee1262997e1b576502family=Robertson, familyi=R., given=Neil, giveni=N.hash=9c86b2da84f8e18c3c5e6194460cbfbbfamily=Seymour, familyi=S., given=P.D., giveni=P. author1hash=c545d7824479095ebb97455d2aba396dfamily=Rossman, familyi=R., given=Benjamin, giveni=B. author1hash=c545d7824479095ebb97455d2aba396dfamily=Rossman, familyi=R., given=Benjamin, giveni=B. author1hash=2b1601ffc02a16c070aba1cb8902d4a6family=Sagan, familyi=S., given=Bruce E., giveni=B. E. author1hash=9e8819a6b30f14d244201901dd2c98f3family=Seppelt, familyi=S., given=Tim, giveni=T. author1hash=9e8819a6b30f14d244201901dd2c98f3family=Seppelt, familyi=S., given=Tim, giveni=T. author1hash=9e8819a6b30f14d244201901dd2c98f3family=Seppelt, familyi=S., given=Tim, giveni=T. author1hash=a61a7c4d8f4b467437038b8bc2aee208family=Stanley, familyi=S., given=Richard P., giveni=R. P.

Symmetric algebraic circuits of polynomial orbit size are bounded-treewidth homomorphism combinations

Anuj Dawar, Benedikt Pago, Tim Seppelt

arxiv:2502.06740; CC BY 4.0

Editorial note. Only full orbit-size/treewidth equivalence counts. Orbit size and total circuit size are different; all author qualifications retained. Complete author body and appendices preserved. Native restate-key declarations converted mechanically to restatable environments; original source retained. Author compiled biblatex bibliography supplied directly as an asset.

Editorial note. Author biblatex3.2 data rendered with a 3.3 compatibility copy: only format marker and datalist context key updated to the key in the generated 3.3 BCF. All bibliographic fields and original bbl retained.

Editorial note. Repeated anchors supplied both in native restate metadata and the theorem body are emitted once in render copy.

Allowed standard prerequisites (editorial).

- Author-cited discrepancy lifting and standard resolution background. All source-local constructions and appendix proofs retained.

Abstract

The central open question of algebraic complexity is whether $\mathsf{VP} \neq \mathsf{VNP}$, which is saying that the permanent cannot be represented by families of polynomial-size algebraic circuits. For symmetric algebraic circuits, this has been confirmed by Dawar and Wilsenach (2020, 2025), who showed exponential lower bounds on the size of symmetric circuits for the permanent. In this work, we set out to develop a more general symmetric algebraic complexity theory. Our main result is that a family of symmetric polynomials admits small symmetric circuits if and only if they can be written as a linear combination of homomorphism counting polynomials of graphs of bounded treewidth. We also establish a relationship between the symmetric complexity of subgraph counting polynomials and the vertex cover number of the pattern graph. As a concrete example, we examine the symmetric complexity of immanant families (a generalisation of the determinant and permanent) and show that a known conditional dichotomy due to Curticapean (2021) holds unconditionally in the symmetric setting.

1 Introduction

The study of *algebraic circuit complexity* (also called *arithmetic circuit complexity*) aims to understand the power of circuits to succinctly express (or compute) polynomials. In short, we are interested in establishing how many operations of addition and multiplication are needed in a circuit that computes a polynomial $p \in \mathbb{F}[\mathcal{X}]$, for some field $\mathbb{F}$ and set of variables $\mathcal{X}$. We are usually interested in how this complexity grows with n for a family of polynomials $(p_n)_{n\in\mathbb{N}}$. The central conjecture in the field (known as $\mathsf{VP} \neq \mathsf{VNP}$) is that there are no polynomial-size circuits for the *permanent* in the way that there are for the *determinant*.

Both the determinant and permanent are examples of polynomials over matrices. That is to say that we can treat the set of variables $\mathcal{X}$ as the entries x_{ij} of a square matrix. Moreover, the permanent is *symmetric* in the sense that permuting the rows and columns of the matrix does not change the polynomial. The determinant has a smaller group of symmetries, being invariant under a permutation applied *simultaneously* to the rows and columns. This motivates the study of *symmetric algebraic circuits* introduced in [20, 21], which are circuits where symmetries of the polynomial computed are reflected in the automorphisms of the circuits. In [20, 21], an exponential gap between the symmetric circuit complexity of the determinant and the permanent was established: There are polynomial-size symmetric circuits (in the sense of invariance under simultaneous row and column permutations) that compute the determinant but any family of symmetric

circuits computing the permanent is of exponential size. It is the latter, exponential lower bound, that is the main technical achievement of that paper. Notably, this result also opens up a new approach to the question of separating VNP from VP and challenges us to find the minimum group of symmetries for a polynomial for which we can prove unconditional lower bounds. Thus, the study of symmetric circuits is motivated by the possibility of proving strong lower bounds which are presently out of reach without the assumption of symmetry.

Among the restrictions of circuit models that have been extensively studied in the literature, symmetry is arguably one of the newest and most interesting ones: Symmetric circuits not only admit provable lower bounds but are still surprisingly powerful, in that they can efficiently compute the determinant. By contrast, for example the restricted model of *multilinear formulas* requires super-polynomial size for both determinant and permanent [47], and the same is true for *bounded-depth multilinear circuits* [48]. Also for the well-studied class of *monotone* circuits, we have exponential lower bounds for the permanent [35]. This does not apply to the determiinant as it is not a monotone function, but with a suitably adapted definition of monotonicity the same methods also yield exponential lower bounds for the determinant [37].

In the present paper, we approach the study of symmetric algebraic circuits more systematically than has been done before. We develop a framework that links the expressive power of such circuits to the theory of graph *homomorphism counting*. Using this, we are able to completely characterise – within a large and interesting class of polynomials – which polynomials admit efficient symmetric circuits and which ones do not. This characterisation is the main result of the paper.

To be more precise, the polynomials we consider are over (not necessarily square) matrices of variables, have rational coefficients and are invariant under arbitrary permutations of the rows and columns. These are families $(p_{n,m})_{n,m\in\mathbb{N}}$ of polynomials where $p_{n,m} \in \mathbb{Q}[\mathcal{X}_{n,m}]$, and $\mathcal{X}_{n,m}$ is the set of variables $\{x_{ij} \mid i \in [n], j \in [m]\}$. The invariance condition we require is that for any pair of permutations $\pi \in \mathbf{Sym}_n$, $\sigma \in \mathbf{Sym}_m$, if $p_{n,m}^{(\pi,\sigma)}$ denotes the polynomial obtained from $p_{n,m}$ by replacing every occurrence of x_{ij} with $x_{\pi(i)\sigma(j)}$, then $p_{n,m} = p_{n,m}^{(\pi,\sigma)}$. Examples of such families are the permanent, for $m = n$, or more generally, for $m \geq n$, the *rectangular* permanent. The latter has been used recently in a lower bound proof for bounded-depth algebraic circuits [28].

The reason for focussing on this class of *matrix-symmetric* polynomials is that they have a very natural semantic interpretation: Every $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric $p_{n,m} \in \mathbb{Q}[\mathcal{X}_{n,m}]$ describes a function from weighted bipartite undirected graphs G with (n, m) vertices (i.e. G has a fixed bipartition $A \uplus B$ with $|A| = n$ and $|B| = m$) to $\mathbb{Q}$. The resulting value $p_{n,m}(G) \in \mathbb{Q}$ is the evaluation of $p_{n,m}$ in the bi-adjacency matrix of G, whose rows are indexed with the vertices in A, and columns with the vertices in B. The symmetry of $p_{n,m}$ ensures that the value $p_{n,m}(G)$ does not depend on the row and column ordering of the bi-adjacency matrix, and hence the $\mathbb{Q}$-valued function expressed by $p_{n,m}$ is in fact an isomorphism-invariant graph parameter.

It follows from a classical result of 'Operations with structures' [42] that the isomorphism-invariant $\mathbb{Q}$-valued functions on n-vertex graphs are precisely the linear combinations of homomorphism counting functions $\sum \alpha_F \hom(F, -)$ for finitely many graphs F and rational coefficients α_F. In other words, the isomorphism-invariant $\mathbb{Q}$-valued functions on graphs are precisely the *non-uniform graph motif parameters*, i.e. sequences $(\sum \alpha_{F,n} \hom(F, -))_{n\in\mathbb{N}}$ of linear combinations of homomorphism counts, one for every input graph size n. In an influential paper [9], Curticapean, Dell, and Marx studied functions which can be uniformly expressed in this way. Their *uniform graph motif parameters* are the functions of the form $\sum \alpha_F \hom(F, -)$ whose domain are all graphs regardless of their size. They show that the fixed-parameter counting complexity of such uniform graph motif parameters is governed by the treewidth of the patterns F with non-zero coefficients α_F.

Characterising matrix-symmetric polynomials admitting small symmetric circuits. We broaden the scope of [9] twofold by precisely characterising the symmetric circuit complexity of all, i.e. not necessarily uniform, graph motif parameters. To that end, we observe that the matrix-symmetric polynomials $p \in \mathbb{Q}[\mathcal{X}_{n,m}]$ are precisely those that can be written as linear combinations of homomorphism polynomials, cf. Lemma 3.1: For a bipartite multigraph F with bipartition $A \uplus B$, the *homomorphism polynomial* $\hom_{F,n,m} \in \mathbb{Q}[\mathcal{X}_{n,m}]$ is defined as

$$\hom_{F,n,m} \coloneqq \sum_{h\colon A\uplus B\to[n]\uplus[m]} \prod_{ab\in E(F)} x_{h(a)h(b)}.$$

The name is justified by the fact that $\hom_{F,n,m}$ evaluates to the number of homomorphisms from F to a (n, m)-vertex bipartite graph G when substituting the bi-adjacency matrix of G for the matrix of variables $\mathcal{X}_{n,m}$.

Our main result asserts that the symmetric complexity of a family $(p_{n,m})$ of matrix-symmetric polynomials is completely governed by the treewidth of the graphs whose homomorphism polynomials arise in linear expansions[1] of $(p_{n,m})$. By symmetric complexity we mean the smallest possible *orbit size* of a family $(C_{n,m})_{n,m\in\mathbb{N}}$ of symmetric circuits representing $(p_{n,m})_{n,m\in\mathbb{N}}$. This is the number of gates of $C_{n,m}$ in the largest orbit of the action of $\mathbf{Sym}_n \times \mathbf{Sym}_m$ on $C_{n,m}$ (see Section 2). Note that lower bounds on orbit size imply lower bounds on the total number of gates in a circuit but not vice versa. We also show that orbit size, not total size, is the correct measure that allows us to build our theory, cf. Theorem 3.2.

To state our main result, let $\mathfrak{T}^k_{n,m}$ denote, for every $k, n, m \in \mathbb{N}$, the collection of all polynomials that can be obtained as linear combinations of homomorphism polynomials $\hom_{F,n,m}$ for graphs F of treewidth less than k.

Theorem 1.1. *For every family of polynomials $p_{n,m} \in \mathbb{Q}[\mathcal{X}_{n,m}]$, the following are equivalent:*

1. *there exists a constant $k \in \mathbb{N}$ such that $p_{n,m} \in \mathfrak{T}^k_{n,m}$ for all $n, m \in \mathbb{N}$,*
2. *the $p_{n,m}$ admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of orbit size polynomial in $n + m$.*

It is instructive to consider again the permanent $\operatorname{perm}_n = \sum_{\pi\in\mathbf{Sym}_n} \prod_{i\in[n]} x_{i\pi(i)}$. This is not a homomorphism polynomial and not even a uniform graph motif parameter but it can be seen as counting, given a bipartite graph G instantiated as a bi-adjacency matrix $\mathcal{X}_{n,n}$, the number of subgraphs isomorphic to an n-matching. By Möbius inversion, cf. Corollary A.3, subgraph counts can be written as a linear combination of homomorphism counts. One consequence of Theorem 1.1 together with the exponential lower bound on symmetric circuits for the permanent [21] is that perm_n cannot be expressed as a linear combination of homomorphism polynomials of bounded treewidth. Alternatively, we can understand one direction of Theorem 1.1 as a vast (in fact, as vast as possible) generalisation of the lower bound on the permanent. Indeed, the theorem captures *all* super-polynomial lower bounds on the orbit size of symmetric circuits for matrix-symmetric polynomials.

Lower bounds via counting width. Theorem 1.1 stipulates that, to derive a super-polynomial lower bound for some family $p_{n,m}$, one must show that there is no constant $k \in \mathbb{N}$ such that $p_{n,m} \in \mathfrak{T}^k_{n,m}$ for all $n, m \in \mathbb{N}$. This, however, seems to be difficult with the presently available techniques. Therefore, our next goal is to give more usable characterisations of $\mathfrak{T}^k_{n,m}$ via a descriptive complexity measure for graph parameters, namely in terms of counting width. The *counting width* [12, 18] of a graph parameter p is defined as the smallest $k \in \mathbb{N}$ such that whenever two graphs G, H are indistinguishable by the k-dimensional Weisfeiler-Leman algorithm, then $p(G) = p(H)$, see also Definition 5.1. The Weisfeiler-Leman indistinguishability on graphs is a standard relaxation of graph isomorphism (see [6, 38]), and coincides with equivalence in the $(k + 1)$-variable fragment of first-order logic with counting quantifiers.

Counting width is well-studied in finite model theory and affords lower bounds, usually via the so-called Cai-Fürer-Immerman construction [6]. For Boolean circuits, a tight relationship between the counting width and orbit size of symmetric circuits is known [1].

The aforementioned exponential lower bound for symmetric circuits for the permanent was in fact established via a counting width lower bound for the number of perfect matchings. In general, unbounded counting width implies super-polynomial symmetric circuit lower bounds (via techniques for example from [1, 21]), so by Theorem 1.1, polynomials with unbounded counting width cannot be in $\mathfrak{T}^k_{n,m}$, for any fixed k. The question is, does the converse hold: Is bounded counting width a sufficient criterion for the existence of orbit-small symmetric circuits? We answer this question positively, at least in certain restricted, but interesting cases, cf. Figure 1:

1. when the $p_{n,m}$ are single homomorphism polynomials (rather than linear combinations of such), cf. Theorem 6.1,
2. when the $p_{n,m}$ are single subgraph polynomials for graphs of sublinear size, i.e. counting the number of subgraphs isomorphic to the pattern, cf. Theorem 3.6, and
3. when the $p_{n,m}$ are linear combinations of homomorphism polynomials of sublinear size, cf. Theorem 3.5.

Moreover, in the case of Items 1 and 2 and when Item 3 is restricted to linear combinations of polynomially many homomorphism counts, bounded counting width characterises the families of matrix-symmetric polynomials admitting symmetric circuits of polynomial size (rather than orbit size). For Items 1 and 2, we give a combinatorial characterisation of the families of patterns whose homomorphism/subgraph polynomials have bounded counting width in terms of treewidth and vertex cover number, respectively.

[1] In stark contrast to uniform graph motif parameters studied by 'Homomorphisms Are a Good Basis for Counting Small Subgraphs' [9], non-uniform graph motif parameters generally do not admit a unique expansion as linear combinations of homomorphism counts.

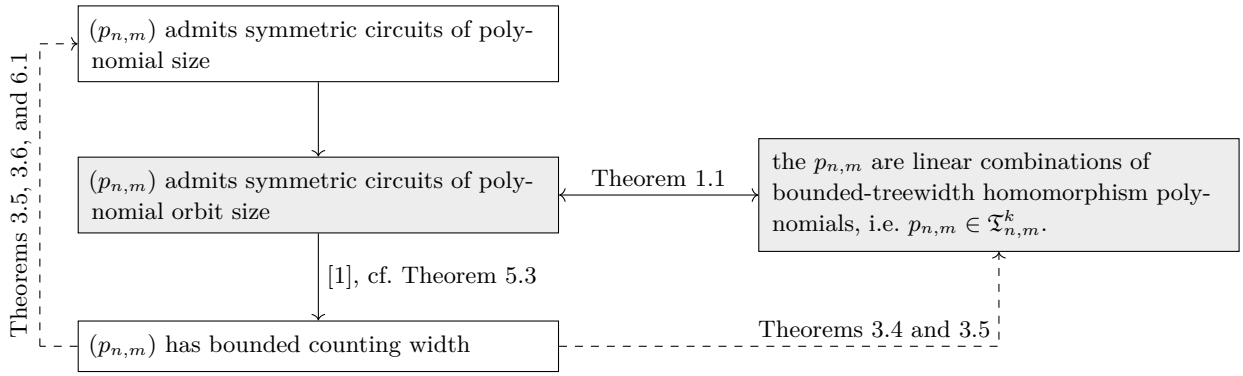


Figure 1: Implications between properties of families of matrix-symmetric polynomials $(p_{n,m})$. The dashed implications hold under additional assumptions.

Finally, to give another concrete example for the merits of the symmetric circuit framework, in Section 8 we show that a complexity dichotomy for the *immanant* families due to [7], whose hard cases are conditional on certain complexity-theoretic assumptions, holds unconditionally for symmetric circuits. The immanants interpolate between the permanent and determinant polynomials, which are in fact the extreme cases. Thus, the symmetric dichotomy for the immanants completes the picture begun in [21].

Related work. **Symmetric circuits** are an emerging area of research in recent years. We have already mentioned the lower bound for symmetric circuits for the permanent from [21]. In a similar vein, it has been shown that the determinant admits no small $\mathbf{Alt}_n \times \mathbf{Alt}_n$-symmetric circuits, even though it does have polynomial size $\mathbf{Sym}_n$-symmetric ones [19]. Other known lower bounds are for example for the symmetric formula complexity of computing the product of permutation matrices [33], and a lower bound for certain types of symmetric AC^0 circuits for the parity function [52]. More recent work establishes lower bounds for symmetric bounded-depth circuits with modular counting gates [36]. In [1], a precise characterisation of the power of uniform polynomial-size symmetric Boolean circuits with threshold gates was given: Such circuit families are equally powerful as fixed-point logic with counting, a formalism that is also related to the Weisfeiler-Leman algorithm. In particular, the graph properties computable with these circuits have bounded counting width, too. Similarly, in [22], rank logic is characterised via a class of symmetric circuits with more involved types of gates.

Another direction that has been explored are symmetric algebraic circuits in the context of algebraic proof complexity, more specifically in the circuit-based *Ideal Proof System* [15]. We hope that this direction will benefit from the framework we develop here, possibly leading to new lower bounds in proof complexity.

Another aspect of our work are the **homomorphism polynomials**. Interestingly, their complexity has been studied before with respect to a different restricted model, namely *monotone* circuits. In [2, 39] it is shown that the monotone circuit complexity of homomorphism polynomials is controlled by the treewidth of the pattern graphs in a similar way as we show this for symmetric circuits (and moreover, the circuit depth relates to the depth of the tree decompositions). It should be mentioned, though, that our setting is more general in two ways: We consider *linear combinations* of homomorphism polynomials, rather than single ones, and our set-up is non-uniform, while in [2, 39], the pattern graph is taken to be the same fixed graph for all sizes of host graphs. Certain types of homomorphism polynomials (defined differently from the ones we study) have also been shown to be VP-complete [25].

Our results also compare nicely to what is known on the *computational* complexity of **homomorphism** and **subgraph counting**. As already mentioned, it was shown in [9] that the computational complexity of graph parameters expressible as linear combinations of homomorphism counts (so-called *graph motif parameters*) depends only on the treewidth of the pattern graphs. The problem is in FPT if the treewidth is bounded, and $\#\mathsf{W}[1]$-hard, otherwise. Our Theorem 1.1 yields the same tractability criterion for the symmetric circuit complexity of the polynomials expressing these parameters but our proof techniques are very different. Moreover, as already mentioned, [9] concerns only uniform graph motif parameters, whereas we also cover the non-uniform case where we are counting different patterns for each host graph size.

The relationship between the complexity of subgraph counting and the *vertex cover number* of the pattern graphs that we obtain in Theorem 7.8 also appears in [9]. In [45, Theorem 1.3], a tight connection between the counting width of subgraph counts (for a fixed constant-size pattern) and the vertex cover number of the

pattern is established. Our Theorem 7.8 generalises this result to the case of sublinearly growing patterns.

Acknowledgments. We acknowledge fruitful discussions with Radu Curticapean, Filip Kučerák, Deepanshu Kush, and Benjamin Rossman.

The first and second author were funded by UK Research and Innovation (UKRI) under the UK government's Horizon Europe funding guarantee: grant number EP/X028259/1. The third author was supported by the European Union (CountHom, 101077083). Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council Executive Agency. Neither the European Union nor the granting authority can be held responsible for them.

2 Preliminaries

We write $\mathbb{N} = \{0, 1, \dots\}$ and $[n] = \{1, \dots, n\}$ for $n \geq 1$. For a tuple $\boldsymbol{x} \in X^k$ and $i \in [k]$, $x \in X$, write $\boldsymbol{x}[i/x] = x_1 \dots x_{i-1} x x_{i+1} \dots x_k \in X^k$ and $\boldsymbol{x}[i/] = x_1 \dots x_{i-1} x_{i+1} \dots x_k \in X^{k-1}$. For a set A, write $\Pi(A)$ for the set of partitions of A. For a partition $\pi \in \Pi(A)$, write A/π for its set of parts. Abusing notation, we also write π for the canonical map $A \to A/\pi$.

2.1 Permutation groups and supports

Let Γ be a group acting on a set X. For $S \subseteq X$, let $\mathbf{Stab}_\Gamma(S) \coloneqq \{\pi \in \Gamma \mid \pi(S) = S\}$ be the *stabiliser* of S and let $\mathbf{Stab}^\bullet_\Gamma(S) \coloneqq \{\pi \in \Gamma \mid \pi(s) = s \text{ for every } s \in S\}$ be the *pointwise stabiliser* of S. The *orbit* of S is denoted $\mathbf{Orb}_\Gamma(S) = \{\pi(S) \mid \pi \in \Gamma\}$. It holds $|\mathbf{Orb}_\Gamma(S)| = |\Gamma|/|\mathbf{Stab}_\Gamma(S)| = [\Gamma : \mathbf{Stab}_\Gamma(S)]$ by the Orbit-Stabiliser Theorem. We drop the subscript Γ if the group is clear from the context.

Let $\Delta \leq \Gamma$ be a subgroup. A set $S \subseteq X$ is a *support* of Δ if $\mathbf{Stab}^\bullet_\Gamma(S) \leq \Delta$. In what follows, we are mainly interested in the case where Γ is the group $\mathbf{Sym}_n$ acting on $[n]$ in the natural way, or it is the group $\mathbf{Sym}_I \times \mathbf{Sym}_J$ acting on $I \uplus J$ where $I = [n], J = [m]$ denote the bipartition of a bipartite graph. For simplicity, we just refer to the group Γ when we mean its action on the relevant set. In these special cases it is known that there always exists a unique minimal support.

Lemma 2.1. *Every subgroup $\Delta \leq \mathbf{Sym}_n$ that has a support of size $< n/2$ has a unique minimal support* $\sup(\Delta)$*. Every subgroup $\Delta \leq \mathbf{Sym}_I \times \mathbf{Sym}_J$ that has a support S with $|S \cap I| < |I|/2$ and $|S \cap J| < |J|/2$ has a unique minimal support* $\sup(\Delta)$*. In both cases $\mathbf{Stab}^\bullet(\sup(\Delta)) \leq \Delta \leq \mathbf{Stab}(\sup(\Delta))$.*

Proof. The existence of a unique minimal support is shown by proving that the intersection of any two supports with the respective size bound is again a support. For $\Delta \leq \mathbf{Sym}_n$, this is proved in Lemma 26 in [3]. For $\Delta \leq \mathbf{Sym}_I \times \mathbf{Sym}_J$, it is easy to check that the proof of that lemma also goes through. The assertion $\mathbf{Stab}^\bullet(\sup(\Delta)) \leq \Delta \leq \mathbf{Stab}(\sup(\Delta))$ is shown in [1]. □

We also use the following fact without explicitly referring to it later.

Lemma 2.2 ([1])**.** *Let $\Delta \leq \Gamma$ for an action of Γ on X such that the minimal support* $\sup(\Delta)$ *is defined. Let $\Delta' \coloneqq \pi \Delta \pi^{-1}$ for $\pi \in \Gamma$. Then* $\sup(\Delta') = \pi(\sup(\Delta))$*.*

When $\Gamma = \mathbf{Sym}_I \times \mathbf{Sym}_J$ acting on $I \uplus J$ we write $\sup_L(\Delta) \coloneqq \sup(\Delta) \cap I$ and $\sup_R(\Delta) \coloneqq \sup(\Delta) \cap J$ to denote the restrictions of the support to the left and right part of the bipartition, respectively. Generally, for any set $S \subseteq I \uplus J$, we write S_L to denote $S \cap I$ and S_R for $S \cap J$.

We also often need to think of supports as ordered tuples rather than unordered sets, in which case we write $\vec{\sup}(\Delta)$, or $\vec{\sup}_L(\Delta)$, $\vec{\sup}_R(\Delta)$. The ordering of the tuples is specified in the respective context.

2.2 Symmetric algebraic circuits

An *algebraic circuit* over a variable set $\mathcal{X}$ and a field $\mathbb{F}$ is a connected DAG with multiedges such that each vertex is labelled with an element of $\mathcal{X} \cup \mathbb{F} \cup \{+, \times\}$. The vertices of a circuit are called *gates*, the edges *wires*. Gates labelled with elements from $\mathcal{X} \cup \mathbb{F}$ are called *input gates* and they are not allowed to have incoming edges. Every element of $\mathcal{X} \cup \mathbb{F}$ is allowed to appear at most once as the label of a gate – this is no restriction because one can simply identify the respective gates. We assume that a circuit contains exactly one gate without outgoing wires, and this is called the *output gate*.

An algebraic circuit represents (or computes) a polynomial in $\mathbb{F}[\mathcal{X}]$ in the obvious way: the gates $+$ and $\times$ are simply interpreted as addition and multiplication on polynomials, and we care about the polynomial

at the output gate. Arrows are directed from a computation result towards the next gate where that result is used as an input. The set of *children* of a gate g, denoted $\mathrm{children}(g)$, is the set of gates h such that (h, g) is a wire of the circuit. Multiedges are allowed so that for example the polynomial x^2 can be represented with just one input and one multiplication gate. The *size* of a circuit C, is defined as the number of gates plus the number of wires, counted with multiplicities. It is denoted $\|C\|$.

Let Γ be a group that acts on $\mathcal{X}$. Then a circuit C is Γ-*symmetric* if the action of every $\pi \in \Gamma$ on the input gates $\mathcal{X}$ extends to an automorphism of the circuit: For every input gate g, let $\ell(g)$ denote its label. This is either a variable or a field element. Field elements are fixed by every $\pi \in \Gamma$. We say that a permutation $\pi \in \Gamma$ *extends to a circuit automorphism* of the circuit C if there exists a $\sigma \in \mathbf{Sym}(V(C))$ such that $\ell(\sigma(g)) = \pi(\ell(g))$ for every input gate, and such that σ is an automorphism of the multigraph structure of C (i.e. it maps edges to edges, non-edges to non-edges, and preserves the operation types of the gates). For more details, see [21].

A Γ-symmetric circuit is called *rigid* if it has no non-trivial circuit automorphism that fixes every input gate. This is equivalent to saying that for every $\pi \in \Gamma$, there is a *unique* circuit automorphism σ that extends π. In that case, we also write $\pi(g)$ for internal gates g, to denote the application of the unique circuit automorphism that extends π. Symmetric circuits can always be assumed to be rigid:

Lemma 2.3. *Let C be a Γ-symmetric circuit. Then there exists a Γ-symmetric* rigid *circuit C' that represents the same polynomial with $\|C'\| \leq \|C\|$.*

Proof. For Boolean circuits, the transformation of a non-rigid into a rigid symmetric circuit is described in the proof of [1, Lemma 7]. It is not hard to see that a similar transformation can be carried out for algebraic circuits, and that it satisfies $\|C'\| \leq \|C\|$. □

Note that in a rigid Γ-symmetric circuit, we can identify Γ with the automorphism group of the circuit. Another advantage of rigid circuits is that we obtain a well-defined notion of *support* of gates, and that there is a strong link between the size of these supports and the orbit sizes of the gates. In the following, we always assume that Γ is $\mathbf{Sym}_n$ or $\mathbf{Sym}_n \times \mathbf{Sym}_m$. In the former case, our variable set is $\mathcal{X}_n = \{x_{ij} \mid i, j \in [n]\}$, and $\pi \in \mathbf{Sym}_n$ acts simultaneously on both indices, so $\pi(x_{ij}) = x_{\pi(i)\pi(j)}$. In the latter case, the variable set is $\mathcal{X}_{n,m} = \{x_{ij} \mid i \in [n], j \in [m]\}$. Then a pair of permutations $(\pi, \pi') \in \mathbf{Sym}_n \times \mathbf{Sym}_m$ applied to x_{ij} yields $x_{\pi(i)\pi'(j)}$. However, the support of a subgroup Δ of Γ is always defined with respect to the action of Γ on $[n]$ in the case of $\mathbf{Sym}_n$ and on $[n] \uplus [m]$ in the case of $\mathbf{Sym}_n \times \mathbf{Sym}_m$.

Complexity measures for symmetric circuits We study the complexity of symmetric polynomials in terms of properties of their symmetric circuits. An important measure for symmetric circuits is $\mathrm{maxOrb}(C)$, the *maximum orbit size* of any gate in C, formally:

$$\mathrm{maxOrb}(C) \coloneqq \max_{g \in V(C)} |\{\sigma(g) \mid \sigma \in \Gamma\}|.$$

If C is rigid, and Γ and C are such that every stabiliser group of a gate admits a unique minimal support, then we can also speak about the *support size* of C. In particular, this is possible if $\Gamma \in \{\mathbf{Sym}_n, \mathbf{Sym}_n \times \mathbf{Sym}_m\}$ and all gates have a support of size $\leq n/2$, because then, Lemma 2.1 applies. We write $\sup(g) \subseteq [n] \uplus [m]$ to denote the *minimal support* of the group $\mathbf{Stab}_{\mathbf{Sym}_n \times \mathbf{Sym}_m}(g)$ in $\mathbf{Sym}_n \times \mathbf{Sym}_m$, and as before, $\sup_L(g)$ and $\sup_R(g)$ are its restrictions to $[n]$ and $[m]$, respectively. Similarly, in a $\mathbf{Sym}_n$-symmetric circuit, $\sup(g) \subseteq [n]$ denotes the minimal support of the group $\mathbf{Stab}_{\mathbf{Sym}_n}(g)$. The *support size* of a circuit C is then:

$$\mathrm{maxSup}(C) \coloneqq \max_{g \in V(C)} |\sup(g)|.$$

We have the following relations between maxOrb and maxSup.

Lemma 2.4. *Let $(\Gamma_{n,m})_{n,m \in \mathbb{N}}$ be either the sequence $(\mathbf{Sym}_n)_{n \in \mathbb{N}}$ or $(\mathbf{Sym}_n \times \mathbf{Sym}_m)_{n,m \in \mathbb{N}}$. Let $(C_{n,m})_{n,m \in \mathbb{N}}$ be a sequence of $\Gamma_{n,m}$-symmetric rigid circuits.*

1. *If $\mathrm{maxOrb}(C_{n,m})$ is polynomially bounded in $n + m$, then there exists a constant $k \in \mathbb{N}$ such that for all large enough $n, m \in \mathbb{N}$, it holds $\mathrm{maxSup}(C_{n,m}) \leq k$.*
2. *If $\mathrm{maxOrb}(C_{n,m})$ is at most $2^{o(n+m)}$, then $\mathrm{maxSup}(C_{n,m}) \leq o(n + m)$.*

Proof. In the case $\Gamma_n = \mathbf{Sym}_n$, this is stated in Theorems 6.2 and 6.3 in [21]. One can check that for $\Gamma_n = \mathbf{Sym}_n \times \mathbf{Sym}_m$, the proofs of these theorems also go through. We refrain from spelling out all details but the core argument is this: If the orbit size of a gate is polynomially bounded, then by the orbit-stabiliser

theorem, the index of its stabiliser group in Γ_n is also polynomial, i.e. can be bounded by $\binom{n}{k}$, for some constant k. Theorem 6.2 in [21] relies on Theorems 4.6 and 4.10 in [22]. The former of them states the following: If $\Delta \leq \mathbf{Sym}_n$ is a subgroup of index at most $\binom{n}{k}$, then Δ contains an alternating group A on $(n-k)$ points. The proof of [22, Theorem 4.10] then shows that if Δ is the stabiliser group of a gate in a circuit, then the gate is stabilised not only by A, but by the full symmetric group on the $n-k$ many points. This means that the remaining k points constitute a support for the gate. Now if $\Delta \leq \mathbf{Sym}_n \times \mathbf{Sym}_m$ is the stabiliser group of a gate, then we can consider $\Delta_1, \Delta_2 \leq \Delta$ defined as the subgroups of Δ consisting of those members of the product that have the neutral element in the first or second component, respectively. Because the orbit size of the gate is polynomially bounded, the index of Δ_1, Δ_2 in $\mathbf{Sym}_n, \mathbf{Sym}_m$, respectively, is at most $\binom{n}{k}$ (or $\binom{m}{k}$), for some constant k. Then the same reasoning as before applies (this is analogous to the argument made, in the context of the alternating group, in [19]). This finishes the first part of the theorem. The second part follows from the first part as shown in the proof of [21, Theorem 6.3], and this is independent of whether $\Gamma_n = \mathbf{Sym}_n$ or $\Gamma_n = \mathbf{Sym}_n \times \mathbf{Sym}_m$. □

The converse of Lemma 2.4, part 1, is also true:

Lemma 2.5. *Let $(C_{n,m})_{n\in\mathbb{N}}$ be a sequence of rigid $\Gamma_{n,m}$-symmetric circuits, for $\Gamma \in \{\mathbf{Sym}_n, \mathbf{Sym}_n \times \mathbf{Sym}_m\}$. Assume that there is a constant $k \in \mathbb{N}$ such that $\mathrm{maxSup}(C_{n,m}) \leq k$, for all $n, m \in \mathbb{N}$. Then $\mathrm{maxOrb}(C_{n,m}) \leq (n+m)^k$.*

Proof. We consider the case $\Gamma = \mathbf{Sym}_n \times \mathbf{Sym}_m$, the other case is similar. Since $C \coloneqq C_{n,m}$ is rigid, every $(\pi, \pi') \in \mathbf{Sym}_n \times \mathbf{Sym}_m$ has a unique extension to a circuit automorphism σ. Let $g \in V(C)$ be a gate. Its stabiliser subgroup in $\mathbf{Sym}_n \times \mathbf{Sym}_m$ contains $\mathbf{Stab}^\bullet_{\mathbf{Sym}_n \times \mathbf{Sym}_m}(\mathrm{sup}(g))$. This means that for any two pairs $(\pi_1, \pi_1'), (\pi_2, \pi_2') \in \mathbf{Sym}_n \times \mathbf{Sym}_m$, it holds: If $(\pi_1, \pi_1')(s) = (\pi_2, \pi_2')(s)$ for every $s \in \mathrm{sup}(g)$, and $\sigma_1, \sigma_2 \in \mathbf{Aut}(C_n)$ are the unique circuit automorphisms that $(\pi_1, \pi_1'), (\pi_2, \pi_2')$ extend to, then $\sigma_1(g) = \sigma_2(g)$. Thus, if $\sigma_1(g) \neq \sigma_2(g)$, then also (π_1, π_1') and (π_2, π_2') must disagree on some $s \in \mathrm{sup}(g)$. Therefore, the number of possible distinct images of g under the circuit automorphisms that the action of $\mathbf{Sym}_n \times \mathbf{Sym}_m$ extends to is at most the number of different images of $\vec{\mathrm{sup}}(g)$, where this denotes $\mathrm{sup}(g)$ written as a tuple in an arbitrary fixed order:

$$\begin{aligned}&|\{\sigma(g) \mid \sigma \text{ is the unique circuit automorphism extending a } (\pi, \pi') \in \mathbf{Sym}_n \times \mathbf{Sym}_m\}| \\ \leq &|\{(\pi, \pi')(\vec{\mathrm{sup}}(g)) \mid (\pi, \pi') \in \mathbf{Sym}_n \times \mathbf{Sym}_m\}| \leq (n+m)^k.\end{aligned}$$

The last inequality is due to the fact that $|\mathrm{sup}(g)| \leq k$, so the number of different k-tuples in $[n] \uplus [m]$ is at most $(n+m)^k$. Since C does not have other circuit automorphisms due to its rigidity, this is indeed an upper bound on the orbit size of the gate. □

If we are considering a sequence $(C_{n,m})_{n,m\in\mathbb{N}}$ of $\Gamma_{n,m}$-symmetric circuits, then we say they have *polynomially bounded orbit size* if there is a polynomial p such that asymptotically, $\mathrm{maxOrb}(C_{n,m}) \leq p(n+m)$. Note that the overall size $\|C_{n,m}\|$ may grow super-polynomially even if the orbit size is polynomially bounded. In that case, symmetry itself is not the reason why the circuits are large: They simply contain a super-polynomial number of distinct orbits.

Relationship between supports of children and parent gates The children of a gate g can be partitioned into orbits with respect to the action of $\mathbf{Stab}(g)$ because any permutation that induces a circuit automorphism fixing g must also fix the children of g setwise. We can also consider the orbit partition of children of g with respect to $\mathbf{Stab}^\bullet(\mathrm{sup}(g)) \leq \mathbf{Stab}(g)$, which may be finer.

Lemma 2.6. *Let C be a $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuit such that $n, m \geq 3 \cdot \mathrm{maxSup}(C)$. Let $g \in V(C)$ an internal gate, h a child of g. Let $O_h \coloneqq \mathbf{Orb}_{\mathbf{Stab}^\bullet(\mathrm{sup}(g))}(h)$ and $S(h) \coloneqq \bigcap_{h' \in O_h} \mathrm{sup}(h')$. Then $S(h) \subseteq \mathrm{sup}(g)$, and for every $h' \in O_h$, $(\mathrm{sup}(h') \setminus S(h)) \cap \mathrm{sup}(g) = \emptyset$.*

Proof. For the first part, let $s \in S(h)$, so in particular, $s \in \mathrm{sup}(h)$. Assume for a contradiction that $s \notin \mathrm{sup}(g)$. Then let $(\pi, \pi') \in \mathbf{Sym}_n \times \mathbf{Sym}_m$ be such that $(\pi, \pi')(\mathrm{sup}(h) \setminus \mathrm{sup}(g))$ does not contain s, and (π, π') fixes $\mathrm{sup}(g)$ pointwise. Such a permutation exists because we are assuming n and m to be large enough compared to the support sizes. Then $h' \coloneqq (\pi, \pi')(h) \in O_h$ but $s \notin \mathrm{sup}(h')$, so $s \notin S(h)$. Contradiction. For the second part, assume for a contradiction that there exists $s \in (\mathrm{sup}(h') \setminus S(h)) \cap \mathrm{sup}(g)$, for some $h' \in O_h$. Since $s \in \mathrm{sup}(g)$, every permutation in $\mathbf{Stab}^\bullet(\mathrm{sup}(g))$ fixes s. So s must then be in $\mathrm{sup}(h')$ for every $h' \in O_h$. So it is in $S(h)$. Contradiction. □

2.3 Graphs and Homomorphisms

A *simple graph* is a tuple $G = (V(G), E(G))$ of a finite set $V(G)$ and a set $E(G) \subseteq \binom{V(G)}{2}$. A *multigraph* is a tuple $G = (V(G), E(G))$ of a finite set $V(G)$ and a multiset $E(G)$ of elements in $\binom{V(G)}{2}$. When a distinction is not necessary, we call both simple graphs and multigraphs *graphs*. For a graph G, write $\|G\| \coloneqq |V(G)| + |E(G)|$.

A *tree decomposition* of a graph F is a pair (T, β) of a tree T and a map $\beta \colon V(T) \to 2^{V(F)}$ such that the following conditions hold:

1. $\bigcup_{t \in V(T)} \beta(t) = V(F)$,
2. for every edge $e \in E(F)$, there exists a node $t \in V(T)$ such that $e \subseteq \beta(t)$,
3. for evert vertex $v \in V(F)$, the subgraph of T induced by the vertices $t \in V(T)$ such that $v \in \beta(t)$ is connected.

The *width* of a tree decomposition (T, β) is $\max_{v \in V(T)} |\beta(v)| - 1$. The *treewidth* $\operatorname{tw}(F)$ of a graph F is the minimum width of a tree decomposition of F.

Let F and G be graphs. A *homomorphism* $h \colon F \to G$ is a map $h \colon V(F) \to V(G)$ such that $h(uv) \in E(G)$ for all $uv \in E(F)$. We write $\hom(F, G)$ for the number of homomorphisms from F to G.

Homomorphism counts behave nicely with respect to graph operations. For two graphs F_1, F_2, write $F_1 + F_2$ for their *disjoint union*, i.e. $V(F_1 + F_2) \coloneqq V(F_1) \uplus V(F_2)$ and $E(F_1 + F_2) \coloneqq E(F_1) \uplus E(F_2)$. For multiple graphs $F_1, \dots, F_n$, we write $\coprod_{i=1}^{n} F_i$ for $F_1 + \cdots + F_n$. For two graphs G_1, G_2, write $G_1 \times G_2$ for their *categorical product*, i.e. $V(G_1 \times G_2) \coloneqq V(G_1) \times V(G_2)$ and (v_1, v_2) and (w_1, w_2) are adjacent in $G_1 \times G_2$ if, and only if, $v_1 w_1 \in E(G_1)$ and $v_2 w_2 \in E(G_2)$. The following identities hold for all graphs F_1, F_2, G_1, G_2, and all connected graphs K, cf. e.g. [41, (5.28)–(5.30)]:

$$\hom(F_1 + F_2, G) = \hom(F_1, G) \hom(F_2, G), \tag{1}$$

$$\hom(F, G_1 \times G_2) = \hom(F, G_1) \hom(F, G_2), \text{ and} \tag{2}$$

$$\hom(K, G_1 + G_2) = \hom(K, G_1) + \hom(K, G_2). \tag{3}$$

For graphs G and H, write $G \boxplus H$ for the graph with vertex set $V(G \boxplus H) \coloneqq V(G) \uplus V(H)$ and $E(G \boxplus H) \coloneqq E(G) \uplus E(H) \uplus \{vw \mid v \in V(G), w \in V(H)\}$. For a graph F and $U \subseteq V(F)$, write $F[U]$ for the *subgraph induced by* U and $F - U$ for the subgraph obtained by deleting U, i.e. $V(F[U]) \coloneqq U$, $E(F[U]) \coloneqq E(F) \cap \binom{U}{2}$, and $F - U \coloneqq F[V(F) \setminus U]$. For every graph F, it holds that

$$\hom(F, G \boxplus H) = \sum_{U \subseteq V(F)} \hom(F[U], G) \hom(F - U, H). \tag{4}$$

For graphs G_1 and G_2, write $G_1 \cdot G_2$ for their *lexicographic product*, i.e. $V(G_1 \cdot G_2) \coloneqq V(G_1) \times V(G_2)$ and (v_1, v_2) and (w_1, w_2) are adjacent in $G_1 \cdot G_2$ if, and only if, $v_1 = w_1$ and $v_2 w_2 \in E(G_2)$, or $v_1 w_1 \in E(G_1)$. For a graph F and a partition $\mathcal{R}$ of $V(F)$, write $F/\mathcal{R}$ for the graph with vertex set $\mathcal{R}$ and $RS \in E(F/\mathcal{R})$ if, and only if, $R \neq S$ and there exist $r \in R$ and $s \in S$ such that $rs \in E(F)$. By [56, Theorem 16], for every graph F,

$$\hom(F, G_1 \cdot G_2) = \sum_{\mathcal{R} \in \Gamma(F)} \hom(F/\mathcal{R}, G_1) \hom(\coprod_{R \in \mathcal{R}} F[R], G_2). \tag{5}$$

where $\Gamma(F)$ denotes the set of all partitions $\mathcal{R}$ of $V(F)$ such that for all $R \in \mathcal{R}$ the graph $F[R]$ is connected.

2.4 Counting Logic and Bijective Games

Counting logic is first-order logic augmented by counting quantifiers of the form $\exists^{\geq i} x$, for every $i \in \mathbb{N}$. A graph (or another finite structure) G satisfies a formula $\exists^{\geq i} x \varphi(x)$, written as $G \models \exists^{\geq i} x \varphi(x)$, if there exist at least i many distinct $v \in V(G)$ such that $G \models \varphi(v)$. The k-variable fragment of counting logic is denoted $\mathcal{C}^k$. More details on counting logic and references may be found in [14]. For our purposes, the main interest is the equivalence relation that this induces on graphs. Two structures G, H are called *$\mathcal{C}^k$-equivalent*, denoted $G \equiv_{\mathcal{C}^k} H$, if G and H satisfy exactly the same sentences of $\mathcal{C}^k$. This family of equivalence relations has many characterisations, in combinatorics, logic, algebra and linear optimisation. In particular, it is known that $G \equiv_{\mathcal{C}^{k+1}} H$ if, and only if, G and H are not distinguished by the k-dimensional Weisfeiler-Leman test for graph isomorphism (see [6]).

For our purposes, a particularly useful characterisation is in terms of *k-pebble bijective games*, introduced by Hella [34]. This is a game played by two players called *Spoiler* and *Duplicator* on the graphs G and H

using k pairs of pebbles $(a_1, b_1), \ldots, (a_k, b_k)$. In a game position, some (or all) of the pebbles $a_1, \ldots, a_k$ are placed on vertices of G while the matching pebbles among $b_1, \ldots, b_k$ are placed on vertices of H. Where it causes no confusion, we do not distinguish notationally between the pebble a_i (or b_i) and the vertex on which it is placed. In each move of the game, Spoiler chooses a pair of pebbles (a_i, b_i) and Duplicator has to respond by giving a bijection $f : V(G) \to V(H)$ which agrees with the map $a_j \mapsto b_j$ for all $j \neq i$. Spoiler chooses $a \in V(G)$ and the pebbles a_i and b_i are placed on a and $f(a)$ respectively. If the resulting partial map from G to H given by $a_i \mapsto b_i$ is not a partial isomorphism, then Spoiler has won the game. We say that Duplicator has a *winning strategy* if, no matter how Spoiler plays, she can play forever without losing. It is known that $G \equiv_{\mathcal{C}^k} H$ if, and only if, Duplicator has a winning strategy in the k-pebble bijective game played on G and H.

'An optimal lower bound on the number of variables for graph identification' [6] showed that for each k there are graphs G and H with $O(k)$ vertices which are not isomorphic but such that $G \equiv_{\mathcal{C}^k} H$. The graphs G and H are constructed from a base graph F which is sufficiently richly connected (in particular it has treewidth larger than k) by replacing each vertex v of F by a small gadget whose size depends on the degree of v. The construction is known as the CFI construction, and the gadgets as the CFI gadgets. There are many variations of the construction in the literature, and we rely on the construction from [49], known as the CFI construction without internal vertices (see also [51]). Details can be found in Appendix C.

3 Overview of results

We start with a summary of our results and proof techniques. Full technical details of all proofs are presented in Sections 4–8.

3.1 Characterisation of symmetric circuit complexity by homomorphism polynomials

We start by observing that any symmetric polynomial is expressible as a linear combination of *homomorphism* or *subgraph polynomials*. Let F be a bipartite multigraph with bipartition $A \uplus B$. For integers $n, m \in \mathbb{N}$, the *homomorphism polynomial* $\hom_{F,n,m} \in \mathbb{Q}[\mathcal{X}_{n,m}]$ and the *subgraph polynomial* $\mathrm{sub}_{F,n,m} \in \mathbb{Q}[\mathcal{X}_{n,m}]$ of F are defined as

$$\hom_{F,n,m} \coloneqq \sum_{h\colon A\uplus B \to [n]\uplus[m]} \prod_{ab\in E(F)} x_{h(a)h(b)}, \qquad \mathrm{sub}_{F,n,m} \coloneqq \frac{1}{|\mathbf{Aut}(F)|} \sum_{h\colon A\uplus B \hookrightarrow [n]\uplus[m]} \prod_{ab\in E(F)} x_{h(a)h(b)}.$$

Here, we think of h as mapping A to $[n]$ and B to $[m]$. For example, $\hom_{K_2,n,m} = \sum_{v\in[n]} \sum_{w\in[m]} x_{vw}$ and $\mathrm{sub}_{K_{n,m},n,m} = \prod_{v\in[n]} \prod_{w\in[m]} x_{vw}$. If $A_G \in \{0,1\}^{[n]\times[m]}$ is the bi-adjacency matrix of an undirected bipartite graph G with bipartition $V(G) = [n] \uplus [m]$, then $\hom_{F,n,m}(A_G)$ evaluates to the number of bipartition-respecting homomorphisms from F to G, and $\mathrm{sub}_{F,n,m}(A_G)$ is the number of occurrences of F as a subgraph in G. We also write $\hom_{F,n,m}(G)$ and $\mathrm{sub}_{F,n,m}(G)$ for the polynomials evaluated in the bi-adjacency matrix of G. Let $\mathfrak{G}_{n,m} \coloneqq \{\sum \alpha_i \hom_{F_i,n,m} \mid \text{bipartite multigraphs } F_i, \alpha_i \in \mathbb{Q}\} \subseteq \mathbb{Q}[\mathcal{X}_{n,m}]$. Then we have:

Lemma 3.1. *For $n, m \in \mathbb{N}$ and $p \in \mathbb{Q}[\mathcal{X}_{n,m}]$, the following are equivalent:*

1. *the polynomial p is $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric,*
2. *$p \in \mathfrak{G}_{n,m}$,*
3. *$p = \sum \alpha_i \mathrm{sub}_{F_i,n,m}$ for some bipartite multigraphs F_i and $\alpha_i \in \mathbb{Q}$.*

For the proof, Item 3 can be concluded from Item 1 by partitioning the monomials in p into their orbits, and showing that each of these orbits is a subgraph polynomial for some pattern F. Via Möbius inversion (Corollary A.3), subgraph polynomials can be expressed as linear combinations of homomorphism polynomials.

Thus, both subgraph and homomorphism polynomials can be seen as spanning sets for the vector space of matrix-symmetric polynomials. We are now interested in the ones that admit polynomial size symmetric circuits. For $k, n, m \in \mathbb{N}$, write

$$\mathfrak{T}^k_{n,m} \coloneqq \left\{ \sum \alpha_i \hom_{F_i,n,m} \;\middle|\; \text{bipartite multigraphs } F_i \text{ such that } \mathrm{tw}(F_i) < k, \alpha_i \in \mathbb{Q} \right\} \subseteq \mathfrak{G}_{n,m}.$$

for the set of finite $\mathbb{Q}$-linear combinations of homomorphism polynomials of bipartite multigraphs of treewidth less than k.

Our main result, Theorem 1.1, states that the families of polynomials in $\mathfrak{T}^k_{n,m}$, for constant k, are precisely the ones that admit symmetric circuits of polynomial orbit size. The full proof can be found in Section 4 but the idea is the following:

Proof sketch of Theorem 1.1. The easier direction is to show that if $p_{n,m} \in \mathfrak{T}^k_{n,m}$, then it has circuits of polynomial orbit size. It turns out that each $\hom_{F,n,m}$ can be represented by a symmetric circuit with support size at most $\mathrm{tw}(F) < k$. The circuit computes $\hom_{F,n,m}$ inductively along a tree decomposition of F, using ideas as in e.g. [26]. Then any polynomial in $\mathfrak{T}^k_{n,m}$ is just a linear combination of such circuits. By Lemma 2.5, a circuit with support size $\leq k$ has orbit size at most $(n+m)^k$, as desired. Note that a polynomial in $\mathfrak{T}^k_{n,m}$ may be a linear combination involving a super-polynomial number of terms. Then only the orbit size of the circuit will be polynomial, but the total number of gates will not be.

The other direction of Theorem 1.1 requires substantial technical effort. It is proved using so-called *labelled* homomorphism polynomials: Imagine that a pattern F has k distinctly labelled vertices $a_1, \ldots, a_k$. Then for any k-tuple of vertices $b_1, \ldots, b_k$ in the host graph G, we can ask how many homomorphisms h exist from F to G such that $h(a_i) = b_i$, for each $i \in [k]$. A labelled homomorphism polynomial is one that counts only these label-preserving homomorphisms.

We want to prove that every polynomial represented by a symmetric circuit of polynomial orbit size is in $\mathfrak{T}^k_{n,m}$, for some $k \in \mathbb{N}$. Lemma 2.4 tells us that the support size of every gate in such a circuit must be bounded by some constant k. We prove by induction on the circuit structure that at every gate g in the circuit, the polynomial computed at g is a linear combination of *labelled* homomorphism polynomials of patterns of treewidth $\leq k$. The crucial insight is that the supports of the gates always correspond exactly to the images of the labels. That is, semantically, the gate g counts homomorphisms that map the labels $a_1, \ldots, a_k$ of the pattern graphs to the vertices $b_1, \ldots, b_k$ enumerating $\sup(g)$. The root gate of a symmetric circuit always has empty support (as it is fixed by every permutation in $\mathbf{Sym}_n \times \mathbf{Sym}_m$), and so, the polynomial computed at the root is a linear combination of homomorphism polynomials for patterns F with $\mathrm{tw}(F) \leq k$ and without labels. Thus, the circuit computes a polynomial in $\mathfrak{T}^k_{n,m}$. □

We also prove in Section 4 that under the common assumption $\mathsf{VP} \neq \mathsf{VNP}$, Theorem 1.1 is best-possible in the sense that the characterisation cannot be improved to capture total circuit size rather than orbit size: There exist polynomials in $\mathfrak{T}^k_{n,m}$, for constant k, that do not admit polynomial size circuits (neither symmetric nor asymmetric), unless $\mathsf{VP} = \mathsf{VNP}$.

Theorem 3.2. *There is a* VNP*-hard family of polynomials* $(p_n)_{n\in\mathbb{N}}$ *such that* $p_n \in \mathfrak{T}^2_{n,n}$ *for all* $n \in \mathbb{N}$.

3.2 Symmetric complexity of homomorphism polynomials

In order to make the characterisation in Theorem 1.1 more usable, we study $\mathfrak{T}^k_{n,m}$ in relation with the class of polynomials of bounded counting width.

Definition 3.3 ([1])**.** Let p be a graph parameter on (n,m)-vertex bipartite graphs. The *counting width of* p *on simple graphs* is the least integer k such that, for all (n,m)-vertex simple bipartite graphs G and H, if G and H are $\mathcal{C}^k$-equivalent, then $p(G) = p(H)$.

It follows with known arguments that whenever a family $(p_{n,m})_{n,m\in\mathbb{N}}$ admits circuits of polynomial orbit size, then its counting width is bounded (see Theorem 5.2). We ask in which cases the converse is true. As a start, we identify where the difficulty lies in this question. Namely, if we only care about the semantics of the polynomials on simple graphs, that is $\{0,1\}$-valued rather than $\mathbb{Q}$-valued matrices, then we do have a conclusive answer. On $\{0,1\}$-assignments, any polynomial p is equivalent to its *multilinearisation*. This is the polynomial obtained from p by replacing every exponent greater than 1 with 1. Up to multilinearisation, bounded counting width is indeed sufficient for the existence of polynomial orbit size symmetric circuits:

Theorem 3.4. *Let* $k, n, m \in \mathbb{N}$*. For* $p \in \mathbb{Q}[\mathcal{X}_{n,m}]$*, the following are equivalent:*

1. *there exists* $q \in \mathfrak{T}^k_{n,m}$ *such that* p *and* q *are equal up to multilinearisation,*
2. *the counting width of* p *on* (n,m)*-vertex simple bipartite graphs is at most* k*.*

Thus, if we only care about evaluating the polynomial p on $\{0,1\}$ (equivalently, we regard it as a parameter on simple, unweighted, bipartite graphs) the symmetric tractability question is completely settled. But this does not settle the general case , where we wish to consider it on all $\mathbb{Q}$-valued assignments (see, e.g. Example 7.10). However, we can characterise it in more specific cases. In particular, we obtain a precise characterisation of the bounded counting width case for linear combinations of homomorphism polynomials with sublinear-size patterns:

Theorem 3.5 (see Theorem 6.9)**.** *For $n, m \in \mathbb{N}$, let $F_{n,m,i}$ be bipartite multigraphs and $\alpha_{n,m,i} \in \mathbb{Q} \setminus \{0\}$ for $i \in [s_{n,m}]$ for some $s_{n,m} \in \mathbb{N}$. Let $p_{n,m} \coloneqq \sum_i \alpha_{n,m,i} \hom_{F_{n,m,i},n,m}$. Suppose that, for all $\nu, \mu \in \mathbb{N}$, there are $n', m' \in \mathbb{N}$ such that $\max_i |F_{\nu n,\mu m,i}| \leq \min\{n, m\}$ for all $n > n'$ and $m > m'$. Then the following are equivalent:*

1. *$\max_i \operatorname{tw}(F_{n,m,i})$ is bounded,*
2. *the counting width of $p_{n,m}$ on (n, m)-vertex simple graphs is bounded,*
3. *the $p_{n,m}$ admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of orbit size polynomial in $n + m$.*

If the linear expansions of the $p_{n,m}$ are comprised of polynomially many homomorphism polynomials, then the above conditions are equivalent to the $p_{n,m}$ admitting $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of total size polynomial in $n + m$. In the case where $p_{n,m} = \hom_{F_{n,m},n,m}$ for all $n, m \in \mathbb{N}$, i.e. we just have a single homomorphism polynomial for each n, m, we obtain the above result also if $|F_{n,m}|$ is not in $o(n + m)$, see Theorem 6.1.

We prove Theorem 6.9 by generalising notions introduced by *Oddomorphisms and homomorphism indistinguishability over graphs of bounded degree* [49] for describing the distinguishing power of graph isomorphism relaxations via homomorphism counts. By [23, 26], two graphs G and H are $\mathcal{C}^k$-equivalent if, and only if, they are *homomorphism indistinguishable* over the graphs of treewidth less than k, i.e. $\hom(F, G) = \hom(F, H)$ for all F such that $\operatorname{tw}(F) < k$. Hence, a graph parameter p has bounded counting width if, and only if, it is determined by homomorphism counts from bounded-treewidth graphs. For uniform graph motif parameters $\sum \alpha_F \hom(F, -)$, it was known [40, 45, 56] that this happens if and only if $\max_F \operatorname{tw}(F)$ is bounded. We generalise this result to non-uniform graph motif parameters, whose patterns and coefficients may depend on the size of the host graph.

3.3 Symmetric complexity of subgraph polynomials

By Lemma 3.1, linear combinations of subgraph and homomorphism polynomials can be transformed into one another. Nevertheless, we also study subgraph polynomials on their own because it seems likely that in this case we find another natural parameter of the pattern graphs that fully controls the counting width and symmetric complexity. Indeed, we show that for patterns of sublinear size, this parameter is $\operatorname{vc}(F)$, the *vertex cover number*.

Theorem 3.6 (see Theorem 7.8)**.** *Let $(F_{n,m})_{n,m\in\mathbb{N}}$ be a family of simple bipartite graphs such that, for all $\nu, \mu \in \mathbb{N}$, there exist $n', m' \in \mathbb{N}$, such that $|F_{\nu n,\mu m}| \leq \min\{n, m\}$ for all $n > n'$ and $m > m'$. The following are equivalent:*

1. *$\operatorname{vc}(F_{n,m})$ is bounded,*
2. *the counting width of $\operatorname{sub}_{F_{n,m},n,m}$ on (n, m)-vertex simple graphs is bounded,*
3. *the $\operatorname{sub}_{F_{n,m},n,m}$ admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of orbit size polynomial in $n + m$,*
4. *the $\operatorname{sub}_{F_{n,m},n,m}$ admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of size polynomial in $n + m$.*

The size bound in this theorem is necessary as it is easy to find examples of subgraph polynomials (of size $\Theta(n + m)$) whose complexity is not governed by the pattern's vertex cover number. Consider e.g. $p_{n,m} = \operatorname{sub}_{K_{n,m},n,m} = \prod_{v\in[n]} \prod_{w\in[m]} x_{vw}$. The natural symmetric circuit for this linear-size pattern has just one product gate that ranges over all variables, so this family $(p_{n,m})_{n,m\in\mathbb{N}}$ has low symmetric circuit complexity (see also Example 7.2). But, $\operatorname{vc}(K_{n,m})$ is clearly unbounded.

More generally, for any function $f\colon \mathbb{N} \to \mathbb{N}$ with $f(n) \leq n$, and the complete bipartite graphs $K_{f(n),f(n)}$ as patterns, we show that the only tractable cases for $\operatorname{sub}_{K_{f(n),f(n)},n,n}$ are when $\min\{f(n), n - f(n)\}$ is constant (Theorem 7.3). This suggests that it might be the minimum vertex cover number of the pattern graph and its complement that determines the symmetric complexity of $\operatorname{sub}_{F_{n,m},n,m}$. We call this parameter the *logical vertex cover number* $\operatorname{lvc}_{n,m}(F_{n,m})$. In particular, we can show for all F that the counting width of $\operatorname{sub}_{F,n,m}$ and $\operatorname{sub}_{\overline{F},n,m}$ is the same (Theorem 7.9), giving further evidence for this hypothesis, which is formally stated as Conjecture 7.7.

3.4 Symmetric complexity of the immanants

So far, we have studied $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric families. There are also polynomials in the variables $\mathcal{X}_{n,n}$ which are symmetric under the simultaneous action of $\mathbf{Sym}_n$ on rows and columns but not under $\mathbf{Sym}_n \times \mathbf{Sym}_n$. One such example is the *determinant*, and more generally, other *immanants*. We have chosen to develop our theory for $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric families but we believe that most of it can be adapted

to the $\mathbf{Sym}_n$-symmetric case. Then, the relevant homomorphism or subgraph polynomials involve *directed* and not necessarily bipartite graphs. Rather than doing that here, we consider the immanants as a concrete $\mathbf{Sym}_n$-symmetric example and show that their symmetric circuit complexity is subject to a dichotomy similar to what is known for their computational complexity [7].

Immanants are families of $\mathbf{Sym}_n$-symmetric polynomials $(p_n)_{n\in\mathbb{N}}$ that are defined via so-called *irreducible characters* of the symmetric group. An irreducible character of $\mathbf{Sym}_n$ is a homomorphism f from $\mathbf{Sym}_n$ to the multiplicative group $\mathbb{C}\setminus\{0\}$ which is constant on the conjugacy classes of $\mathbf{Sym}_n$. Given such an f, the corresponding immanant is defined as $\operatorname{imm}_f = \sum_{\pi\in\mathbf{Sym}_n} f(\pi)\prod_{i\in[n]} x_{i,\pi(i)}$. If f is constantly 1, then imm_f is the permanent. The determinant is obtained by letting $f = \operatorname{sgn}$. While all immanents are $\mathbf{Sym}_n$-symmetric, the symmetry groups of particular members of this family may be larger. In particular, the permanent is, in fact, $\mathbf{Sym}_n \times \mathbf{Sym}_n$ while the determinant is symmetric under a subgroup of this of index 2 consisting of those pairs of permutations (π,σ) for which $\operatorname{sgn}(\pi) = \operatorname{sgn}(\sigma)$. The $\mathbf{Sym}_n$ symmetric circuit complexity of the permanent and the determinant was studied in [21] and for the larger symmetry groups in [19].

In [7], Curticapean shows a complexity dichotomy for the immanants: In the tractable case, imm_f is in VP (i.e. admits polynomial size algebraic circuits) and is computable in polynomial time. In the intractable case, it is not in VP, unless $\mathsf{VFPT} = \mathsf{VW}[1]$, and also, it is not computable in polynomial time, unless $\mathsf{FPT} = \#\mathsf{W}[1]$. The complexity is controlled by a certain parameter of f.

We show that the same parameter of f produces the analogous dichotomy with regards to the complexity of $\mathbf{Sym}_n$-symmetric algebraic circuits for the immanants, but without the need for complexity-theoretic assumptions for the hardness part. This generalizes the resuls of [21]. The irreducible characters f correspond naturally to partitions of $[n]$ (see, for example, [53, Prop. 1.10.1]). If λ is a partition of $[n]$, we denote by $\operatorname{imm}_\lambda$ the immanant defined by the irreducible character corresponding to λ. Let $b(\lambda) \coloneqq n - s$, where s denotes the number of parts and n the size of the set that is partitioned by λ. This is the parameter that controls the complexity of $\operatorname{imm}_\lambda$.

A family Λ of partitions is said to *support growth* $g\colon \mathbb{N}\to\mathbb{N}$ if for every $n\in\mathbb{N}$ there is a partition $\lambda^{(n)}$ in Λ with $b(\lambda^{(n)}) \geq g(n)$ and size $\Theta(n)$. According to the dichotomy from [7], $(\operatorname{imm}_\lambda)_{\lambda\in\Lambda}$ is tractable if there is a constant that bounds $b(\lambda)$ for all $\lambda\in\Lambda$. Otherwise, if Λ supports growth $\omega(1)$, then the immanant family is (conditionally) intractable. If Λ even supports growth $\omega(n^k)$, then $(\operatorname{imm}_\lambda)_{\lambda\in\Lambda}$ is VNP-complete. Our analogous result for symmetric circuits is the following.

Theorem 3.7. *Let Λ be a family of partitions.*

1. *If there exists a $k\in\mathbb{N}$ such that $b(\lambda)\leq k$ for all $\lambda\in\Lambda$, then the $(\operatorname{imm}_\lambda)_{\lambda\in\Lambda}$ admit $\mathbf{Sym}_n$-symmetric circuits of polynomial size in n.*
2. *If Λ supports growth $g\in\omega(1)$, then the counting width of $(\operatorname{imm}_\lambda)_{\lambda\in\Lambda}$ on directed $\mathbb{Q}$-edge-weighted n-vertex graphs is unbounded, where n is the integer that is partitioned by the respective λ. Thus, all $\mathbf{Sym}_n$-symmetric circuits for the immanant family have super-polynomial size in n.*
3. *If Λ supports growth $g\in\omega(n^k)$ for a constant k, then the counting width of $(\operatorname{imm}_\lambda)_{\lambda\in\Lambda}$ on directed $\mathbb{Q}$-edge-weighted n-vertex graphs is $\Omega(n)$, so all $\mathbf{Sym}_n$-symmetric circuits have size $2^{\Omega(n)}$.*

4 Circuits with Bounded Support and Homomorphism Polynomials

In this section, we prove Theorem 1.1 which characterises the polynomials which admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of polynomial orbit size. To that end, we first observe that the polynomials which admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of arbitrary orbit size are precisely the linear combinations of subgraph or homomorphism polynomials.

Let F be a bipartite multigraph with bipartition $A\uplus B$. For integers $n,m\in\mathbb{N}$, the *homomorphism polynomial* $\hom_{F,n,m}\in\mathbb{Q}[\mathcal{X}_{n,m}]$ and the *subgraph polynomial* $\operatorname{sub}_{F,n,m}\in\mathbb{Q}[\mathcal{X}_{n,m}]$ of F are defined as

$$\hom_{F,n,m} \coloneqq \sum_{h\colon A\uplus B\to[n]\uplus[m]} \prod_{ab\in E(F)} x_{h(a)h(b)},$$

$$\operatorname{sub}_{F,n,m} \coloneqq \frac{1}{|\mathbf{Aut}(F)|} \sum_{h\colon A\uplus B\hookrightarrow[n]\uplus[m]} \prod_{ab\in E(F)} x_{h(a)h(b)}.$$

Here, h ranges over maps taking A to $[n]$ and B to $[m]$. For example, $\hom_{K_2,n,m} = \sum_{v\in[n]}\sum_{w\in[m]} x_{vw}$ and $\operatorname{sub}_{K_{n,m},n,m} = \prod_{v\in[n]}\prod_{w\in[m]} x_{vw}$. If $A_G\in\{0,1\}^{[n]\times[m]}$ is the bi-adjacency matrix of an undirected bipartite

graph G with bipartition $V(G) = [n] \uplus [m]$, then $\hom_{F,n,m}(A_G)$ evaluates to the number of bipartition-respecting homomorphisms from F to G, and $\mathrm{sub}_{F,n,m}(A_G)$ is the number of occurrences of F as a subgraph in G. We also write $\hom_{F,n,m}(G)$ and $\mathrm{sub}_{F,n,m}(G)$ for the polynomials evaluated in the bi-adjacency matrix of G.

For every multigraph F, the coefficients of all monomials in $\mathrm{sub}_{F,n,m}$ are 1. If F is a simple graph, then $\mathrm{sub}_{F,n,m}$ is multilinear. This is generally not the case for $\hom_{F,n,m}$, not even when F is simple.

As a first step, we show that the set

$$\mathfrak{G}_{n,m} \coloneqq \left\{ \sum \alpha_i \hom_{F_i,n,m} \mid \text{bipartite multigraphs } F_i, \alpha_i \in \mathbb{Q} \right\} \subseteq \mathbb{Q}[\mathcal{X}_{n,m}].$$

contains precisely the $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric polynomials, and we also get the same collection if we take linear combinations of subgraph polynomials instead.

Lemma 3.1. *For $n, m \in \mathbb{N}$ and $p \in \mathbb{Q}[\mathcal{X}_{n,m}]$, the following are equivalent:*

1. *the polynomial p is $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric,*
2. *$p \in \mathfrak{G}_{n,m}$,*
3. *$p = \sum \alpha_i \mathrm{sub}_{F_i,n,m}$ for some bipartite multigraphs F_i and $\alpha_i \in \mathbb{Q}$.*

Proof. For $1 \Rightarrow 3$, partition the monomial set of p into its $\mathbf{Sym}_n \times \mathbf{Sym}_m$-orbits. We aim to show that each orbit is the polynomial $\mathrm{sub}_{F,n,m}$, for some F, multiplied with some non-zero constant. Let Ω be such an orbit, i.e. a set of monomials. Let $q \in \Omega$ denote an arbitrary representative. It describes a bipartite graph F with bipartition $V(F) = \{i \in [n] \mid \text{ there exists } j \text{ s.t. } x_{ij} \text{ divides } q\} \uplus \{j \in [m] \mid \text{ there exists } i \text{ s.t. } x_{ij} \text{ divides } q\}$. An edge ij appears in F with the multiplicity given by the degree of x_{ij} in q.

Now q encodes the identity embedding $\iota_{\mathrm{id}} \colon V \hookrightarrow [n] \uplus [m]$. For every $(\pi, \sigma) \in \mathbf{Sym}_n \times \mathbf{Sym}_m$, we have $(\pi, \sigma)(q) = \prod_{vw \in E(F)} x_{\pi(\iota_{\mathrm{id}}(v))\sigma(\iota_{\mathrm{id}}(w))}$. Each embedding $\iota \colon V \hookrightarrow [n] \uplus [m]$ can be written as $(\pi, \sigma) \circ \iota_{\mathrm{id}}$ for some $(\pi, \sigma) \in \mathbf{Sym}_n \times \mathbf{Sym}_m$. Therefore, the orbit Ω is indeed $\mathrm{sub}_{F,n,m}$, multiplied with some constant.

The implication $3 \Rightarrow 2$ follows from Corollary A.3. For $2 \Rightarrow 1$, let F be a bipartite multigraph and $(\pi, \sigma) \in \mathbf{Sym}_n \times \mathbf{Sym}_m$. Then

$$(\pi, \sigma)(\hom_{F,n,m}) = \sum_{h \colon A \uplus B \to [n] \uplus [m]} \prod_{ab \in E(F)} x_{\pi(h(a))\sigma(h(b))} = \sum_{h \colon A \uplus B \to [n] \uplus [m]} \prod_{ab \in E(F)} x_{h(a))h(b)} = \hom_{F,n,m}$$

observing that the set of maps $h \colon A \uplus B \to [n] \uplus [m]$ is in bijection with the set of maps $(\pi, \sigma) \circ h$ for $h \colon A \uplus B \to [n] \uplus [m]$. □

Remark 4.1. One can similarly characterise the $\mathbf{Sym}_n$-symmetric polynomials in terms of homomorphism/subgraph polynomials for *directed* and not necessarily bipartite graphs. A $\mathbf{Sym}_n$-symmetric polynomial is one that is fixed by every $\pi \in \mathbf{Sym}_n$ acting on $\mathcal{X}_{n,n}$ by mapping x_{ij} to $x_{\pi(i)\pi(j)}$. In fact, we believe that all or most of the theory we develop in this paper for $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric polynomials applies to $\mathbf{Sym}_n$-symmetric ones as well, but we prove our results only for the $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric setting.

Even though the definition of $\mathfrak{G}_{n,m}$ involves homomorphism polynomials of infinitely many graphs, it follows from Lemma 3.1 that, for every $n, m \in \mathbb{N}$, finitely many such polynomials already generate $\mathfrak{G}_{n,m}$ as a $\mathbb{Q}$-algebra:

Corollary 4.2. *For every $n, m \in \mathbb{N}$, there exist finitely many bipartite multigraphs $F_1, \ldots, F_r$ such that, for every $p \in \mathfrak{G}_{n,m}$, there exists a polynomial $q \in \mathbb{Q}[y_1, \ldots, y_r]$ such that $p = q(\hom_{F_1,n,m}, \ldots, \hom_{F_r,n,m})$.*

Proof. By the invariant theory of finite groups, cf. e.g. [24, Proposition 3.0.1], the set of $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric polynomials in $\mathbb{Q}[\mathcal{X}_{n,m}]$ is finitely generated as a $\mathbb{Q}$-algebra. By Lemma 3.1, each of its generators is a finite linear combination of homomorphism polynomials from some bipartite multigraphs. Hence, the set of $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric polynomials in $\mathbb{Q}[\mathcal{X}_{n,m}]$ is equal to the $\mathbb{Q}$-algebra generated by some finite set of homomorphism polynomials of bipartite multigraphs. □

We now turn to the characterisation of symmetric polynomials admitting symmetric circuits of polynomial orbit size. To that end, for $k, n, m \in \mathbb{N}$, write

$$\mathfrak{T}^k_{n,m} \coloneqq \left\{ \sum \alpha_i \hom_{F_i,n,m} \;\middle|\; \text{bipartite multigraphs } F_i \text{ such that } \mathrm{tw}(F_i) < k, \alpha_i \in \mathbb{Q} \right\} \subseteq \mathfrak{G}_{n,m}.$$

for the set of finite $\mathbb{Q}$-linear combinations of homomorphism polynomials of bipartite multigraphs of treewidth less than k. Our main theorem is the following:

Theorem 1.1. *For every family of polynomials $p_{n,m} \in \mathbb{Q}[\mathcal{X}_{n,m}]$, the following are equivalent:*

1. *there exists a constant $k \in \mathbb{N}$ such that $p_{n,m} \in \mathfrak{T}^k_{n,m}$ for all $n, m \in \mathbb{N}$,*
2. *the $p_{n,m}$ admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of orbit size polynomial in $n + m$.*

As a first step towards Theorem 1.1, we remark that homomorphism polynomials of graphs of bounded treewidth admit polynomial-size symmetric circuits.

Theorem 4.3. *Let $k, n, m \in \mathbb{N}$. Let F be a bipartite multigraph of treewidth less than k. Then there exists a $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric rigid circuit C_F for $\hom_{F,n,m}$ of size at most $5k!^2k \cdot (n+m)^{k+1} \cdot \|F\|^2$ satisfying $\operatorname{maxSup}(C_F) \leq k$.*

Theorem 4.3 yields the forward implication in Theorem 1.1: If $p \in \mathfrak{T}^k_{n,m}$, then it is a linear combination of polynomials $\hom_{F,n,m}$, where $\operatorname{tw}(F) \leq k$ for each F in the linear combination. With Theorem 4.3, we get a rigid symmetric circuit C_F with $\operatorname{maxSup}(C_F) \leq k$ for each of these $\hom_{F,n,m}$. To obtain p, we just take a linear combination of these circuits and rigidify it again using Lemma 2.3. This yields a rigid circuit C representing p that still satisfies $\operatorname{maxSup}(C) \leq k$. By Lemma 2.5, it has orbit size at most $(n+m)^k$.

The remainder of this section is concerned with the backward implication in Theorem 1.1 and the proof of Theorem 4.3.

4.1 Homomorphism Polynomials of Labelled Graphs

The main tool for proving Theorem 1.1 are homomorphism polynomials of labelled graphs.

Definition 4.4. Let $\ell, r \in \mathbb{N}$. An *(ℓ, r)-labelled bipartite graph* $\boldsymbol{F} = (F, \boldsymbol{a}, \boldsymbol{b})$ is a tuple of a bipartite multigraph F with bipartition $A \uplus B = V(F)$, an ℓ-tuple $\boldsymbol{a} \in A^\ell$, and an r-tuple $\boldsymbol{b} \in B^r$. Write $\mathcal{G}(\ell, r)$ for the class of all (ℓ, r)-labelled bipartite graphs.

For $n, m \in \mathbb{N}$ and tuples $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$, write

$$\boldsymbol{F}_{n,m}(\boldsymbol{v}, \boldsymbol{w}) \coloneqq \sum_{\substack{h\colon A \uplus B \to [n] \uplus [m] \\ h(\boldsymbol{a}) = \boldsymbol{v} \\ h(\boldsymbol{b}) = \boldsymbol{w}}} \prod_{ab \in E(F)} x_{h(a)h(b)} \in \mathbb{Q}[\mathcal{X}_{n,m}]$$

for the *homomorphism polynomial of $\boldsymbol{F}$ at $\boldsymbol{v}, \boldsymbol{w}$*. The map $\boldsymbol{F}_{n,m}\colon [n]^\ell \times [m]^r \to \mathbb{Q}[\mathcal{X}_{n,m}]$ given by $(\boldsymbol{v}, \boldsymbol{w}) \mapsto \boldsymbol{F}_{n,m}(\boldsymbol{v}, \boldsymbol{w})$ is the *homomorphism polynomial map* of $\boldsymbol{F}$. Write

$$\mathfrak{G}_{n,m}(\ell, r) \coloneqq \left\{ \sum \alpha_i \boldsymbol{F}^i_{n,m} \;\middle|\; \alpha_i \in \mathbb{Q}, \boldsymbol{F}^i \in \mathcal{G}(\ell, r) \right\} \subseteq \mathbb{Q}[\mathcal{X}_{n,m}]^{[n]^\ell \times [m]^r}$$

for the set of finite $\mathbb{Q}$-linear combinations of homomorphism polynomial maps of (ℓ, r)-labelled bipartite graphs.

Note that for $\ell = 0 = r$, the homomorphism polynomial map $\boldsymbol{F}_{n,m}$ simplifies to the homomorphism polynomial $\hom_{F,n,m}$, i.e. essentially $\mathfrak{G}_{n,m}(0,0) = \mathfrak{G}_{n,m}$.

Example 4.5. Let $\boldsymbol{F} = (F, a, b)$ denote the $(1,1)$-labelled bipartite graph with $V(F) = \{a, b\}$ and $E(F) = \{ab\}$. For $n, m \in \mathbb{N}$, $v \in [n]$, and $w \in [m]$, we have $\boldsymbol{F}_{n,m}(v, w) = x_{vw} \in \mathbb{Q}[\mathcal{X}_{n,m}]$.

Next, we define what it means for a labelled bipartite graph to have bounded treewidth. Furthermore, we generalise $\mathfrak{T}^k_{n,m}$.

Definition 4.6. Let $\ell, r, k \in \mathbb{N}$. An *$(\ell, r)$-labelled bipartite graph* $\boldsymbol{F} = (F, \boldsymbol{a}, \boldsymbol{b})$ has *treewidth* less than k if there exists a tree decomposition (T, β) of F of width less than k with a vertex $s \in V(T)$ such that $a_1, \ldots, a_\ell, b_1, \ldots, b_r \in \beta(s)$. Write $\mathcal{T}^k(\ell, r) \subseteq \mathcal{G}(\ell, r)$ for the class of all (ℓ, r)-labelled bipartite graphs of treewidth less than k. For $n, m \in \mathbb{N}$, write

$$\mathfrak{T}^k_{n,m}(\ell, r) \coloneqq \left\{ \sum \alpha_i \boldsymbol{F}^i_{n,m} \;\middle|\; \alpha_i \in \mathbb{Q}, \boldsymbol{F}^i \in \mathcal{T}^k(\ell, r) \right\} \subseteq \mathfrak{G}_{n,m}(\ell, r)$$

for the set of $\mathbb{Q}$-linear combinations of homomorphism polynomial maps of (ℓ, r)-labelled bipartite graphs of treewidth less than k.

To reiterate, the k in $\mathfrak{T}^k_{n,m}(\ell,r)$ indicates the maximal admissible size of a bag in a tree decomposition rather than the treewidth, which is defined as the maximal bag size minus one. This convention simplifies subsequent arguments.

Note that essentially $\mathfrak{T}^k_{n,m} = \mathfrak{T}^k_{n,m}(0,0)$. For $\boldsymbol{v} \in [n]^\ell, \boldsymbol{w} \in [m]^r$, we also define the set of *instantiated homomorphism polynomial maps* as $\mathfrak{T}^k_{n,m}(\boldsymbol{v},\boldsymbol{w}) \coloneqq \left\{\varphi(\boldsymbol{v},\boldsymbol{w}) \;\middle|\; \varphi \in \mathfrak{T}^k_{n,m}(\ell,r)\right\} \subseteq \mathbb{Q}[\mathcal{X}_{n,m}]$.

Example 4.7. For $\ell, r \in \mathbb{N}$, let $\boldsymbol{J} = (J, \boldsymbol{a}, \boldsymbol{b})$ be the (ℓ,r)-labelled edge-less (ℓ,r)-vertex bipartite graphs with labels residing on distinct vertices. Clearly, $\boldsymbol{J} \in \mathcal{T}^{\ell+r}(\ell,r)$. For all $n,m \in \mathbb{N}$, $\boldsymbol{v} \in [n]^\ell$, and $\boldsymbol{w} \in [m]^r$,

$$\boldsymbol{J}_{n,m}(\boldsymbol{v},\boldsymbol{w}) \coloneqq \sum_{\substack{h\colon A \uplus B \to [n] \uplus [m] \\ h(\boldsymbol{a}) = \boldsymbol{v} \\ h(\boldsymbol{b}) = \boldsymbol{w}}} 1 = 1$$

That is, $\boldsymbol{J}_{n,m} \in \mathfrak{T}^{\ell+r}_{n,m}(\ell,r)$ is the homomorphism polynomial map which maps all tuples to 1.

4.2 Operations on Labelled Graphs and Homomorphism Polynomial Maps

Fix $n,m,\ell,r,k \in \mathbb{N}$ throughout this section. In this section, we prove various closure properties of $\mathfrak{T}^k_{n,m}(\ell,r)$.

4.2.1 Swapping Left and Right Labels

We first note that the left and right labels can be interchanged. In order to simplify notation, we subsequently focus on operations involving only the left labels. The following purely syntactic lemma justifies this approach. To that end, define for $\boldsymbol{F} = (F, \boldsymbol{a}, \boldsymbol{b}) \in \mathcal{G}(\ell,r)$ the (r,ℓ)-labelled bipartite graph $\boldsymbol{F}^* \coloneqq (F^*, \boldsymbol{b}, \boldsymbol{a}) \in \mathcal{G}(r,\ell)$ where F^* is the bipartite multigraph obtained from F by interchanging the parts of the bipartition. Analogously, for $\varphi = \sum \alpha_i \boldsymbol{F}^i_{n,m} \in \mathfrak{G}_{n,m}(\ell,r)$, define $\varphi^* \coloneqq \sum \alpha_i (\boldsymbol{F}^i)^*_{m,n} \in \mathfrak{G}_{m,n}(r,\ell)$. The following is then immediate.

Lemma 4.8. *Let $\boldsymbol{F} \in \mathcal{G}(\ell,r)$ and $\varphi \in \mathfrak{G}_{n,m}(\ell,r)$.*

1. *If $\boldsymbol{F} \in \mathcal{T}^k(\ell,r)$, then $\boldsymbol{F}^* \in \mathcal{T}^k(r,\ell)$.*
2. *For $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$, it holds that $\boldsymbol{F}_{n,m}(\boldsymbol{v},\boldsymbol{w}) = \boldsymbol{F}^*_{m,n}(\boldsymbol{w},\boldsymbol{v})$.*
3. *If $\varphi \in \mathfrak{T}^k_{n,m}(\ell,r)$, then $\varphi^* \in \mathfrak{T}^k_{m,n}(r,\ell)$.*

4.2.2 Unlabelling and Sums

Next, we consider the unlabelling operation. For $\ell \geq 1$, $\boldsymbol{F} = (F, \boldsymbol{a}, \boldsymbol{b}) \in \mathcal{G}(\ell,r)$, and $i \in [\ell]$, define $\Sigma_i \boldsymbol{F} \coloneqq (F, \boldsymbol{a}[i/], \boldsymbol{b}) \in \mathcal{G}(\ell-1,r)$ as the graph obtained by dropping the i-th left label. Analogously, for $\varphi \in \mathfrak{G}_{n,m}(\ell,r)$, define $\Sigma_i \varphi \in \mathfrak{G}_{n,m}(\ell-1,r)$ via $(\Sigma_i\varphi)(\boldsymbol{v}[i/], \boldsymbol{w}) \coloneqq \sum_{v \in [n]} \varphi(\boldsymbol{v}[i/v], \boldsymbol{w})$ for all $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$.

Lemma 4.9. *For $\ell \geq 1$, let $\boldsymbol{F} \in \mathcal{G}(\ell,r)$ and $\varphi \in \mathfrak{G}_{n,m}(\ell,r)$.*

1. *If $\boldsymbol{F} \in \mathcal{T}^k(\ell,r)$, then $\Sigma_i \boldsymbol{F} \in \mathcal{T}^k(\ell-1,r)$.*
2. *For $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$, it is*

$$(\Sigma_i \boldsymbol{F})_{n,m}(\boldsymbol{v}[i/], \boldsymbol{w}) = \sum_{v \in [n]} \boldsymbol{F}_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w}).$$

3. *If $\varphi \in \mathfrak{T}^k_{n,m}(\ell,r)$, then $\Sigma_i \varphi \in \mathfrak{T}^k_{n,m}(\ell-1,r)$.*

4.2.3 Disjoint Union and Tensor Products

The *disjoint union* of labelled graphs is defined as follows: For elements $\boldsymbol{F} = (F, \boldsymbol{a}, \boldsymbol{b}) \in \mathcal{G}(\ell,r)$ and $\boldsymbol{F}' = (F', \boldsymbol{a}', \boldsymbol{b}') \in \mathcal{G}(\ell',r')$, define $\boldsymbol{F} \otimes \boldsymbol{F}' \in \mathcal{G}(\ell+\ell', r+r')$ as $(F \uplus F', \boldsymbol{a}\boldsymbol{a}', \boldsymbol{b}\boldsymbol{b}')$. Analogously, for $\varphi \in \mathfrak{G}(\ell,r)$ and $\psi \in \mathfrak{G}(\ell',r')$, define $\varphi \otimes \psi \in \mathfrak{G}(\ell+\ell', r+r')$ by $(\varphi \otimes \psi)(\boldsymbol{v},\boldsymbol{w}) \coloneqq \varphi(v_1 \dots v_\ell, w_1 \dots w_\ell) \cdot \psi(v_{\ell+1} \dots v_{\ell+\ell'}, w_{r+1} \dots w_{r+r'})$ for all $\boldsymbol{v} \in [n]^{\ell+\ell'}$ and $\boldsymbol{w} \in [m]^{r+r'}$.

We use $\otimes$ to denote disjoint unions in order to be aligned with the notation of [43, 55] for homomorphism tensors where the disjoint union of labelled graphs corresponds to the Kronecker product of their homomorphism tensors.

Lemma 4.10. *For $\ell,\ell',r,r',k,k',n,m \in \mathbb{N}$, let $\boldsymbol{F} \in \mathcal{G}(\ell,r)$, $\boldsymbol{F}' \in \mathcal{G}(\ell',r')$, $\varphi \in \mathfrak{G}(\ell,r)$, and $\psi \in \mathfrak{G}(\ell',r')$.*

1. *If $\boldsymbol{F} \in \mathcal{T}^k(\ell,r)$ and $\boldsymbol{F}' \in \mathcal{T}^{k'}(\ell',r')$, then $\boldsymbol{F} \otimes \boldsymbol{F}' \in \mathcal{T}^{k''}(\ell+\ell', r+r')$ for $k'' \coloneqq \max\{k, k', \ell+\ell'+r+r'\}$.*

2. *For* $\boldsymbol{v} \in [n]^{\ell+\ell'}$ *and* $\boldsymbol{w} \in [m]^{r+r'}$,

$$(\boldsymbol{F} \otimes \boldsymbol{F}')_{n,m}(\boldsymbol{v}, \boldsymbol{w}) = \boldsymbol{F}_{n,m}(v_1 \dots v_\ell, w_1 \dots w_r) \cdot \boldsymbol{F}'_{n,m}(v_{\ell+1} \dots v_{\ell+\ell'}, w_{r+1} \dots w_{r+r'}).$$

3. *If* $\varphi \in \mathfrak{T}^k(\ell, r)$ *and* $\varphi' \in \mathfrak{T}^{k'}(\ell', r')$*, then* $\varphi \otimes \varphi' \in \mathfrak{T}^{k''}(\ell + \ell', r + r')$ *for* $k'' \coloneqq \max\{k, k', \ell + \ell' + r + r'\}$.

Proof. For the first assertion, note that a tree decomposition for $\boldsymbol{F} \otimes \boldsymbol{F}'$ can be constructed by making the root nodes of the tree decompositions of $\boldsymbol{F}$ and $\boldsymbol{F}'$ adjacent to a fresh node containing the vertices labelled in $\boldsymbol{F} \otimes \boldsymbol{F}'$. This yields a tree decomposition with bags of size at most $k'' \coloneqq \max\{k, k', \ell + \ell' + r + r'\}$. □

4.2.4 Gluing and Point-wise Products

The final elementary operation is gluing. For elements $\boldsymbol{F} = (F, \boldsymbol{a}, \boldsymbol{b})$ and $\boldsymbol{F}' = (F', \boldsymbol{a}', \boldsymbol{b}')$ of $\mathcal{G}(\ell, r)$, define $\boldsymbol{F} \odot \boldsymbol{F}' \in \mathcal{G}(\ell, r)$ by taking the disjoint union of F and F' and identifying for $i \in [\ell]$ and $j \in [r]$ the vertices a_i with a'_i and b_i with b'_i. Since left/right labels come from the left/right side of the bipartitions the resulting graph is indeed bipartite. If $\ell = 0 = r$, then gluing amounts to taking a disjoint union.

For $\varphi, \varphi' \in \mathfrak{G}_{n,m}(\ell, r)$, define the point-wise product $\varphi \odot \varphi' \in \mathfrak{G}_{n,m}(\ell, r)$ via $(\varphi \odot \varphi')(\boldsymbol{v}, \boldsymbol{w}) \coloneqq \varphi(\boldsymbol{v}, \boldsymbol{w})\varphi'(\boldsymbol{v}, \boldsymbol{w})$. When convenient, we write $\varphi\varphi'$ for the point-wise product $\varphi \odot \varphi'$.

Lemma 4.11. *Let* $\boldsymbol{F}, \boldsymbol{F}' \in \mathcal{G}(\ell, r)$*. Let* $\varphi, \varphi' \in \mathfrak{T}^k_{n,m}(\ell, r)$.

1. *If* $\boldsymbol{F}, \boldsymbol{F}' \in \mathcal{T}^k(\ell, r)$*, then* $\boldsymbol{F} \odot \boldsymbol{F}' \in \mathcal{T}^k(\ell, r)$.
2. *For* $\boldsymbol{v} \in [n]^\ell$ *and* $\boldsymbol{w} \in [m]^r$,

$$(\boldsymbol{F} \odot \boldsymbol{F}')_{n,m}(\boldsymbol{v}, \boldsymbol{w}) = \boldsymbol{F}_{n,m}(\boldsymbol{v}, \boldsymbol{w}) \cdot \boldsymbol{F}'_{n,m}(\boldsymbol{v}, \boldsymbol{w}).$$

3. *The point-wise product* $\varphi \odot \varphi'$ *of* φ *and* φ' *is in* $\mathfrak{T}^k_{n,m}(\ell, r)$.

Proof. For a detailed proof, see [31, Lemma 6.13] and [55, Lemma 3.2.13]. Intuitively, for the first claim, if (T, β) and (T', β') are tree decompositions of $\boldsymbol{F}$ and $\boldsymbol{F}'$ of width less than k with all labelled vertices occurring in the bags of vertices $r \in V(T)$ and $r' \in V(T')$, then are tree decomposition for $\boldsymbol{F} \odot \boldsymbol{F}'$ can be constructed by taking the disjoint union of T, T', and a fresh vertex x and connecting x to r and r'. The bags of the new decomposition are induced by β on T and by β' on T'. The bag associated with x contains the labelled vertices in $\boldsymbol{F} \odot \boldsymbol{F}'$. □

Lemma 4.11 shows that $\mathfrak{T}^k_{n,m}(\ell, r)$ forms a $\mathbb{Q}$-algebra for all $k, n, m, \ell, r \in \mathbb{N}$. If $\ell + r \leq k$, then this algebra is unital by Example 4.7.

4.2.5 Restricted Sums

As a modification of Lemma 4.9, we consider the following restricted sum operation. It is needed when we analyse the semantics of summation gates in dependence of their support. Let $i \in [\ell]$ and $J \subseteq [\ell] \setminus \{i\}$. For $\varphi \in \mathfrak{G}_{n,m}(\ell, r)$, define $\Sigma_{i,J}\varphi \in \mathfrak{G}_{n,m}(\ell - 1, r)$ via

$$(\Sigma_{i,J}\varphi)(\boldsymbol{v}[i/], \boldsymbol{w}) \coloneqq \sum_{v \in [n] \setminus \{v_j \mid j \in J\}} \varphi(\boldsymbol{v}[i/v], \boldsymbol{w}).$$

for all $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$. Note that $\Sigma_i\varphi = \Sigma_{i,\emptyset}\varphi$.

Lemma 4.12. *Suppose that* $\ell + r \leq k$*. For* $\varphi \in \mathfrak{T}^k_{n,m}(\ell, r)$*,* $i \in [\ell]$*, and* $J \subseteq [\ell] \setminus \{i\}$*, we have that* $\Sigma_{i,J}(\varphi) \in \mathfrak{T}^k_{n,m}(\ell - 1, r)$.

Proof. For distinct $i, j \in [\ell]$, let $\boldsymbol{D}^{i,j} \in \mathcal{T}^k(\ell, r)$ be the (ℓ, r)-labelled edge-less $(\ell-1, r)$-vertex bipartite graph whose labels are placed such that the i-th and the j-th left label reside on the same vertex while all other labels are carried by distinct vertices. Note that, for all $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$,

$$\boldsymbol{D}^{i,j}_{n,m}(\boldsymbol{v}, \boldsymbol{w}) = \begin{cases} 1, & v_i = v_j, \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Let $p \in \mathbb{Q}[x]$ be a polynomial such that $p(0) = 1$ and $p(1) = \dots = p(\ell) = 0$. By applying Lemma 4.11 and Example 4.7, define $\delta \coloneqq p(\sum_{j \in J} \boldsymbol{D}^{i,j}_{n,m}) \in \mathfrak{T}^k_{n,m}(\ell, r)$. Note that, for all $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$,

$$\delta(\boldsymbol{v}, \boldsymbol{w}) = \begin{cases} 1, & v_i \notin \{v_j \mid j \in J\}, \\ 0, & \text{otherwise.} \end{cases}$$

Hence, $\sum_{i,J} \varphi = \sum_i (\delta \odot \varphi)$. The claim follows from Lemmas 4.9 and 4.11. □

4.2.6 Products

Suppose that $\ell \geq 1$. In this section, we consider the following product operator: For $\varphi \in \mathfrak{G}_{n,m}(\ell, r)$ and $i \in [\ell]$, define $\Pi_i\varphi \in \mathfrak{G}_{n,m}(\ell-1, r)$ via

$$(\Pi_i\varphi)(\boldsymbol{v}[i/], \boldsymbol{w}) \coloneqq \prod_{v\in[n]} \varphi(\boldsymbol{v}[i/v], \boldsymbol{w})$$

for all $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^k$. Note that, in contrast to Σ_i, and $\odot$, which are linear and bilinear, respectively, the operator Π_i is not linear. Despite that, the following Theorem 4.13 shows that applying Π_i to a linear combination of homomorphism polynomial maps yields a linear combination of homomorphism polynomial maps. This theorem is the most important technical novelty in the proof of Theorem 1.1.

Theorem 4.13. *For $\varphi \in \mathfrak{T}^k_{n,m}(\ell, r)$ and $i \in [\ell]$, we have that $\Pi_i\varphi \in \mathfrak{T}^k_{n,m}(\ell-1, r)$.*

Theorem 4.13 is proved in several steps. We first show in Lemma 4.15 that $\Pi_i \boldsymbol{F}_{n,m} \in \mathfrak{T}^k_{n,m}(\ell-1, r)$ for all $\boldsymbol{F} \in \mathcal{T}^k(\ell, r)$. To that end, consider the following lemma.

Lemma 4.14. *Let $\boldsymbol{F}^1, \dots, \boldsymbol{F}^n \in \mathcal{T}^k(\ell, r)$ and $i \in [\ell]$. For a partition π of $[n]$, define the $(\ell-1, r)$-labelled bipartite graph $\Pi^\pi_i(\boldsymbol{F}^1, \dots, \boldsymbol{F}^n) \coloneqq \bigodot_{P\in[n]/\pi} \Sigma_i(\bigodot_{v\in P} \boldsymbol{F}^v) \in \mathcal{G}(\ell-1, r)$. Then*

$$(\Pi^\pi_i(\boldsymbol{F}^1, \dots, \boldsymbol{F}^n))_{n,m}(\boldsymbol{v}[i/], \boldsymbol{w}) = \sum_{h\colon [n]/\pi\to[n]} \prod_{v\in[n]} \boldsymbol{F}^v_{n,m}(\boldsymbol{v}[i/(h\circ\pi)(v)], \boldsymbol{w})$$

for all $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$. Furthermore, $\Pi^\pi_i(\boldsymbol{F}^1, \dots, \boldsymbol{F}^n) \in \mathcal{T}^k(\ell-1, r)$.

Proof. The final assertion follows from Lemmas 4.9 and 4.11. For the first assertion, observe that, by Lemmas 4.9 and 4.11,

$$\begin{aligned}
(\Pi^\pi_i(\boldsymbol{F}^1, \dots, \boldsymbol{F}^n))_{n,m}(\boldsymbol{v}[i/], \boldsymbol{w}) &= \prod_{P\in[n]/\pi} \sum_{v'\in[n]} \prod_{v\in P} \boldsymbol{F}^v_{n,m}(\boldsymbol{v}[i/v'], \boldsymbol{w}) \\
&= \sum_{h\colon [n]/\pi\to[n]} \prod_{P\in[n]/\pi} \prod_{v\in P} \boldsymbol{F}^v_{n,m}(\boldsymbol{v}[i/h(P)], \boldsymbol{w}) \\
&= \sum_{h\colon [n]/\pi\to[n]} \prod_{v\in[n]} \boldsymbol{F}^v_{n,m}(\boldsymbol{v}[i/(h\circ\pi)(v)], \boldsymbol{w}).
\end{aligned}$$

□

Lemma 4.14 is used to prove the following Lemma 4.15. Here, the $\boldsymbol{F}^1, \dots, \boldsymbol{F}^n$ are taken to be all the same (ℓ, r)-labelled bipartite graph $\boldsymbol{F}$. We abbreviate $\Pi^\pi_i \boldsymbol{F} \coloneqq \Pi^\pi_i(\boldsymbol{F}, \dots, \boldsymbol{F})$. Lemma 4.14 then yields

$$(\Pi^\pi_i \boldsymbol{F})_{n,m}(\boldsymbol{v}[i/], \boldsymbol{w}) = \sum_{h\colon [n]/\pi\to[n]} \prod_{v\in[n]} \boldsymbol{F}_{n,m}(\boldsymbol{v}[i/(h\circ\pi)(v)], \boldsymbol{w}) \tag{7}$$

Lemma 4.15. *For $\boldsymbol{F} \in \mathcal{T}^k(\ell, r)$ and $i \in [\ell]$, $\Pi_i(\boldsymbol{F}_{n,m}) \in \mathfrak{T}^k_{n,m}(\ell-1, r)$.*

Proof. Write μ for the Möbius function of the partition lattice of $[n]$, cf. Equation (17). More concretely, we show that $\Pi_i \boldsymbol{F}_{n,m} = \frac{1}{n!} \sum_{\pi\vdash n} \mu_\pi (\Pi^\pi_i \boldsymbol{F})_{n,m}$. By Lemma 4.14, $\Pi^\pi_i \boldsymbol{F} \in \mathcal{T}^k(\ell-1, r)$. Hence, it remains to show that

$$\prod_{v\in[n]} \boldsymbol{F}_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w}) = \frac{1}{n!} \sum_{\pi\vdash n} \mu_\pi (\Pi^\pi_i \boldsymbol{F})_{n,m}(\boldsymbol{v}[i/], \boldsymbol{w})$$

for all $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$. Note that, crucially, the coefficients in the above expression do not depend on $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$.

The left hand-side expression can be rewritten as follows noting that, for every bijection $h\colon [n] \hookrightarrow [n]$, it is $\prod_{v\in[n]} \boldsymbol{F}_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w}) = \prod_{v\in[n]} \boldsymbol{F}_{n,m}(\boldsymbol{v}[i/h(v)], \boldsymbol{w})$ by commutativity of the product. The second equality follows from Lemma A.2, the third from Equation (7).

$$\begin{aligned}
\prod_{v\in[n]} \boldsymbol{F}_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w}) &= \frac{1}{n!} \sum_{h\colon [n]\hookrightarrow[n]} \prod_{v\in[n]} \boldsymbol{F}_{n,m}(\boldsymbol{v}[i/h(v)], \boldsymbol{w}) \\
&= \frac{1}{n!} \sum_{\pi\vdash n} \mu_\pi \sum_{h\colon [n]/\pi\to[n]} \prod_{v\in[n]} \boldsymbol{F}_{n,m}(\boldsymbol{v}[i/(h\circ\pi)(v)], \boldsymbol{w}) \\
&= \frac{1}{n!} \sum_{\pi\vdash n} \mu_\pi (\Pi^\pi_i \boldsymbol{F})_{n,m}(\boldsymbol{v}[i/], \boldsymbol{w}).
\end{aligned}$$

□

Now we consider the general case: applying the product operator to a linear combination of labelled homomorphism polynomials. In preparation, we note the following Fact 4.16.

Fact 4.16. *Let Γ denote a finite group acting on a finite set X. For an element $x \in X$, write β_x for the number of pairs $(z, \gamma) \in X \times \Gamma$ such that $\gamma(z) = x$. If $x, y \in X$ are in the same orbit under Γ, then $\beta_x = \beta_y$.*

Proof. Suppose that $\iota(x) = y$ for some $\iota \in \Gamma$. Then the map $(z, \gamma) \mapsto (z, \iota\gamma)$ sends a pair (z, γ) such that $\gamma(z) = x$ to the pair $(z, \iota\gamma)$ satisfying $\iota\gamma(z) = \iota(x) = y$. This map is bijective. □

Proof of Theorem 4.13. Write $\varphi = \sum_{t \in T} \alpha_t \boldsymbol{F}^t_{n,m}$ for some finite set T, coefficients $\alpha_t \in \mathbb{Q}$ and $\boldsymbol{F}^t \in \mathcal{T}^k(\ell, r)$. The symmetric group $\mathbf{Sym}_n$ acts on the set of functions $f\colon [n] \to T$ by composition. The orbits of this action are in bijection with maps $\lambda\colon T \to \{0, \dots, n\}$ such that $\sum_{t \in T} \lambda(t) = n$. Write Λ for the set of all such functions. A function $f\colon [n] \to T$ belongs to the orbit O_λ for $\lambda \in \Lambda$ if $|f^{-1}(t)| = \lambda(t)$ for all $t \in T$.

Let $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$. First, product and sum are interchanged. The resulting sum over all functions $f\colon [n] \to T$ is then grouped by orbits:

$$\begin{aligned}
(\Pi_i \varphi)(\boldsymbol{v}[i/], \boldsymbol{w}) &= \prod_{v \in [n]} \sum_{t \in T} \alpha_t \boldsymbol{F}^t_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w}) \\
&= \sum_{f\colon [n] \to T} \prod_{v \in [n]} \alpha_{f(v)} \boldsymbol{F}^{f(v)}_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w}) \\
&= \sum_{\lambda \in \Lambda} \sum_{f \in O_\lambda} \prod_{v \in [n]} \alpha_{f(v)} \boldsymbol{F}^{f(v)}_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w}) \\
&= \sum_{\lambda \in \Lambda} \alpha_\lambda \sum_{f \in O_\lambda} \prod_{v \in [n]} \boldsymbol{F}^{f(v)}_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w}).
\end{aligned}$$

where $\alpha_\lambda \coloneqq \prod_{t \in T} \alpha_t^{\lambda(t)} \in \mathbb{Q}$.

From now on, we consider each orbit $\lambda \in \Lambda$ separately. By Fact 4.16, there exists a positive number $\beta_\lambda \in \mathbb{N}$ such that, for every $f \in O_\lambda$, the number of pairs $g\colon [n] \to T$ and $h \in \mathbf{Sym}_n$ such that $g \circ h = f$ is β_λ. In any such pair, we necessarily have $g \in O_\lambda$. Towards applying Möbius inversion, we denote permutations $h \in \mathbf{Sym}_n$ as injective maps $[n] \hookrightarrow [n]$.

$$\begin{aligned}
\beta_\lambda \sum_{f \in O_\lambda} \prod_{v \in [n]} \boldsymbol{F}^{f(v)}_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w}) &= \sum_{g \in O_\lambda} \sum_{h\colon [n] \hookrightarrow [n]} \prod_{v \in [n]} \boldsymbol{F}^{(g \circ h)(v)}_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w}) \\
&= \sum_{g \in O_\lambda} \sum_{h\colon [n] \hookrightarrow [n]} \prod_{v \in [n]} \boldsymbol{F}^{g(v)}_{n,m}(\boldsymbol{v}[i/h(v)], \boldsymbol{w}) \\
&= \sum_{g \in O_\lambda} \sum_{\pi \vdash n} \mu_\pi \sum_{h\colon [n]/\pi \to [n]} \prod_{v \in [n]} \boldsymbol{F}^{g(v)}_{n,m}(\boldsymbol{v}[i/(h \circ \pi)(v)], \boldsymbol{w}) \\
&= \sum_{g \in O_\lambda} \sum_{\pi \vdash n} \mu_\pi (\Pi^\pi_i(\boldsymbol{F}^{g(1)}, \dots, \boldsymbol{F}^{g(n)}))_{n,m}(\boldsymbol{v}[i/], \boldsymbol{w})
\end{aligned}$$

Here, the first equality follows from Fact 4.16. The second equality is obtained by first replacing the product over $v \in [n]$ by a product over $h(v)$ for $v \in [n]$ and then replacing the sum over $h\colon [n] \hookrightarrow [n]$ by a sum over h^{-1} for $h\colon [n] \hookrightarrow [n]$. The third equality follows from Lemma A.2. The final equality is implied by Lemma 4.14.

Combining the identities above, we obtain

$$(\Pi_i \varphi)(\boldsymbol{v}[i/], \boldsymbol{w}) = \sum_{\lambda \in \Lambda} \frac{\alpha_\lambda}{\beta_\lambda} \sum_{g \in O_\lambda} \sum_{\pi \vdash n} \mu_\pi \Pi^\pi_i(\boldsymbol{F}^{g(1)}, \dots, \boldsymbol{F}^{g(n)})_{n,m}(\boldsymbol{v}[i/], \boldsymbol{w}). \tag{8}$$

Note that neither the α_λ nor β_λ depend on $\boldsymbol{v}$. Hence, $\Pi_i \varphi \in \mathfrak{T}^k_{n,m}(\ell - 1, r)$. □

Remark 4.17. Theorem 4.13 indeed generalises Lemma 4.15. If $T = \{\boldsymbol{F}\}$ is a singleton and there are no coefficients, then Λ consists of a single map λ. Then, $\alpha_\lambda = 1$ and $\beta_\lambda = n!$. The orbit O_λ contains only one map g and $\Pi^\pi_i(\boldsymbol{F}^{g(1)}, \dots, \boldsymbol{F}^{g(n)}) = \Pi^\pi_i \boldsymbol{F}$.

4.2.7 Restricted Products

Finally, we derive a corollary of Theorem 4.13 concerned with the restricted product operator $\Pi_{i,J}$ which is defined as follows: Let $i \in [\ell]$ and $J \subseteq [\ell] \setminus \{i\}$. For $\varphi \in \mathfrak{G}_{n,m}(\ell, r)$, define $\Pi_{i,J} \varphi \in \mathfrak{G}_{n,m}(\ell - 1, r)$ via

$$(\Pi_{i,J} \varphi)(\boldsymbol{v}[i/], \boldsymbol{w}) \coloneqq \prod_{v \in [n] \setminus \{v_j \mid j \in J\}} \varphi(\boldsymbol{v}[i/v], \boldsymbol{w}).$$

for all $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$. Note that $\Pi_i \varphi = \Pi_{i,\emptyset} \varphi$.

Corollary 4.18. *Suppose that $\ell + r \leq k$. For $\varphi \in \mathfrak{T}^k_{n,m}(\ell, r)$, $i \in [\ell]$ and $J \subseteq [\ell] \setminus \{i\}$, $\Pi_{i,J}(\varphi) \in \mathfrak{T}^k_{n,m}(\ell - 1, r)$.*

Proof. For $J \subseteq [\ell]$, write $V_J \coloneqq [n] \setminus \{v_j \mid j \in J\}$. The proof is by induction on $|J|$. The base case $|J| = 0$ is established in Theorem 4.13.

For the inductive step, let $j \in J$ and write $J' \coloneqq J \setminus \{j\}$. We first distinguish cases based on the form of the tuple $\boldsymbol{v}$. This is a priori problematic since the desired expression is required not to depend on $\boldsymbol{v}$. We overcome this problem by implementing the case distinction itself using homomorphism polynomial maps.

Write $\varphi = \sum_{t \in T} \alpha_t \boldsymbol{F}^t$ and suppose first that $v_j \notin \{v_{j'} \mid j' \in J'\}$, i.e. $V_{J'} = V_J \uplus \{v_j\}$. Recall the definition of $\boldsymbol{D}^{i,j} \in \mathcal{T}^k(\ell, r)$ from the proof of Lemma 4.12. We rewrite $\Pi_{i,J}(\varphi)$ in terms of expressions to which the inductive hypothesis applies by considering the following polynomial:

$$\begin{aligned}
&\prod_{v \in V_{J'}} \left(\varphi(\boldsymbol{v}[i/v], \boldsymbol{w}) + \boldsymbol{D}^{i,j}_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w})\right) \\
&= \left(\varphi(\boldsymbol{v}[i/v_j], \boldsymbol{w}) + \boldsymbol{D}^{i,j}_{n,m}(\boldsymbol{v}[i/v_j], \boldsymbol{w})\right) \prod_{v \in V_J} \left(\varphi(\boldsymbol{v}[i/v], \boldsymbol{w}) + \boldsymbol{D}^{i,j}_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w})\right) \\
&\overset{(6)}{=} \left(\varphi(\boldsymbol{v}[i/v_j], \boldsymbol{w}) + 1\right) \prod_{v \in V_J} \varphi(\boldsymbol{v}[i/v], \boldsymbol{w}) \\
&= \prod_{v \in V_{J'}} \varphi(\boldsymbol{v}[i/v], \boldsymbol{w}) + \prod_{v \in V_J} \varphi(\boldsymbol{v}[i/v], \boldsymbol{w}).
\end{aligned}$$

If otherwise $v_j \in \{v_{j'} \mid j' \in J'\}$, then the expression for J equals the expression for J'. Formally,

$$\prod_{v \in V_{J'}} \varphi(\boldsymbol{v}[i/v], \boldsymbol{w}) = \prod_{v \in V_J} \varphi(\boldsymbol{v}[i/v], \boldsymbol{w}).$$

In order to combine the two cases considered above, observe that, for every $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$, by Lemma 4.9,

$$\frac{1}{n} \sum_{j' \in J'} (\Sigma_i \boldsymbol{D}^{j',j})_{n,m}(\boldsymbol{v}[i/], \boldsymbol{w}) = \frac{1}{n} \sum_{v \in [n]} \sum_{j' \in J'} \boldsymbol{D}^{j',j}_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w}) = \sum_{j' \in J'} \boldsymbol{D}^{j',j}_{n,m}(\boldsymbol{v}, \boldsymbol{w})$$

is equal to the number of indices $j' \in J'$ such that $v_j = v_{j'}$. This is a value from $\{0, \ldots, \ell\}$. Here, it is crucial that $i \notin J$, as assumed. Let $p \in \mathbb{Q}[x]$ denote a polynomial such that $p(0) = 0$ and $p(1) = p(2) = \cdots = p(\ell) = 1$. By Lemmas 4.9 and 4.11, $\psi \coloneqq p(\frac{1}{n} \sum_{j' \in J'} (\Sigma_i \boldsymbol{D}^{j',j})_{n,m}) \in \mathfrak{T}^k_{n,m}(\ell - 1, r)$. Then

$$\psi(\boldsymbol{v}[i/], \boldsymbol{w}) = \begin{cases} 1, & v_j \in \{v_{j'} \mid j' \in J'\}, \\ 0, & \text{otherwise.} \end{cases}$$

Combining the above, it follows that the expression $\prod_{v \in V_J} \varphi(\boldsymbol{v}[i/v], \boldsymbol{w})$ of interest is equal to

$$\begin{aligned}
&\psi(\boldsymbol{v}[i/], \boldsymbol{w}) \prod_{v \in V_{J'}} \varphi(\boldsymbol{v}[i/v], \boldsymbol{w}) \\
&+ (1 - \psi(\boldsymbol{v}[i/], \boldsymbol{w})) \left(\prod_{v \in V_{J'}} \left(\varphi(\boldsymbol{v}[i/v], \boldsymbol{w}) + \boldsymbol{D}^{i,j}_{n,m}(\boldsymbol{v}[i/v], \boldsymbol{w})\right) - \prod_{v \in V_{J'}} \varphi(\boldsymbol{v}[i/v], \boldsymbol{w}) \right).
\end{aligned}$$

This expression depends uniformly on $\boldsymbol{v}$ and $\boldsymbol{w}$. In other words,

$$\Pi_{i,J}(\varphi) = \psi \; \Pi_{i,J'}(\varphi) + (1 - \psi) \left(\Pi_{i,J'}(\varphi + \boldsymbol{D}^{i,j}_{n,m}) - \Pi_{i,J'}(\varphi)\right).$$

Here, we regard 1 as the element of $\mathfrak{T}^k_{n,m}(\ell - 1, r)$ which is uniformly 1, cf. Example 4.7. The induction hypothesis applies to $\Pi_{i,J'}$. Hence, $\Pi_{i,J'}(\varphi), \Pi_{i,J'}(\varphi + \boldsymbol{D}^{i,j}_{n,m}) \in \mathfrak{T}^k_{n,m}(\ell - 1, r)$. By Lemma 4.11, $\mathfrak{T}^k_{n,m}(\ell - 1, r)$ is closed under pointwise products. Hence, $\Pi_{i,J}(\varphi) \in \mathfrak{T}^k_{n,m}(\ell - 1, r)$. □

4.3 Homomorphism Polynomials from Symmetric Circuits

In this section we prove the direction 2 ⇒ 1 of Theorem 1.1 using the results we have established so far. For the proof, we start with a rigid symmetric circuit C with $\mathrm{maxSup}(C) \leq k$, and show via induction from the inputs to the output that at each gate g, the computed polynomial is in $\mathfrak{T}^{2k}_{n,m}(\vec{\sup}_L(g), \vec{\sup}_R(g))$. Recall that $\vec{\sup}_L(g)$ and $\vec{\sup}_R(g)$ denote ordered tuples whose entries are the elements of $\sup_L(g)$ and $\sup_R(g)$ (see Section 2.1). The ordering of the tuples represents the assignment of the labels to the support elements: The label $i \in [\ell]$ is mapped to the i-th entry of $\vec{\sup}_L(g)$, and analogously for $\vec{\sup}_R(g)$.

For the inductive step of the proof, we assume that g is a multiplication or summation gate such that every child h of g computes a polynomial in $\mathfrak{T}^{2k}_{n,m}(\vec{\sup}_L(h), \vec{\sup}_R(h))$. Then we show that the polynomial computed at g is in $\mathfrak{T}^{2k}_{n,m}(\vec{\sup}_L(g), \vec{\sup}_R(g))$. In the following, if g is a gate, then p_g denotes the polynomial that this gate outputs.

Observation 4.19. *Let $n, m, \ell, r \in \mathbb{N}$. Let $\boldsymbol{F} \in \mathcal{G}(\ell, r)$ and let $(\pi, \sigma) \in \mathbf{Sym}_n \times \mathbf{Sym}_m$. Then $(\pi, \sigma)(\boldsymbol{F}_{n,m}(\boldsymbol{v}, \boldsymbol{w})) = \boldsymbol{F}_{n,m}(\pi(\boldsymbol{v}), \sigma(\boldsymbol{w}))$ for all $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$.*

Proof. For $\boldsymbol{F} = (F, \boldsymbol{a}, \boldsymbol{b})$, we have $(\pi, \sigma)(\{h \colon A \uplus B \to [n] \uplus [m] \mid h(\boldsymbol{a}) = \boldsymbol{v}, h(\boldsymbol{b}) = \boldsymbol{w}\}) = \{h \colon A \uplus B \to [n] \uplus [m] \mid h(\boldsymbol{a}) = \pi(\boldsymbol{v}), h(\boldsymbol{b}) = \sigma(\boldsymbol{v})\}$. □

Lemma 4.20. *Let C be a rigid $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuit and $g \in V(C)$ a gate such that the polynomial computed at g is of the form $p_g = \varphi(\vec{\sup}_L(g), \vec{\sup}_R(g))$ for some $\varphi \in \mathfrak{T}^{2k}_{n,m}(\ell, r)$. Let $g' \in \mathbf{Orb}_{\mathbf{Sym}_n \times \mathbf{Sym}_m}(g)$. Then the polynomial computed by g' is $p_{g'} = \varphi(\vec{\sup}_L(g'), \vec{\sup}_R(g'))$.*

Proof. Let $(\pi, \sigma) \in \mathbf{Sym}_n \times \mathbf{Sym}_m$ such that $(\pi, \sigma)(g) = g'$. So the polynomial computed by g' is $(\pi, \sigma)(p)$. Then the lemma follows from Observation 4.19. □

The next lemma shows the inductive step in case that the children of the gate g form a single orbit.

Lemma 4.21. *Let $k \in \mathbb{N}$. Let C be a rigid $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuit, and $g \in V(C)$ a multiplication or a summation gate with $|\sup(g)| \leq k$. Assume that for every $h \in \mathrm{children}(g)$, it also holds that $|\sup(h)| \leq k$ and the polynomial computed at h is $p_h = \varphi(\vec{\sup}_L(h), \vec{\sup}_R(h))$ for some $\varphi \in \mathfrak{T}^{2k}_{n,m}(|\vec{\sup}_L(h)|, |\vec{\sup}_R(h)|)$. Let $h \in \mathrm{children}(g)$ and let $O_h \coloneqq \mathbf{Orb}_{\mathbf{Stab}^\bullet(\sup(g))}(h)$.*
Then $\prod_{h' \in O_h} p_{h'}$ and $\sum_{h' \in O_h} p_{h'}$ are in $\mathfrak{T}^{2k}_{n,m}(\vec{\sup}_L(g), \vec{\sup}_R(g))$.

Proof. As in Lemma 2.6, let $S(h) = \bigcap_{h' \in O_h} \sup(h')$, and $S_L(h) = S(h) \cap [n]$, $S_R(h) = S(h) \cap [m]$. Note that for every $h' \in O_h$, the tuples $\vec{\sup}_L(h')$ and $\vec{\sup}_R(h')$ have the elements of $S(h)$ in the same positions because $S(h) \subseteq \sup(g)$ by Lemma 2.6. Since all these gates are in the same $\mathbf{Stab}^\bullet(\sup(g))$-orbit, the positions of $S(h)$ in $\vec{\sup}_L(h')$ and $\vec{\sup}_R(h')$ are the same. In the following, when we write $\vec{S}_L(h)$ and $\vec{S}_R(h)$, we refer to the ordering of $S_L(h)$ and $S_R(h')$ as in $\vec{\sup}_L(h')$ and $\vec{\sup}_R(h')$, respectively.

For simplicity, assume in the following that the first $|S_L(h)|$ positions in each tuple $\vec{\sup}_L(h')$ are occupied by the elements of $S_L(h)$ and the first $|S_R(h)|$ positions in $\vec{\sup}_R(h')$ are occupied by $S_R(h)$. Since $\mathbf{Orb}_{\mathbf{Stab}^\bullet(\sup(g))}(h)$ is a subset of $\mathbf{Orb}_{\mathbf{Sym}_n \times \mathbf{Sym}_m}(h)$, by Lemma 4.20, all these gates compute the same instantiated homomorphism polynomial map, just with different instantiation of the labels. Let $\varphi \in \mathfrak{T}^{2k}_{n,m}(\ell, r)$ denote this homomorphism polynomial map, where $\ell = |\sup_L(h)|, r = |\sup_R(h)|$. Recall the definition of $\boldsymbol{J}$ from Example 4.7. First consider $\prod_{h' \in O_h} p_{h'}$.

$$
\begin{aligned}
\prod_{h' \in O_h} p_{h'} &= \prod_{h' \in O_h} \varphi(\vec{\sup}_L(h'), \vec{\sup}_R(h')) \\
&= \prod_{\boldsymbol{v} \in ([n] \setminus \sup_L(g))^{\ell - |S_L(h)|}} \prod_{\boldsymbol{w} \in ([m] \setminus \sup_R(g))^{r - |S_R(h)|}} \varphi(\vec{S}_L(h)\boldsymbol{v}, \vec{S}_R(h)\boldsymbol{w}) \\
&= \prod_{\boldsymbol{v} \in ([n] \setminus \sup_L(g))^{\ell - |S_L(h)|}} \prod_{\boldsymbol{w} \in ([m] \setminus \sup_R(g))^{r - |S_R(h)|}} \varphi(\vec{S}_L(h)\boldsymbol{v}, \vec{S}_R(h)\boldsymbol{w}) \cdot \boldsymbol{J}_{n,m}(\vec{\sup}_L(g) \setminus \vec{S}_L(h), \vec{\sup}_R(g) \setminus \vec{S}_R(h)) \\
&= \prod_{\boldsymbol{v} \in ([n] \setminus \sup_L(g))^{\ell - |S_L(h)|}} \prod_{\boldsymbol{w} \in ([m] \setminus \sup_R(g))^{r - |S_R(h)|}} (\varphi \otimes \boldsymbol{J}_{n,m})(\vec{\sup}_L(g)\boldsymbol{v}, \vec{\sup}_R(g)\boldsymbol{w}).
\end{aligned}
$$

The second equality is true because by Lemma 2.6, for each $h' \in O_h$, the set $\sup(h') \setminus S(h)$ does not intersect $\sup(g)$, and O_h is the orbit of h under the pointwise stabiliser of $\sup(g)$ in $\mathbf{Sym}_n \times \mathbf{Sym}_m$. Hence, there is a bijection between the gates in O_h and the set $\{\boldsymbol{v}\boldsymbol{w} \mid \boldsymbol{v} \in ([n] \setminus \sup_L(g))^{\ell - |S_L(h)|}, \boldsymbol{w} \in ([m] \setminus \sup_R(g))^{r - |S_R(h)|}\}$. The third equality holds because by Example 4.7, $\boldsymbol{J}_{n,m}(\vec{\sup}_L(g) \setminus \vec{S}_L(h), \vec{\sup}_R(g) \setminus \vec{S}_R(h)) = 1$. The last

equality is due to the definition of $\otimes$ on homomorphism polynomial maps, cf. Lemma 4.10. Analogously, it holds that

$$\sum_{h' \in O_h} p_{h'} = \sum_{\boldsymbol{v} \in ([n] \setminus \sup_L(g))^{\ell - |S_L(h)|}} \sum_{\boldsymbol{w} \in ([m] \setminus \sup_R(g))^{r - |S_R(h)|}} (\varphi \otimes \boldsymbol{J}_{n,m})(\vec{\sup}_L(g)\boldsymbol{v}, \vec{\sup}_R(g)\boldsymbol{w}).$$

By Lemma 4.10, $(\varphi \otimes \boldsymbol{J}) \in \mathfrak{T}^{2k}_{n,m}(|\sup_L(g)| + |\sup_L(h) \setminus \sup_L(g)|, |\sup_R(g)| + |\sup_R(h) \setminus \sup_R(g)|)$. Here it is important that $|\sup(g)| + |\sup(h)| \leq 2k$. By repeatedly applying Corollary 4.18 to the last expression above, it follows that $\prod_{h' \in O_h} p_{h'} \in \mathfrak{T}^{2k}_{n,m}(\vec{\sup}_L(g), \vec{\sup}_R(g))$. If g is a summation gate, then similarly, with Lemma 4.12, $\sum_{h' \in O_h} p_{h'} \in \mathfrak{T}^{2k}_{n,m}(\vec{\sup}_L(g), \vec{\sup}_R(g))$. □

We now generalise the above lemma so that it covers all orbits of children of g.

Lemma 4.22. *Let $k \in \mathbb{N}$. Let C be a rigid $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuit. Let $g \in V(C)$ be a multiplication or summation gate with $|\sup(g)| \leq k$. Assume that for every $h \in \mathrm{children}(g)$, the polynomial p_h is in $\mathfrak{T}^{2k}_{n,m}(\vec{\sup}_L(h), \vec{\sup}_R(h))$. Then the polynomial computed at g is in $\mathfrak{T}^{2k}_{n,m}(\vec{\sup}_L(g), \vec{\sup}_R(g))$.*

Proof. We treat the case when g is a multiplication gate in detail. The other case is analogous. Partition the set $\mathrm{children}(g)$ into orbits under $\mathbf{Stab}^\bullet(\sup(g))$. To each such orbit O_h, we apply Lemma 4.21, which shows that $p_{O_h} \coloneqq \prod_{h' \in O_h} p_{h'} \in \mathfrak{T}^{2k}_{n,m}(\vec{\sup}_L(g), \vec{\sup}_R(g))$. Let $\Omega \coloneqq \{O_h \mid h \in \mathrm{children}(g)\}$ and let $\varphi_{O_h}(\vec{\sup}_L(g), \vec{\sup}_R(g))$ be the instantiated homomorphism polynomial map in $\mathfrak{T}^{2k}_{n,m}(\vec{\sup}_L(g), \vec{\sup}_R(g))$ that is equal to p_{O_h}. For each orbit $O_h \in \Omega$, let $m(O_h) \in \mathbb{N}$ denote the multiplicity of the wire between g and each gate $h' \in O_h$. The polynomial computed at g is

$$p_g = \prod_{O_h \in \Omega} p_{O_h}^{m(O_h)} = \prod_{O_h \in \Omega} \varphi_{O_h}(\vec{\sup}_L(g), \vec{\sup}_R(g))^{m(O_h)}.$$

This polynomial is in $\mathfrak{T}^{2k}_{n,m}(\vec{\sup}_L(g), \vec{\sup}_R(g))$ by Lemma 4.11. □

4.4 Proof of the backward direction of Theorem 1.1

Proof. Let $(C_{n,m})_{n,m \in \mathbb{N}}$ be a family of circuits such that each $C_{n,m}$ is $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric, and $\mathrm{maxOrb}(C_{n,m})$ is polynomially bounded in $(n+m)$. By Lemma 2.3, we can assume that the circuits are rigid. Then by Lemma 2.4, there exists a constant $k \in \mathbb{N}$ such that $\mathrm{maxSup}(C_{n,m}) \leq k$ for all $n, m \in \mathbb{N}$. Consider now fixed $n, m \in \mathbb{N}$. By induction on the structure of $C_{n,m}$, we show that for every gate $g \in V(C_{n,m})$, we have $p_g \in \mathfrak{T}^{2k}_{n,m}(\vec{\sup}_L(g), \vec{\sup}_R(g))$. If g is an input gate labelled with a constant, then g is fixed by every permutation, so $\sup(g) = \emptyset$. Indeed, any constant is a polynomial in $\mathfrak{T}^{2k}_{n,m}(0,0)$ by Example 4.7. If g is an input gate labelled with a variable x_{ij}, then $\sup_L(g) = \{i\}, \sup_R(g) = \{j\}$. It is clear that the polynomial x_{ij} is in $\mathfrak{T}^{2k}_{n,m}(i,j)$ because $x_{ij} = \boldsymbol{F}_{n,m}(i,j)$ where $\boldsymbol{F}$ is the graph that consists of a single edge. This finishes the base cases. In the inductive step, g is either a summation or multiplication gate. Both these cases are handled by Lemma 4.22. Finally, the output gate of $C_{n,m}$ is stabilised by $\mathbf{Sym}_n \times \mathbf{Sym}_m$, so its minimal support is empty, which means that the polynomial computed by $C_{n,m}$ is in $\mathfrak{T}^{2k}_{n,m}$. □

4.5 Symmetric Circuits for Homomorphism Polynomials

In this section, we prove Theorem 4.3. Let F be a graph with a tree decomposition (T, β) and $r \in V(T)$. For $s \in V(T)$, write T^s for the subtree of T induced by s and all its descendents in the rooted tree (T, r). Write F^s for the subgraph of F induced by $\bigcup_{t \in V(T^s)} \beta(t)$.

Lemma 4.23. *Let $k, n, m \in \mathbb{N}$. Let F be a bipartite multigraph with bipartition $A \uplus B$, tree decomposition (T, β), and $r \in V(T)$ such that*

1. *$|\beta(t)| = k$ for all $t \in V(T)$,*
2. *$|\beta(s) \cap \beta(t)| = k - 1$ for all $st \in E(T)$,*
3. *every vertex in the rooted tree (T, r) has out-degree at most k.*

There exists a rigid $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuit of size at most $4k!^2 k \cdot (n+m)^{k+1} \cdot |V(T)| \cdot \|F\|$ and support size at most k which contains, for every $s \in V(T)$, $\boldsymbol{a} \in A^\ell$, $\boldsymbol{b} \in B^r$, $\ell + r = k$ such that $\beta(s) = \{a_1, \ldots, a_\ell, b_1, \ldots, b_r\}$, and $\boldsymbol{v} \in [n]^\ell$, $\boldsymbol{w} \in [m]^r$, a gate which computes $\boldsymbol{F}^s_{n,m}(\boldsymbol{v}, \boldsymbol{w})$ where $\boldsymbol{F}^s \coloneqq (F^s, \boldsymbol{a}, \boldsymbol{b}) \in \mathcal{T}^k(\ell, r)$.

Proof. We build the circuit by induction on the depth of s in (T, r). For the base case, consider all leaves of T. For every leaf $s \in V(T)$, all vertices in $\boldsymbol{F}^s$ are labelled. Hence, $\boldsymbol{F}^s_{n,m}(\boldsymbol{v}, \boldsymbol{w})$ can be represented by a single multiplication gate whose fan-in equals the number of edges in F^s. For every leaf, this yields a subcircuit of size at most $\|F\| n^\ell m^r k! \leq \|F\|(n+m)^k k!$.

For the inductive step, suppose that a circuit has been constructed which contains the required gates for all $s \in V(T)$ of depth $> d$. Let $s \in V(T)$ be a vertex at depth d. Fix $\boldsymbol{v} \in [n]^\ell$, $\boldsymbol{w} \in [m]^r$, $\boldsymbol{a} \in A^\ell$, $\boldsymbol{b} \in B^r$, $\ell + r = k$ such that $\beta(s) = \{a_1, \dots, a_\ell, b_1, \dots, b_r\}$.

Let $t \in V(T)$ be a child of s. Write u for the unique vertex in $\beta(s) \setminus \beta(t)$. Suppose wlog that $u = a_\ell \in A$. Write K^t for the subgraph of F induced by $\beta(s) \cup \bigcup_{t' \in V(T^t)} \beta(t')$. Let $\boldsymbol{K}^t \coloneqq (K^t, \boldsymbol{a}, \boldsymbol{b})$. We first construct a circuit for $\boldsymbol{K}^t_{n,m}(\boldsymbol{v}, \boldsymbol{w})$. Let u' denote the unique vertex in $\beta(t) \setminus \beta(s)$. Distinguish cases:

1. If $u' \in A$, let $\boldsymbol{a}' \coloneqq \boldsymbol{a}[\ell/u']$ and define $\boldsymbol{F}^t \coloneqq (F^t, \boldsymbol{a}', \boldsymbol{b})$. Then

$$\boldsymbol{K}^t_{n,m}(\boldsymbol{v}, \boldsymbol{w}) = \sum_{v \in [n]} \boldsymbol{F}^t_{n,m}(\boldsymbol{v}[\ell/v], \boldsymbol{w}) \prod_{j \in [r]} \prod_{\substack{ab \in E(F) \\ a = a_\ell \\ b = b_j}} x_{v_\ell w_j}. \tag{9}$$

2. If $u' \in B$, let $\boldsymbol{a}' \coloneqq \boldsymbol{a}[\ell/]$ and define $\boldsymbol{F}^t \coloneqq (F^t, \boldsymbol{a}', \boldsymbol{b}u')$. Then

$$\boldsymbol{K}^t_{n,m}(\boldsymbol{v}, \boldsymbol{w}) = \sum_{w \in [m]} \boldsymbol{F}^t_{n,m}(\boldsymbol{v}[\ell/], \boldsymbol{w}w) \prod_{j \in [r]} \prod_{\substack{ab \in E(F) \\ a = a_\ell \\ b = b_j}} x_{v_\ell w_j}. \tag{10}$$

In both cases, the inductive hypothesis applies to $\boldsymbol{F}^t$. In order to construct $\boldsymbol{K}^t_{n,m}(\boldsymbol{v}, \boldsymbol{w})$, we require one summation gate of fan-in $\max\{n, m\}$ and a product gate of fan-in at most $\|F\| + 1$.

Let now $t_1, \dots, t_\ell$ for $\ell \leq k$ denote the children of s. By Lemma 4.11,

$$\boldsymbol{F}^s_{n,m}(\boldsymbol{v}, \boldsymbol{w}) = \prod_{i=1}^{\ell} \boldsymbol{K}^{t_i}_{n,m}(\boldsymbol{v}, \boldsymbol{w}). \tag{11}$$

This polynomial can be realised by one additional product gate of fan-in at most k.

Thus, in order to construct the additional gates for $s \in V(T)$, we require at most

$$n^\ell m^r \ell! r! (4 + k + \max\{n, m\} + \|F\|) \leq 4(n+m)^{k+1} k!^2 k \cdot \|F\|$$

gates and edges. It follows that the overall circuit has the desired size. In the end, we invoke Lemma 2.3 to make the circuit rigid, which allows us to speak about the supports of gates.

Let $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$. At any stage, the subcircuit computing $\boldsymbol{F}^s_{n,m}(\boldsymbol{v}, \boldsymbol{w})$ is by construction $\mathbf{Stab}^\bullet_{\mathbf{Sym}_n \times \mathbf{Sym}_m}(\boldsymbol{v}, \boldsymbol{w})$-symmetric, so $\boldsymbol{v}\boldsymbol{w}$ is a support of the gate computing $\boldsymbol{F}^s_{n,m}(\boldsymbol{v}, \boldsymbol{w})$. Note that $|\boldsymbol{v}\boldsymbol{w}| \leq k$.

The overall circuit is $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric, since, by Observation 4.19, $(\pi, \sigma)(\boldsymbol{F}^s_{n,m}(\boldsymbol{v}, \boldsymbol{w})) = \boldsymbol{F}^s_{n,m}(\pi(\boldsymbol{v}), \sigma(\boldsymbol{w}))$ is computed as well for $(\pi, \sigma) \in \mathbf{Sym}_n \times \mathbf{Sym}_m$. □

The lemma implies the theorem:

Theorem 4.3. *Let $k, n, m \in \mathbb{N}$. Let F be a bipartite multigraph of treewidth less than k. Then there exists a $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric rigid circuit C_F for $\hom_{F,n,m}$ of size at most $5k!^2 k \cdot (n+m)^{k+1} \cdot \|F\|^2$ satisfying $\operatorname{maxSup}(C_F) \leq k$.*

Proof. By [54, Lemma 6] and its proof, there exists a tree decomposition (T, β) of F with a vertex $r \in V(T)$ possessing the properties required by Lemma 4.23. Note that $|V(T)| \leq |V(F)|$.

Write $A \uplus B$ for the bipartition of F. Let $\ell, r \in \mathbb{N}$ be such that there exists $\boldsymbol{a} \in A^\ell$ and $\boldsymbol{b} \in B^r$ such that $\beta(r) = \{a_1, \dots, a_\ell, b_1, \dots, b_r\}$. Let $\boldsymbol{F} \coloneqq (F, \boldsymbol{a}, \boldsymbol{b}) \in \mathcal{T}^k(\ell, r)$. By Lemma 4.23, for $n, m \in \mathbb{N}$, there exists a $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuit computing $\boldsymbol{F}_{n,m}(\boldsymbol{v}, \boldsymbol{w})$ for every $\boldsymbol{v} \in [n]^\ell$ and $\boldsymbol{w} \in [m]^r$. The circuit is of size at most $4k!^2 k \cdot (n+m)^{k+1} \cdot \|F\|^2$ and support size at most k. By Lemma 4.9, the desired circuit can be constructed from this circuit by adding one additional summation gate of fan-in $(n+m)^k$. That is,

$$\hom_{F,n,m} = \sum_{\boldsymbol{v} \in [n]^\ell} \sum_{\boldsymbol{w} \in [m]^r} \boldsymbol{F}_{n,m}(\boldsymbol{v}, \boldsymbol{w}).$$ □

4.6 Polynomial Orbit Size Versus Polynomial Size

Theorem 1.1 gives a characterisation for when symmetric circuits with polynomial *orbit size*, rather than polynomial total size, exist. A super-polynomial lower bound on the orbit size of course implies a super-polynomial lower bound on the circuit size itself, but for upper bounds, this is not true: There can exist symmetric circuits with polynomial orbit size but whose total size is super-polynomial. The following theorem shows that under the common assumption $\mathsf{VP} \neq \mathsf{VNP}$, this situation indeed arises: There exist polynomials in $\mathfrak{T}^k_{n,m}$ that do not admit polynomial size circuits (neither symmetric nor asymmetric), unless $\mathsf{VP} = \mathsf{VNP}$. Thus, Theorem 1.1 is best-possible, and obtaining a characterisation of *total* circuit size rather than orbit size can be expected to be as difficult for symmetric circuits as for general ones.

Theorem 3.2. *There is a* VNP*-hard family of polynomials* $(p_n)_{n\in\mathbb{N}}$ *such that* $p_n \in \mathfrak{T}^2_{n,n}$ *for all* $n \in \mathbb{N}$.

Proof. We consider a family of polynomials that was shown to be VNP-hard in [10, Theorem 1], and modify it. For a partition $\lambda \vdash d$, write R for the sets of parts of λ, and, for $i \in [d]$, write s_i for the number of size-i parts in λ. Let $n \geq d$ be an integer. Consider the *monomial symmetric polynomial*:

$$m_\lambda(y_1, \dots, y_n) \coloneqq \alpha_\lambda \sum_{f\colon R \hookrightarrow [n]} \prod_{r\in R} y_{f(r)}^{|r|}.$$

Here, $\alpha_\lambda \coloneqq \prod_{i=1}^{d} \frac{1}{s_i!}$ is such that all coefficients in m_λ are 1.

In [10, Theorem 1], the existence of polynomial functions $r, s\colon \mathbb{N} \to \mathbb{N}$ and of partitions $\lambda_n \vdash r(n)$, $n \in \mathbb{N}$, is established such that the family of polynomials $m_{\lambda_n}(y_1, \dots, y_{s(n)})$ is VNP-complete.

Given $m_\lambda(y_1, \dots, y_n)$, we define a polynomial $p_\lambda \in \mathbb{Q}[\mathcal{X}_{n,n}]$ by substituting, for every $v \in [n]$, the expression $\sum_{w\in[n]} x_{vw}$ for y_v. Clearly, m_λ can be recovered from p_λ by substituting $\frac{1}{n} y_v$ for x_{vw} for all $v, w \in [n]$. Hence, the VNP-complete family in [10, Theorem 1] gives rise to a VNP-complete family of polynomials of the form $p_\lambda \in \mathbb{Q}[\mathcal{X}_{n,n}]$. It remains to prove membership in $\mathfrak{T}^2_{n,n}$.

To that end, let $\lambda \vdash d$ and $n \geq d$ be arbitrary. Recall that R denotes the set of parts of λ and apply Möbius inversion to the outer sum. By Lemma A.2,

$$\begin{aligned} m_\lambda(y_1, \dots, y_n) &= \alpha_\lambda \sum_{\pi\in\Pi(R)} \mu_\pi \sum_{f\colon R/\pi \to [n]} \prod_{r\in R} y_{f\pi(r)}^{|r|} \\ &= \alpha_\lambda \sum_{\pi\in\Pi(R)} \mu_\pi \sum_{f\colon R/\pi \to [n]} \prod_{s\in R/\pi} y_{f(s)}^{\sum_{r\in s}|r|} \\ &= \alpha_\lambda \sum_{\pi\in\Pi(R)} \mu_\pi \prod_{s\in R/\pi} \sum_{v\in[n]} y_v^{\sum_{r\in s}|r|}. \end{aligned}$$

For $\varphi(v) \coloneqq \sum_{w\in[n]} x_{vw}$ where $v \in [n]$, we have $\varphi \in \mathfrak{T}^2_{n,n}(1, 0)$ by Lemma 4.9 and Example 4.5. Similarly, $\sum_{v\in[n]} \varphi(v)^\ell \in \mathfrak{T}^2_{n,n}$ for every integer $\ell \geq 1$ by Lemmas 4.9 and 4.11. Hence, the desired polynomial $p_\lambda = m_\lambda(\varphi(1), \dots, \varphi(n))$ is a linear combination of polynomials in $\mathfrak{T}^2_{n,m}$ and thus in $\mathfrak{T}^2_{n,m}$ itself. □

5 Counting Width and Orbit Size Lower Bounds

Theorem 1.1, the main result of the preceding section, characterises the polynomials which are computed by symmetric circuits of polynomial orbit size. Ultimately, the goal of the symmetric circuit programme is to show super-polynomial lower bounds on the orbit and hence circuit size, for polynomials for which this is not currently possible in the non-symmetric circuit setting. However, Theorem 1.1 itself does not give us an immediate tool to prove such lower bounds in general.

In this section and in the following ones, we consider more restricted families of polynomials with the goal of determining their orbit size. To the best of our knowledge, the only available tool for proving orbit size lower bounds is the *counting width* as introduced by 'Definability of semidefinite programming and Lasserre lower bounds for CSPs' [18]. It describes the descriptive complexity of a graph parameter.

In order to define counting width, we view (n, m)-vertex bipartite graphs as elements of $\{0, 1\}^{n\times m}$. More generally, an *edge-weighted* (n, m)-vertex bipartite graph is an element of $\mathbb{Q}^{n\times m}$.

Let $\tau \coloneqq \{A, B\} \cup \{R_q \mid q \in \mathbb{Q}\}$ denote the infinite relational signature with two unary symbols A and B and one binary symbol R_q for every $q \in \mathbb{Q}$. We view elements $G \in \mathbb{Q}^{n\times m}$ as τ-structures with universe $n \times m$ by interpreting $A^G \coloneqq [n]$, $B^G \coloneqq [m]$, and $R_q^G = \{ij \in n \times m \mid G(ij) = q\}$ for $q \in \mathbb{Q}$. An *isomorphism* between $G, H \in \mathbb{Q}^{n\times m}$ is a bijection $h\colon [n] \uplus [m] \to [n] \uplus [m]$ such that $G \circ h = H$.

Definition 5.1 ([18])**.** Let $n, m \in \mathbb{N}$. Let p be an $\mathbf{Sym}_n \times \mathbf{Sym}_m$-invariant function on domain $\mathbb{Q}^{n\times m}$.

1. The *counting width of p on (n,m)-vertex simple graphs* is the least integer k such that, for all $G, H \in \{0,1\}^{n\times m}$, if G and H are $\mathcal{C}^k$-equivalent, then $p(G) = p(H)$.
2. The *counting width of p on (n,m)-vertex edge-weighted graphs* is the least integer k such that, for all $G, H \in \mathbb{Q}^{n\times m}$, if G and H are $\mathcal{C}^k$-equivalent, then $p(G) = p(H)$.

Since p is assumed to be isomorphism-invariant and every edge-weighted graph can be defined in $\mathcal{C}^{n+m}$ up to isomorphism, the counting width on (n,m)-vertex graphs is at most $n+m$ and well-defined. Clearly, the counting width on edge-weighted graphs is an upper bound on the counting width on simple graphs.

The following theorem establishes the desired connection between counting width and orbit size. It is essentially due to 'On Symmetric Circuits and Fixed-Point Logics' [1].

Theorem 5.2. *Let $(p_{n,m})_{n,m\in\mathbb{N}}$ be a family of polynomials.*

1. *If the $p_{n,m}$ admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of orbit size polynomial in $n+m$, then there is a constant $k \in \mathbb{N}$ that bounds the counting width of $p_{n,m}$ on (n,m)-vertex edge-weighted graphs, for all $n, m \in \mathbb{N}$.*
2. *If the $p_{n,m}$ admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of orbit size $2^{o(n)}$, then the counting width of $p_{n,m}$ on (n,m)-vertex edge-weighted graphs is at most $o(n)$.*

Proof. We prove the contrapositive of the first statement. Suppose the counting width of $p_{n,m}$ on (n,m)-vertex edge-weighted graphs is not bounded by a constant. Then for every $k \in \mathbb{N}$, there exist $n, m \in \mathbb{N}$ such that we find $\mathcal{C}^k$-equivalent (n,m)-vertex edge-weighted graphs G, H with $p_{n,m}(G) \neq p_{n,m}(H)$. The graphs being $\mathcal{C}^k$-equivalent means that Duplicator has a winning strategy in the bijective k-pebble played on (G, H). Then by [19, Theorem 4.8], any $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuit whose support sizes are bounded by $k/2$ yields the same output on the adjacency matrices of G and H (the game in that theorem is not the standard bijective k-pebble game, but an equivalent one). Since $p_{n,m}(G) \neq p_{n,m}(H)$, no such circuit can represent $p_{n,m}$. Since this argument can be made for every $k \in \mathbb{N}$, this shows that there do not exist $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits with constant support size for all $p_{n,m}$. Then by the first part of Lemma 2.4 (and the fact that circuits can always be assumed to be rigid), there do not exist symmetric circuits with polynomial orbit size.

The second statement can be shown similarly: Suppose the counting width of $p_{n,m}$ on (n,m)-vertex edge-weighted graphs is $\Omega(n)$. Then for every function $f(n) \in o(n)$, there exist $n, m \in \mathbb{N}$ such that we find $\mathcal{C}^{f(n)}$-equivalent (n,m)-vertex edge-weighted graphs G, H with $p_{n,m}(G) \neq p_{n,m}(H)$. With the same reasoning as before, using the second part of Lemma 2.4 now, we conclude that there do not exist $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits for $p_{n,m}$ with orbit size $2^{o(n)}$. □

By Theorem 5.2, if a family of polynomials $p_{n,m} \in \mathbb{Q}[\mathcal{X}_{n,m}]$ admits $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of orbit size polynomial in $n+m$, then the counting width of $p_{n,m}$ is bounded on edge-weighted bipartite graphs. In fact, the counting width corresponds precisely to the treewidth of the graphs in the homomorphism representation:

Theorem 5.3. *Let $k, n, m \in \mathbb{N}$. If $p \in \mathfrak{T}^k_{n,m}$, then the counting width of p on edge-weighted bipartite (n,m)-vertex graphs is at most k.*

Proof. Since p is a linear combination of homomorphism polynomials, it suffices to show that, if $G, H \in \mathbb{Q}^{n\times m}$ are $\mathcal{C}^k$-equivalent, then $\hom_{F,n,m}(G) = \hom_{F,n,m}(H)$ for every bipartite multigraph F with $\mathrm{tw}(F) < k$. To that end, consider the following claim.

For relational structures G and H over the same universe U, we say that $(G, \boldsymbol{v})$ and $(H, \boldsymbol{w})$ are $\mathcal{C}^k$-*equivalent* for $\boldsymbol{v}, \boldsymbol{w} \in U^\ell$ and $\ell \leq k$ if, for every $\mathcal{C}^k$-formula φ with ℓ free variables, $G \models \varphi(\boldsymbol{v}) \iff H \models \varphi(\boldsymbol{w})$. We prove the following:

Claim 5.3a. Let $\ell, r \in \mathbb{N}$ such that $\ell + r \leq k$ and $\boldsymbol{F} \in \mathcal{T}^k(\ell, r)$. Let $\boldsymbol{v}, \boldsymbol{v}' \in [n]^\ell$ and $\boldsymbol{w}, \boldsymbol{w}' \in [m]^r$. If $G, G' \in \mathbb{Q}^{n\times m}$ are such that $(G, \boldsymbol{v}, \boldsymbol{w})$ and $(G', \boldsymbol{v}', \boldsymbol{w}')$ are $\mathcal{C}^k$-equivalent, then $\boldsymbol{F}_{n,m}(\boldsymbol{v}, \boldsymbol{w})(G) = \boldsymbol{F}_{n,m}(\boldsymbol{v}', \boldsymbol{w}')(G')$.

Proof of Claim. Write $\boldsymbol{F} = (F, \boldsymbol{a}, \boldsymbol{b})$ and $V(F) = A \uplus B$ for its bipartition. By Definition 4.6 and [4, Lemma 7], there exists a tree decomposition (T, β) of F such that

1. there exists a vertex $q \in V(T)$ such that $\beta(q) = \{a_1, \ldots, a_\ell, b_1, \ldots, b_r\}$,
2. for every $s \in V(T)$ which has two children $t_1, t_2 \in V(T)$ in the rooted tree (T, q), we have $\beta(t_1) = \beta(s) = \beta(t_2)$,

3. for every $s \in V(T)$ which has a single child $t \in V(T)$ in the rooted tree (T, q), we have $\beta(t) \subseteq \beta(s)$ and $|\beta(s)| = |\beta(t)| + 1$ or $\beta(s) \subseteq \beta(t)$ and $|\beta(t)| = |\beta(s)| + 1$,
4. for every $s \in V(T)$, $|\beta(s)| \leq k$.

We prove the claim by induction on the height of the rooted tree (T, q).

For the base case, suppose that q is the only vertex in T. Since all vertices of F are labelled, $\boldsymbol{F}_{n,m}(\boldsymbol{v}, \boldsymbol{w})(G)$ depends solely on the isomorphism type of the edge-weighted bipartite subgraph of G induced by $\boldsymbol{v}$ and $\boldsymbol{w}$. Since $\ell + r \leq k$, this isomorphism type can be defined in $\mathcal{C}^k$. This implies the desired statement.

For the inductive step, consider three cases:

1. The vertex q has a single child $t \in V(T)$, $\beta(t) \subseteq \beta(q)$, and $|\beta(q)| = |\beta(t)| + 1$.
   Define F' as the subgraph of F induced by $\bigcup_{s \in V(T) \setminus \{q\}} \beta(s)$. Suppose without loss of generality that $\beta(q) \cap B = \beta(t) \cap B$. Let $i \in [\ell]$ be such that $a_i \notin \beta(t)$. Then $\boldsymbol{F}' \coloneqq (F', \boldsymbol{a}[i/], \boldsymbol{b})$ is a graph to which the inductive hypothesis applies.
   If $(G, \boldsymbol{v}, \boldsymbol{w})$ and $(G', \boldsymbol{v}', \boldsymbol{w}')$ are $\mathcal{C}^k$-equivalent, then so are $(G, \boldsymbol{v}[i/], \boldsymbol{w})$ and $(G', \boldsymbol{v}'[i/], \boldsymbol{w}')$. By the inductive hypothesis, $\boldsymbol{F}'_{n,m}(\boldsymbol{v}[i/], \boldsymbol{w})(G) = \boldsymbol{F}'_{n,m}(\boldsymbol{v}'[i/], \boldsymbol{w}')(G')$.
   Now $\boldsymbol{F}$ differs from $\boldsymbol{F}'$ only in the vertex a_i and the potential edges connecting a_i to the vertices in $\boldsymbol{b}$. More precisely,
   $$\boldsymbol{F}_{n,m}(\boldsymbol{v}, \boldsymbol{w})(G) = \boldsymbol{F}'_{n,m}(\boldsymbol{v}[i/], \boldsymbol{w})(G) \cdot \prod_{a_i b_j \in E(F)} G(v_i w_j).$$
   Since $(G, \boldsymbol{v}, \boldsymbol{w})$ and $(G', \boldsymbol{v}', \boldsymbol{w}')$ are $\mathcal{C}^k$-equivalent, $G(v_i w_j) = G'(v'_i w'_j)$ for all $i \in [\ell]$ and $j \in [r]$. This implies the claim.
2. The vertex q has a single child $t \in V(T)$, $\beta(t) \supseteq \beta(q)$, and $|\beta(q)| + 1 = |\beta(t)|$.
   Without loss of generality $\beta(t) \cap B = \beta(q) \cap B$. Write a for the unique element of $\beta(t) \setminus \beta(q)$. The inductive hypothesis applies to $\boldsymbol{F}' \coloneqq (F, \boldsymbol{a}a, \boldsymbol{b})$. Here, we insert the vertex a at the end of the tuple only to ease notation. If $(G, \boldsymbol{v}, \boldsymbol{w})$ and $(G', \boldsymbol{v}', \boldsymbol{w}')$ are $\mathcal{C}^k$-equivalent and $\boldsymbol{v}$ and $\boldsymbol{w}$ are tuples of total length at most $k-1$, then there exists a bijection $\pi \colon [n] \to [n]$ such that $(G, \boldsymbol{v}v, \boldsymbol{w})$ and $(G', \boldsymbol{v}'\pi(v), \boldsymbol{w}')$ are $\mathcal{C}^k$-equivalent for all $v \in [n]$. By the inductive hypothesis, $\boldsymbol{F}'_{n,m}(\boldsymbol{v}v, \boldsymbol{w})(G) = \boldsymbol{F}'_{n,m}(\boldsymbol{v}'\pi(v), \boldsymbol{w}')(G')$ for all $v \in [n]$. Hence, by Lemma 4.9,
   $$\boldsymbol{F}_{n,m}(\boldsymbol{v}, \boldsymbol{w})(G) = \sum_{v \in [n]} \boldsymbol{F}'_{n,m}(\boldsymbol{v}v, \boldsymbol{w})(G) = \sum_{v \in [n]} \boldsymbol{F}'_{n,m}(\boldsymbol{v}'v', \boldsymbol{w}')(G') = \boldsymbol{F}_{n,m}(\boldsymbol{v}', \boldsymbol{w}')(G').$$
3. The vertex q has several children $t_1, \ldots, t_\ell$.
   For $i \in [\ell]$, define F^i as the subgraph of F induced by $\beta(t_i)$, and $\beta(s)$ for all descendents s of t. Then the inductive hypothesis applies to $\boldsymbol{F}^i \coloneqq (F^i, \boldsymbol{a}, \boldsymbol{b})$. By Lemma 4.11, $\boldsymbol{F}_{n,m}(\boldsymbol{v}, \boldsymbol{w}) = \prod_{i=1}^{\ell} \boldsymbol{F}^i_{n,m}(\boldsymbol{v}, \boldsymbol{w})$. The claim follows. ⌟

The claim implies the theorem with $\ell = 0 = r$. □

Theorems 1.1 and 5.3 yield an alternative proof of the first claim of Theorem 5.2. We are going to establish the converse of Theorem 5.3 in various restricted cases, as summarised in Figure 1. To start with, we note that if we only care about the semantics of the polynomials as functions on simple graphs, then counting width and treewidth indeed coincide.

5.1 Characterisation of polynomials up to multilinearisation

The *multilinearisation* of a polynomial p is obtained from p by setting all non-zero exponents to 1. If we consider polynomials only up to multilinearisation, i.e. up to their evaluation under $\{0,1\}$-assignments, then the following converse of Theorem 5.3 holds.

Theorem 3.4. *Let $k, n, m \in \mathbb{N}$. For $p \in \mathbb{Q}[\mathcal{X}_{n,m}]$, the following are equivalent:*

1. *there exists $q \in \mathfrak{T}^k_{n,m}$ such that p and q are equal up to multilinearisation,*
2. *the counting width of p on (n,m)-vertex simple bipartite graphs is at most k.*

Proof. For the forward direction, observe that, by Theorem 5.3, the desired statement holds for q. The polynomials p and q assume the same values on the Boolean cube $\{0,1\}^{n \times m}$. Hence, the counting width of p on (n,m)-vertex simple bipartite graphs is at most k.

For the converse direction, we write the function $p \colon \{0,1\}^{n \times m} \to \mathbb{Q}$ as a linear combination of $\mathcal{C}^k$-type indicator functions. This argument requires interpolation and thus relies on the fact that there are only finitely many simple (n,m)-vertex bipartite graphs.

Let $C_1 \uplus \cdots \uplus C_r = \{0,1\}^{n\times m}$ be an enumeration of the $\mathcal{C}^k$-types of simple (n,m)-vertex bipartite graphs. Pick representatives $G_1, \dots, G_r$. By [26], for $i \neq j$, there exists a bipartite graph F_{ij} of treewidth at most $k-1$ such that $\hom(F_{ij}, G_i) \neq \hom(F_{ij}, G_j)$. For $i \in [r]$, define the polynomial

$$q_i \coloneqq \prod_{j\neq i} \frac{\hom_{F_{ij},n,m} - \hom(F_{ij}, G_j)}{\hom(F_{ij}, G_i) - \hom(F_{ij}, G_j)} \in \mathfrak{T}^k_{n,m}.$$

Here, $\hom_{F_{ij},n,m}$ is a polynomial in $\mathfrak{T}^k_{n,m}$ while $\hom(F_{ij}, G_j)$ is merely an integer. For all $G \in \{0,1\}^{n\times m}$, $q_i(G) = 1$ if, and only if, $G \in C_i$, and $q_i(G) = 0$ otherwise. Since p is constant on $\mathcal{C}^k$-types, it is

$$p = \sum_{i=1}^{r} \alpha_i q_i$$

as functions on $\{0,1\}^{n\times m}$ for some coefficients $\alpha_i \in \mathbb{Q}$. Hence, p is equal to the polynomial $\sum_{i=1}^r \alpha_i q_i \in \mathfrak{T}^k_{n,m}$ up to multilinearisation. □

We see in Example 7.10 that the statements in Theorem 3.4 do not imply that the counting width of p on (n,m)-vertex *edge-weighted* graphs is bounded. The argument yielding Theorem 3.4 is bound to fail when multilinearisation is not applied:

Example 5.4. Let F be a graph and $n, m \in \mathbb{N}$. Write $C = \{\hom(F,G) \mid G \in \{0,1\}^{n\times m}\}$ for the finite set of homomorphism counts from F to any simple (n,m)-vertex graph. Consider the polynomial

$$\begin{aligned} p \coloneqq \hom_{F,n,m} \prod_{c\in C} (\hom_{F,n,m} - c) &= \hom_{F,n,m} \sum_{C'\subseteq C} \hom_{|C\setminus C'|F,n,m} \prod_{c\in C'} (-1)^{|C'|}\cdot c \\ &= \sum_{C'\subseteq C} \hom_{(|C\setminus C'|+1)F,n,m} \prod_{c\in C'} (-1)^{|C'|}\cdot c. \end{aligned}$$

Here, kF for a graph F and an integer $k \geq 1$ denotes the disjoint union of k copies of F. Then p is a linear combination of homomorphism polynomials and $p(G) = 0$ for all $G \in \{0,1\}^{n\times m}$. Hence, the counting width of p on simple (n,m)-vertex graphs is zero. However, $p \notin \mathfrak{T}^k_{n,m}$ for $k < \operatorname{tw}(F) - 1$ when $n, m \geq |E(F)|$.

6 Symmetric complexity of homomorphism polynomials

Towards a converse of Theorem 5.3, we characterise the homomorphism polynomials and the linear combinations of homomorphism polynomials of sublinear patterns that have bounded counting width.

6.1 Single homomorphism polynomials

We first consider families of single homomorphism polynomials and show the following sufficient and necessary criterion for tractability.

Theorem 6.1. *For every family $(F_{n,m})_{n,m\in\mathbb{N}}$ of bipartite multigraphs, the following are equivalent:*

1. *the treewidth $\operatorname{tw}(F_{n,m})$ is bounded,*
2. *the counting width of $\hom_{F_{n,m},n,m}$ on (n,m)-vertex simple graphs is bounded,*
3. *the counting width of $\hom_{F_{n,m},n,m}$ on (n,m)-vertex edge-weighted graphs is bounded,*
4. *the $\hom_{F_{n,m},n,m}$ admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of orbit size polynomial in $n+m$,*
5. *the $\hom_{F_{n,m},n,m}$ admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of size polynomial in $\|F_{n,m}\| + n + m$ and orbit size polynomial in $n+m$.*

Given the results of the preceding section, it remains to show '2 ⇒ 1'. To that end, we prove a counting width lower bound for homomorphism counts of unbounded treewidth.

6.1.1 The Non-Uniform Homomorphism Distinguishing Closure

Since it is conceivable that a more general statement will contribute to future work, we formulate our counting width lower bound in the language of homomorphism indistinguishability. Two graphs[2] G and H are

[2]In Sections 6.1.1 and 6.1.2, all graphs are simple. Considering homomorphism counts from simple graphs suffices in the context of Theorems 6.1 and 6.9 since we prove lower bounds on the counting width on simple graphs.

homomorphism indistinguishable over a graph class $\mathcal{F}$, in symbols $G \equiv_{\mathcal{F}} H$, if $\hom(F, G) = \hom(F, H)$ for all $F \in \mathcal{F}$. By [23, 26], two graphs are $\mathcal{C}^k$-equivalent if, and only if, they are homomorphism indistinguishable over all graphs of treewidth less than k. Similar characterisations are known for many graph isomorphism relaxations from finite model theory [27, 30], quantum information theory [43], and category theory [16, 44] etc., cf. the monograph [55]. The distinguishing power of such relations is well-understood via the so called *homomorphism distinguishing closure* [49] of a graph class $\mathcal{F}$ defined as

$$\mathrm{cl}(\mathcal{F}) \coloneqq \{F \mid \forall G, H.\ G \equiv_{\mathcal{F}} H \implies \hom(F, G) = \hom(F, H)\}.$$

For example, it is known [45] that the class $\mathcal{TW}_k$ of all graphs of treewidth $\leq k$ is *homomorphism distinguishing closed*, i.e. $\mathrm{cl}(\mathcal{TW}_k) = \mathcal{TW}_k$. By [26], this means that the function $\hom(F, -)$ has counting width at most k if, and only if, $\mathrm{tw}(F) < k$.

In order to treat homomorphism counts $\hom_{F_n,n}$ of patterns F_n depending on the host size n, we introduce the *non-uniform homomorphism distinguishing closure* of a graph class. For $n \in \mathbb{N}$, let

$$\mathrm{cl}_n(\mathcal{F}) \coloneqq \{F \mid \forall G, H \text{ on } n \text{ vertices}.\ G \equiv_{\mathcal{F}} H \implies \hom(F, G) = \hom(F, H)\}.$$

Our motivation for introducing this notion is the observation that $\hom_{F_n,n}$ has counting width at most k if, and only if, $F_n \in \mathrm{cl}_n(\mathcal{TW}_{k-1})$ for all $n \in \mathbb{N}$. Note that $\mathrm{cl}_n(\mathcal{F})$ differs from $\mathrm{cl}(\mathcal{F})$ since e.g. $K_{n+1} \in \mathrm{cl}_n(\mathcal{F})$ for every graph class $\mathcal{F}$, which this is not the case for $\mathrm{cl}(\mathcal{F})$. See Section E for further properties of $\mathrm{cl}_n(\mathcal{F})$.

The following theorem determines the non-uniform homomorphism distinguishing closure of $\mathcal{TW}_k$. It also applies to the class of all graphs of pathwidth $\leq k$ [55], treedepth $\leq k$ [27], or the class of all planar graphs [49]. All these graph classes satisfy the following Proviso 6.2.

Proviso 6.2. *Let $\mathcal{F}$ be a graph class such that*

1. *$\mathcal{F}$ is closed under taking minors,*
2. *$\mathcal{F}$ is non-empty, i.e. contains the one-vertex graph K_1,*
3. *$\mathcal{F}$ is closed under disjoint unions, i.e. if $F_1, F_2 \in \mathcal{F}$, then $F_1 + F_2 \in \mathcal{F}$, and*
4. *$\mathcal{F}$ is closed under weak oddomorphisms, i.e. if $F \to G$ is a weak oddomorphism, cf. Definition C.3, and $F \in \mathcal{F}$, then $G \in \mathcal{F}$.*

Theorem 6.3. *For every graph class $\mathcal{F}$ satisfying Proviso 6.2, there exists a function $f\colon \mathbb{N} \to \mathbb{N}$ such that, for every graph F with chromatic number $\chi(F)$, if $F \in \mathrm{cl}_{f(\chi(F))}(\mathcal{F})$, then $F \in \mathcal{F}$.*

Proof. By the Robertson–Seymour theorem [50], $\mathcal{F}$ is characterised by a finite a set of forbidden minors. Let N denote the maximum number of vertices in any such minor. Let $f\colon \mathbb{N} \to \mathbb{N}$ be the function $c \mapsto c(N2^{N-1}+1)$.

By contraposition, suppose that $F \notin \mathcal{F}$. Write $c \coloneqq \chi(F)$. Hence, F contains a minor on at most N vertices which does not belong to $\mathcal{F}$. In particular, F contains an induced minor P on at most N vertices which does not belong to $\mathcal{F}$. An *induced minor* of a simple graph F is a graph which can be obtained from an induced subgraph of F by contracting edges. By [5, Lemma 4], P can be assumed to be connected.

Let G'_0 and G'_1 denote the CFI graphs of P (see also Appendix C). Since $\mathcal{F}$ is closed under weak oddomorphisms, by Theorem C.4, G'_0 and G'_1 are homomorphism indistinguishable over $\mathcal{F}$. Construct G_0 and G_1 by making the lexicographic products $G'_0 \cdot K_c$ and $G'_1 \cdot K_c$, respectively, adjacent with a copy of K_c, i.e. $G_i \coloneqq (G'_i \cdot K_c) \boxplus K_c$ for $i \in \{0, 1\}$. The resulting graphs have at most $N2^{N-1}c + c = f(c)$ vertices.

By [56, Theorem 14], $G'_0 \cdot K_c$ and $G'_1 \cdot K_c$ are homomorphism indistinguishable over $\mathcal{F}$. By applying the argument yielding [56, Theorem 14] to Equation (4), it follows that G_0 and G_1 are also homomorphism indistinguishable over $\mathcal{F}$.

We proceed by showing that $\hom(F, G_0) \neq \hom(F, G_1)$. For $i \in \{0, 1\}$,

$$\begin{aligned}\hom(F, G_i) &\overset{(4)}{=} \sum_{U \subseteq V(F)} \hom(F[U], G'_i \cdot K_c) \hom(F - U, K_c) \\ &\overset{(5)}{=} \sum_{U \subseteq V(F)} \sum_{\mathcal{R} \in \Gamma(F[U])} \hom(F[U]/\mathcal{R}, G'_i) \hom\Big(\coprod_{R \in \mathcal{R}} F[R], K_c\Big) \hom(F - U, K_c). \qquad (12)\end{aligned}$$

Recall that $\Gamma(K)$ for a graph K denotes the set of all partitions $\mathcal{R}$ of $V(K)$ such that $K[R]$ is connected for all $R \in \mathcal{R}$.

By Theorem C.4, $\hom(F[U]/\mathcal{R}, G'_0) \geq \hom(F[U]/\mathcal{R}, G'_1)$ for all U and $\mathcal{R}$. Furthermore, $\hom(\coprod_{R \in \mathcal{R}} F[R], K_c) > 0$ for all $\mathcal{R}$ and $\hom(F - U, K_c) > 0$ for all U by construction. Moreover, $\hom(F[U]/\mathcal{R}, G'_0) > \hom(F[U]/\mathcal{R}, G'_1)$ for U and $\mathcal{R}$ such that $F[U]/\mathcal{R} \cong P$ by Theorem C.4. This choice of U and $\mathcal{R}$ exists as P is an induced minor of F. It follows that $\hom(F, G_0) \neq \hom(F, G_1)$. □

6.1.2 Homomorphisms Respecting Bipartitions

In order to apply Theorem 6.3 to prove Theorem 6.1, we need to handle the technicality that the graph parameters this article is concerned with do not count general homomorphisms but homomorphisms respecting fixed bipartitions.

For a bipartite graph F without distinguished bipartition and a bipartition $V(F) = A \uplus B$, we write $F_{A,B}$ for the bipartite graph F with distinguished bipartition $A \uplus B$. The number of all graph homomorphisms from F to G which do not necessarily respect any fixed bipartition is denoted by $\hom(F, G)$ while $\hom(F_{A,B}, G_{X,Y})$ denotes the number of homomorphisms $h\colon F \to G$ such that $h(A) \subseteq X$ and $h(B) \subseteq Y$.

For a class $\mathcal{F}$ of graphs with fixed bipartitions, consider

$$\mathrm{cl}^{\mathrm{bip}}_{n,m}(\mathcal{F}) \coloneqq \{F \mid \forall G, H \text{ on } (n,m) \text{ vertices. } G \equiv_{\mathcal{F}} H \implies \hom(F,G) = \hom(F,H)\}\,.$$

Here, F, G, and H range over graphs with fixed bipartitions. We identify a class $\mathcal{F}$ of bipartite graphs with the class of all graphs $F_{A,B}$ with fixed bipartition for $F \in \mathcal{F}$ and any bipartition A, B of F. We show the following:

Proposition 6.4. *For every class $\mathcal{F}$ of bipartite graphs and $n \in \mathbb{N}$, it holds that $\mathrm{cl}^{\mathrm{bip}}_{n,n}(\mathcal{F}) \subseteq \mathrm{cl}_n(\mathcal{F})$.*

Towards Proposition 6.4, we make the following observations:

Fact 6.5. *For every connected bipartite graph F, $\hom(F, K_2) = 2$.*

In other words, every connected bipartite graph admits a bipartition which is unique up to flipping the sides.

Lemma 6.6. *Let F and G be bipartite graphs with bipartitions $V(F) = A \uplus B$ and $V(G) = X \uplus Y$. Suppose that F is connected. Then $\hom(F, G) = \hom(F_{A,B}, G_{X,Y}) + \hom(F_{A,B}, G_{Y,X})$.*

Proof. First observe that $\hom(F, G) \geq \hom(F_{A,B}, G_{X,Y}) + \hom(F_{A,B}, G_{Y,X})$. This is because bipartition-respecting homomorphisms are homomorphisms and the sets of homomorphisms counted by $\hom(F_{A,B}, G_{X,Y})$ and $\hom(F_{A,B}, G_{Y,X})$ are disjoint.

For the converse inequality, observe that for every homomorphism $h\colon F \to G$, it holds $h(A) \cap h(B) = \emptyset$. Indeed, if there are $a \in A$ and $b \in B$ such that $h(a) = h(b)$, then the odd-length path connecting a and b in F is mapped to an odd-length cycle in G, contradicting that G is bipartite.

Let G' denote the subgraph of G containing the edges and vertices in the image of F under h. As the image of a connected graph, G' is connected. As subgraph of G, G' is bipartite. Hence, the bipartition of G' is unique up to flipping sides by Fact 6.5.

This bipartition is, on the one hand, given by $(X \cap V(G')) \uplus (Y \cap V(G'))$ and, on the other hand, given by $h(A) \uplus h(B)$. It follows that $h(A) \subseteq X$ and $h(B) \subseteq Y$, or $h(B) \subseteq X$ and $h(A) \subseteq Y$, as desired. □

The *bipartite double cover* of a graph G is the graph $G \times K_2$ where $\times$ denotes the categorical graph product. Its vertex set is $V(G) \times \{0,1\}$ and (v,i) and (w,j) are adjacent if, and only if, $vw \in E(G)$ and $i \neq j$.

Lemma 6.7. *Let F and G be bipartite graphs. Let H denote the bipartite double cover of G. Let $V(F) = A \uplus B$ be the bipartition of F and $V(H) = X \uplus Y \coloneqq (V(G) \times \{0\}) \uplus (V(G) \times \{1\})$ be the canonical bipartition of the bipartite double cover of G. Then $\hom(F_{A,B}, H_{X,Y}) = \hom(F_{A,B}, H_{Y,X})$.*

Proof. Write $V(H) = V(G) \times \{0,1\}$. The automorphism φ of H given by $(v,i) \mapsto (v, 1-i)$ for all $v \in V(G)$ and $i \in \{0,1\}$ induces a bijection between the set of homomorphisms $F_{A,B} \to H_{X,Y}$ and the set of homomorphisms $F_{A,B} \to H_{Y,X}$. □

Corollary 6.8. *Let F be a bipartite graph with bipartition $V(F) = A \uplus B$. Let G be a graph. Write H for the bipartite double cover of G and write $X \uplus Y = V(H)$ for its canonical bipartition. Then $\hom(F, G) = \hom(F_{A,B}, H_{X,Y})$.*

Proof. Write $F = F^1 + \cdots + F^r$ for the connected components of F. Then

$$\hom(F,G) = \prod_{i=1}^{r} \hom(F_i, G) = \prod_{i=1}^{r} \frac{\hom(F_i, H)}{2} = \prod_{i=1}^{r} \hom((F_i)_{A,B}, H_{X,Y}) = \hom(F_{A,B}, H_{X,Y}).$$

Here, the first identity follows from Equation (1), the second from Fact 6.5 together with Equation (2), the third from Lemmas 6.6 and 6.7, and the fourth from Equation (1). □

We now prove Proposition 6.4:

Proof of Proposition 6.4. Let G and H be graphs on n vertices such that $G \equiv_{\mathcal{F}} H$. By Corollary 6.8, the bipartite double covers $G'_{W,X}$ and $H'_{Y,Z}$ of G and H are homomorphism indistinguishable over $\mathcal{F}$ as bipartite graphs with fixed bipartitions. Let $F_{A,B} \in \mathrm{cl}^{\mathrm{bip}}_{n,n}(\mathcal{F})$. By Corollary 6.8, $\hom(F, G) = \hom(F_{A,B}, G'_{W,X}) = \hom(F_{A,B}, H'_{Y,Z}) = \hom(F, H)$, since $G'_{W,X}$ and $H'_{Y,Z}$ have (n, n) vertices. Hence, $F \in \mathrm{cl}_n(\mathcal{F})$. □

6.1.3 Proof of Theorem 6.1

Proof of Theorem 6.1. Theorem 4.3 yields that 1 implies 5. That 5 implies 4 is immediate. That 4 implies 3 follows from Theorem 5.2. By Definition 5.1, it holds that 3 implies 2. Thus, it remains to show that 2 implies 1.

Write $\mathcal{B}$ for the class of all bipartite graphs and fix a family $(F_{n,m})_{n,m\in\mathbb{N}}$ of bipartite multigraphs, as in Theorem 6.1. Analogous to arguments in [26], if $\hom_{F_{n,m},n,m}$ has counting width at most $k+1$ on (n,m)-vertex simple bipartite graphs, then $F_{n,m} \in \mathrm{cl}^{\mathrm{bip}}_{n,m}(\mathcal{TW}_k \cap \mathcal{B})$ for all $n, m \in \mathbb{N}$. By Proposition 6.4 and Theorem 6.3, for $\ell := \min\{n, m\}$ sufficiently larger than k,

$$F_{n,m} \in \mathrm{cl}^{\mathrm{bip}}_{n,m}(\mathcal{TW}_k \cap \mathcal{B}) \cap \mathcal{B} \subseteq \mathrm{cl}^{\mathrm{bip}}_{\ell,\ell}(\mathcal{TW}_k \cap \mathcal{B}) \cap \mathcal{B} \subseteq \mathrm{cl}_\ell(\mathcal{TW}_k \cap \mathcal{B}) \cap \mathcal{B} \subseteq \mathrm{cl}_\ell(\mathcal{TW}_k) \cap \mathcal{B} \subseteq \mathcal{TW}_k.$$

Hence, the $F_{n,m}$ have bounded treewidth. □

6.2 Linear Combinations of Homomorphism Polynomials for Patterns of Sublinear Size

In general, families of symmetric polynomials are not single homomorphism polynomials but linear combinations of homomorphism polynomials (see Lemma 3.1). In this section, we tackle this general situation, at least for pattern graphs of sublinear size:

Theorem 6.9. *For $n, m \in \mathbb{N}$, let $F_{n,m,i}$ be bipartite multigraphs and $\alpha_{n,m,i} \in \mathbb{Q} \setminus \{0\}$. Let $p_{n,m} := \sum_i \alpha_{n,m,i} \hom_{F_{n,m,i},n,m}$. Suppose that, for all $\nu, \mu \in \mathbb{N}$, there are $n', m' \in \mathbb{N}$ such that $\max_i |F_{\nu n,\mu m,i}| \leq \min\{n, m\}$ for all $n > n'$ and $m > m'$. Then the following are equivalent:*

1. *$\max_i \mathrm{tw}(F_{n,m,i})$ is bounded,*
2. *the counting width of $p_{n,m}$ on (n,m)-vertex simple graphs is bounded,*
3. *the counting width of $p_{n,m}$ on (n,m)-vertex edge-weighted graphs is bounded,*
4. *the $p_{n,m}$ admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of orbit size polynomial in $n+m$.*

We prove the theorem by establishing a descriptive complexity monotonicity for non-uniform graph motif parameters whose patterns are of sublinear size. This monotonicity resembles a similar statement proven by 'Homomorphisms Are a Good Basis for Counting Small Subgraphs' [9] for graph motif parameters and fixed-parameter computations. They show that a linear combination $\sum \alpha_F \hom(F, -)$ is as hard for FPT computations as the hardest term $\hom(F, -)$ occurring in it. We prove a similar monotonicity for the descriptive complexity of homomorphism counts whose patterns may depend on the size of the host, thus generalising [56, Lemma 6]. In order to minimise the technical overhead, we first consider the case of not necessarily bipartite graphs.

Theorem 6.10 (Descriptive Complexity Monotonicity)**.** *Let $p_n(-) = \sum_i \alpha_{n,i} \hom(F_{n,i}, -)$ for graphs $F_{n,i}$ and non-zero rational coefficients $\alpha_{n,i}$. Suppose that $v(n) := \max_i |F_{n,i}| \in o(n)$ and that $\max_i \chi(F_{n,i}) \in O(1)$. Let $\mathcal{F}$ be as in Proviso 6.2 and suppose that*

$$G \equiv_{\mathcal{F}} H \implies p_n(G) = p_n(H)$$

for all graphs G, H on n vertices. Then $F_{n,i} \in \mathcal{F}$ for all sufficiently large n and all i.

Proof. By Proviso 6.2 and since $\hom(F + K_1, G) = n \hom(F, G)$ if G has n vertices, we may suppose that none of the patterns in p_n contains isolated vertices. Under this assumption,

$$p_n(G + \ell K_1) = p_n(G) \tag{13}$$

for all $(n-\ell)$-vertex graphs G and $n, \ell \in \mathbb{N}$. The proof is conducted in two steps. We first show the following:

Claim 6.10a. For every $\nu \in \mathbb{N}$, it holds that $F_{\nu n,i} \in \mathrm{cl}_\nu(\mathcal{F})$ for all $n \gg \nu$ and all i.

Proof of Claim. Let G and H be ν-vertex graphs such that $G \equiv_{\mathcal{F}} H$. Since $v(n) \in o(n)$, there exists $n' \in \mathbb{N}$ such that $v(\nu n) \leq n$ for all $n > n'$. Let K be an arbitrary graph on $\ell \leq n$ vertices. By Equation (2), $G \times K \equiv_{\mathcal{F}} H \times K$. By Proviso 6.2 and [56, Theorem 7], $G \times K + \nu(n-\ell)K_1 \equiv_{\mathcal{F}} H \times K + \nu(n-\ell)K_1$. Hence, by assumption, noting that $G \times K + \nu(n-\ell)K_1$ has νn vertices,

$$p_{\nu n}(G \times K) \overset{(13)}{=} p_{\nu n}(G \times K + \nu(n-\ell)K_1) = p_{\nu n}(H \times K + \nu(n-\ell)K_1) \overset{(13)}{=} p_{\nu n}(H \times K).$$

By Equation (2),

$$p_{\nu n}(G \times K) = \sum_i \alpha_{\nu n,i} \hom(F_{\nu n,i}, G \times K) = \sum_i \alpha_{\nu n,i} \hom(F_{\nu n,i}, G) \hom(F_{\nu n,i}, K)$$

By [42], cf. [56, Lemma 6], the matrix $(\hom(F,K))_{F,K}$ where F, K range over graphs on at most n vertices is invertible. Hence,

$$\alpha_{\nu n,i} \hom(F_{\nu n,i}, G) = \alpha_{\nu n,i} \hom(F_{\nu n,i}, H)$$

for all i. In other words, $F_{\nu n,i} \in \mathrm{cl}_\nu(\mathcal{F})$ for all i. ⌟

In order to deduce the theorem from the claim, fix by Theorem 6.3 an integer $\nu \in \mathbb{N}$ such that $F \in \mathrm{cl}_\nu(\mathcal{F})$ implies $F \in \mathcal{F}$ for all graphs F whose chromatic number does not exceed $\max_n \max_i \chi(F_{n,i})$.

By Claim 6.10a and Theorem 6.3, $F_{\nu n,i} \in \mathcal{F}$ for all sufficiently large n and all i. It remains to take care of the $F_{n,i}$ whose first index is not a multiple of ν. We do so by padding the p_n.

To that end, for every $j < \nu$, consider the graph parameter q^j defined via $q^j_n(G) \coloneqq p_{n+j}(G + jK_1) = p_{n+j}(G)$ for every n-vertex graph G, by Equation (13). By Proviso 6.2 and [56, Theorem 7], for all graphs G and H on n vertices,

$$G \equiv_{\mathcal{F}} H \implies G + jK_1 \equiv_{\mathcal{F}} H + jK_1 \implies p_{n+j}(G + jK_1) = p_{n+j}(H + jK_1) \implies q^j_n(G) = q^j_n(H).$$

Hence, q^j satisfies the assumptions of the theorem and it is $F_{\nu n+j,i} \in \mathrm{cl}_\nu(\mathcal{F})$ for all i and all $n \gg \nu$ by Claim 6.10a. Since the $\nu n + j$ for $j < \nu$ and $n \gg \nu$ cover all indices sufficiently larger than ν, it follows that $F_{n,i} \in \mathrm{cl}_\nu(\mathcal{F})$ for all $n \gg \nu$ and all i. The theorem follows by Theorem 6.3. □

For graphs with fixed bipartitions, the proof is analogous.

Theorem 6.11 (Descriptive Complexity Monotonicity, Bipartite Case)**.**
Let $p_{n,m}(-) = \sum_i \alpha_{n,m,i} \hom(F_{n,m,i}, -)$ for bipartite graphs $F_{n,m,i}$ and non-zero rational coefficients $\alpha_{n,m,i}$. Suppose that, for all $\nu, \mu \in \mathbb{N}$, there exist $n', m' \in \mathbb{N}$, such that $\max_i |F_{\nu n,\mu m,i}| \leq \min\{n, m\}$ for all $n > n'$ and $m > m'$. Let $\mathcal{F}$ be a class of bipartite graphs as in Proviso 6.2 and suppose that

$$G \equiv_{\mathcal{F}} H \implies p_{n,m}(G) = p_{n,m}(H)$$

for all bipartite graphs G, H on (n, m) vertices with fixed bipartitions. Then $F_{n,m,i} \in \mathcal{F}$ for all sufficiently large n, m, and all i.

Proof. As in the proof of Theorem 6.10, it may be supposed that the graphs $F_{n,m,i}$ do not contain isolated vertices. The proof is conducted in two steps. We first show the following:

Claim 6.11a. For every $\nu, \mu \in \mathbb{N}$, it holds that $F_{\nu n,\mu m,i} \in \mathrm{cl}^{\mathrm{bip}}_{\nu,\mu}(\mathcal{F})$ for all $n \gg \nu$, $m \gg \mu$, and all i.

Proof of Claim. Let G and H be (ν, μ)-vertex bipartite graphs such that $G \equiv_{\mathcal{F}} H$. By assumption, there exist n' and m' depending on ν and μ such that $\max_i |F_{\nu n,\mu m,i}| \leq \min\{n, m\}$ for all $n > n'$ and $m > m'$. Let K be an arbitrary bipartite graph on (n'', m'') vertices for $n'' \leq n$ and $m'' \leq n$. By Equation (2), $G \times K \equiv_{\mathcal{F}} H \times K$. By Proviso 6.2 and [56, Theorem 7],

$$G \times K + \nu(n-n'')K_1 + \mu(m-m'')K_1 \equiv_{\mathcal{F}} H \times K + \nu(n-n'')K_1 + \mu(m-m'')K_1.$$

By assumption,

$$\begin{aligned} p_{\nu n,\mu m}(G \times K) &= p_{\nu n,\mu m}(G \times K + \nu(n-n'')K_1 + \mu(m-m'')K_1) \\ &= p_{\nu n,\mu m}(H \times K + \nu(n-n'')K_1 + \mu(m-m'')K_1) \\ &= p_{\nu n,\mu m}(H \times K). \end{aligned}$$

By Equation (2),

$$p_{\nu n,\mu m}(G \times K) = \sum_i \alpha_{\nu n,\mu m,i} \hom(F_{\nu n,\mu m,i}, G \times K) = \sum_i \alpha_{\nu n,\mu m,i} \hom(F_{\nu n,\mu m,i}, G) \hom(F_{\nu n,\mu m,i}, K)$$

By [42], cf. [56, Lemma 6], the matrix $(\hom(F,K))_{F,K}$ where F, K range over (n'',m'')-vertex bipartite graphs for $n'' \leq n$ and $m'' \leq m$ is invertible.[3] Hence,

$$\alpha_{\nu n,\mu m,i} \hom(F_{\nu n,\mu m,i}, G) = \alpha_{\nu n,\mu m,i} \hom(F_{\nu n,\mu m,i}, H)$$

for all i. In other words, $F_{\nu n,\mu m,i} \in \mathrm{cl}^{\mathrm{bip}}_{\nu,\mu}(\mathcal{F})$ for all i. ⌟

In order to deduce the theorem from the claim, fix by Theorem 6.3 and Proposition 6.4 an integer $\nu \in \mathbb{N}$ such that $F \in \mathrm{cl}^{\mathrm{bip}}_{\nu,\nu}(\mathcal{F})$ implies $F \in \mathcal{F}$ for all bipartite graphs F. For every $j, \ell < \nu$, consider the graph parameter $q^{j,\ell}$ defined via $q^{j,\ell}_{n,m}(G) \coloneqq p_{n+j,m+\ell}(G + jK_1 + \ell K_1)$ for every (n,m)-vertex bipartite graph G. As we may assume that the patterns in p do not contain isolated vertices, it holds that $q^{j,\ell}_{n,m}(G) = p_{n+j,m+\ell}(G + jK_1 + \ell K_1) = \sum \alpha_{n+j,m+\ell,i} \hom(F_{n+j,m+\ell,i}, G)$ and hence $q^{j,\ell}$ is as in the assumption of the theorem.

By Proviso 6.2 and [56, Theorem 7],

$$\begin{aligned} G \equiv_{\mathcal{F}} H &\implies G + jK_1 + \ell K_1 \equiv_{\mathcal{F}} H + jK_1 + \ell K_1 \\ &\implies p_{n+j,m+\ell}(G + jK_1 + \ell K_1) = p_{n+j,m+\ell}(H + jK_1 + \ell K_1) \\ &\implies q^{j,\ell}_{n,m}(G) = q^{j,\ell}_{n,m}(H). \end{aligned}$$

Hence, $q^{j,\ell}$ satisfies the assumptions of the theorem and it is $F_{\nu n+j,\nu m+\ell,i} \in \mathrm{cl}^{\mathrm{bip}}_{\nu,\nu}(\mathcal{F})$ for all i and all $n, m \gg \nu$ by the claim. It follows that $F_{n,m,i} \in \mathrm{cl}^{\mathrm{bip}}_{\nu,\nu}(\mathcal{F})$ for all $n, m \gg \nu$ and all i. □

Proof of Theorem 6.9. Given Theorem 1.1, it suffices to prove that 2 implies 1. This follows from Theorem 6.11. □

7 Symmetric complexity of subgraph polynomials

Theorem 1.1 characterises the polynomials admitting symmetric circuits of polynomial orbit size in terms of their representation as linear combinations of homomorphism polynomials. By Lemma 3.1, all symmetric polynomials also admit an alternative presentation as linear combinations of *subgraph polynomials*. The results from this section suggest that in this subgraph representation, $\mathfrak{T}^k_{n,m}$ can be characterised via another natural graph parameter, namely a variant of the *vertex cover number* that is closed under graph complementation. We establish this characterisation at least in the case of subgraph polynomials for single pattern graphs of sublinear size.

As a first step, we give a sufficient criterion for a subgraph polynomial to be in $\mathfrak{T}^k_{n,m}$. This criterion involves the *hereditary treewidth* $\mathrm{hdtw}(F)$ of a bipartite multigraph F defined as the maximum of the treewidth $\mathrm{tw}(F')$ of all bipartite multigraphs F' which admit a vertex- and edge-surjective bipartition-respecting homomorphism $F \to F'$. As already observed in [9], these graphs F' are precisely those that appear when writing the function $\mathrm{sub}(F,-)$ as a linear combination of functions $\hom(F',-)$ via Möbius inversion, cf. Corollary A.3.

Theorem 7.1. *Let F be a bipartite multigraph and $n, m \in \mathbb{N}$. Then $\mathrm{sub}_{F,n,m} \in \mathfrak{T}^{k+1}_{n,m}$ for $k \coloneqq \mathrm{hdtw}(F)$.*

Proof. By Corollary A.3, $\mathrm{sub}_{F,n,m}$ can be written as linear combination of homomorphism polynomials $\mathrm{sub}_{F',n,m}$ for quotients F' of F. By definition of hereditary treewidth, $\mathrm{tw}(F') \leq \mathrm{hdtw}(F) = k$. Hence, $\mathrm{sub}_{F,n,m} \in \mathfrak{T}^{k+1}_{n,m}$ since the $k+1$ in $\mathfrak{T}^{k+1}_{n,m}$ denotes the size of the largest bag rather than the treewidth. □

The expression for $\mathrm{sub}_{F,n,m}$ derived in Theorem 7.1 does not depend on n, m. For this reason, Theorem 7.1 fails to characterise membership of subgraph polynomials in $\mathfrak{T}^k_{n,m}$, as illustrated by the following example which is provided by the machinery developed in Section 4.2.

Example 7.2. For all $n, m \in \mathbb{N}$, $\mathrm{sub}_{K_{n,m},n,m} \in \mathfrak{T}^2_{n,m}$.

[3] Strictly speaking, this requires the theorem of 'Operations with structures' [42] for the category of graphs with fixed bipartitions and bipartition-respecting homomorphisms. One may verify that the original proof goes through in this setting or consult a more category-theoretical reference such as [46].

Proof. Since $\mathrm{sub}_{K_{n,m},n,m} = \prod_{v\in[n]}\prod_{w\in[m]} x_{vw}$, this polynomial is in $\mathfrak{T}^2_{n,m}$ by Example 4.5 and Theorem 4.13. □

Example 7.2 shows that $\mathfrak{T}^k_{n,m}$ contains subgraph polynomials of dense graphs. Hence, a characterisation of $\mathfrak{T}^k_{n,m}$ in terms of subgraph polynomials cannot involve a monotone graph parameter such as (hereditary) treewidth. Towards such a characterisation, we make the following observation:

Theorem 7.3. *For a function $f\colon \mathbb{N}\to\mathbb{N}$ such that $f(n)\leq n$ for all $n\in\mathbb{N}$, the following are equivalent:*

1. *there exists a constant $k\in\mathbb{N}$ such that $\min\{f(n), n-f(n)\}\leq k$ for all $n\in\mathbb{N}$,*
2. *the counting width of $\mathrm{sub}_{K_{f(n),f(n)},n,n}$ on (n,n)-vertex simple graphs is bounded,*
3. *the counting width of $\mathrm{sub}_{K_{f(n),f(n)},n,n}$ on (n,n)-vertex edge-weighted graphs is bounded,*
4. *the $\mathrm{sub}_{K_{f(n),f(n)},n,n}$ admit $\mathbf{Sym}_n\times\mathbf{Sym}_n$-symmetric circuits of orbit size polynomial in n,*
5. *there exists a constant $k\in\mathbb{N}$ such that $\mathrm{sub}_{K_{f(n),f(n)},n,n}\in\mathfrak{T}^k_{n,n}$ for all $n\in\mathbb{N}$,*
6. *the polynomials $\mathrm{sub}_{K_{f(n),f(n)},n,n}$ admit $\mathbf{Sym}_n\times\mathbf{Sym}_n$-symmetric circuits of size polynomial in n.*

Proof. First consider the implication $1\Rightarrow 6$: Observe that

$$\mathrm{sub}_{K_{k,k},n,n} = \sum_{\substack{A\subseteq[n]\\|A|=k}}\sum_{\substack{B\subseteq[n]\\|B|=k}}\prod_{a\in A}\prod_{b\in B} x_{ab},$$

$$\mathrm{sub}_{K_{n-k,n-k},n,n} = \sum_{\substack{A\subseteq[n]\\|A|=k}}\sum_{\substack{B\subseteq[n]\\|B|=k}}\prod_{a\in[n]\setminus A}\prod_{b\in[n]\setminus B} x_{ab}.$$

These formulas represent $\mathbf{Sym}_n\times\mathbf{Sym}_n$-symmetric circuits of size $O(n^{2k})$.

The implications $6\Rightarrow 4\Leftrightarrow 5$ follow from Theorem 1.1. The implications $4\Rightarrow 3\Rightarrow 2$ follow from Theorem 5.2 and Definition 5.1. The remaining implication $2\Rightarrow 1$ is proved by contraposition.

Let $k\geq 2$ be arbitrary. By assumption, there exists $n\in\mathbb{N}$ large enough such that

$$\min\{f(n), n-f(n)\}\geq k2^{k-1}-k.$$

In particular, $f(n)\geq k$. Let $F\coloneqq K_{f(n),f(n)}$ and write $P\subseteq V(F)$ for a set of vertices such that $F[P]\cong K_{k,k}$. Blow up the vertices in P to CFI gadgets (see Appendix C) and call the resulting bipartite graphs G^P_0 and G^P_1. Note that both graphs have $k2^{k-1}+f(n)-k\leq n$ vertices on each side. It follows from Corollary C.9 that G^P_0 and G^P_1 are $\mathcal{C}^k$-equivalent as bipartite graphs with fixed bipartitions.

Write $\rho\colon G^P_i\to F$ for the projection map, cf. Section C. We count embeddings $\mathrm{emb}(F, G^P_i)$. For a homomorphism $h\colon F\to F$, write $\mathrm{emb}_h(F, G^P_i)$ for the number of embeddings $e\colon F\to G^P_i$ such that $\rho\circ e = h$. Consider the following claims:

Claim 7.3a. If $h\colon F\to F$ is not surjective onto P, then $\mathrm{emb}_h(F, G^P_0) = \mathrm{emb}_h(F, G^P_1)$.

Proof of Claim. For $u\in V(P)$, write G^P_u for the CFI graph where the vertex u carries the odd weight. By e.g. [45, Lemma 11], for every $u,v\in P$, there exists an isomorphism $\varphi\colon G^P_u\to G^P_v$ such that $\rho\circ\varphi=\rho$. Hence, $\mathrm{emb}_h(F, G^P_1) = \mathrm{emb}_h(F, G^P_u)$ for every vertex $u\in P$. The graphs obtained from G^P_0 and G^P_u by removing the gadgets corresponding to u are isomorphic. Hence, $\mathrm{emb}_h(F, G^P_0) = \mathrm{emb}_h(F, G^P_u) = \mathrm{emb}_h(F, G^P_1)$ for every $u\in P$ such that $u\notin h(V(F))$. ⌟

Claim 7.3b. If $h\colon F\to F$ is surjective onto P and non-injective, then $\mathrm{emb}_h(F, G^P_0) = 0 = \mathrm{emb}_h(F, G^P_1)$.

Proof of Claim. By assumption, there exist distinct $a,b\in V(F)$ such that $h(a)=h(b)=:v$. If $v\notin P$, then $\mathrm{emb}_h(F, G^P_0) = 0 = \mathrm{emb}_h(F, G^P_1)$ since any map $e\colon F\to G^P_i$ such that $\rho\circ e = h$ is non-injective and thus not an embedding. Hence, $v\in P$.

Let $e\colon F\to G^P_i$ be an embedding such that $\rho\circ e = h$. Since h is surjective onto P, both $e(a)$ and $e(b)$ have a shared neighbour in every CFI gadget on the opposing side of the bipartition. This implies that $e(a)=e(b)$ by Definition C.1, contradicting that e is injective. ⌟

By Claims 7.3a and 7.3b,

$$\begin{aligned}\mathrm{emb}(F, G^P_0) - \mathrm{emb}(F, G^P_1) &= \sum_{h\colon F\to F}\mathrm{emb}_h(F, G^P_0) - \sum_{h\colon F\to F}\mathrm{emb}_h(F, G^P_1)\\ &= \sum_{\substack{h\colon F\to F\\ \text{surjective onto } P\\ \text{injective}}}\mathrm{emb}_h(F, G^P_0) - \sum_{\substack{h\colon F\to F\\ \text{surjective onto } P\\ \text{injective}}}\mathrm{emb}_h(F, G^P_1)\end{aligned}$$

Here, all h are bipartition-respecting. For $i \in \{0,1\}$,

$$\sum_{\substack{h\colon F\to F \\ \text{surjective onto } P \\ \text{injective}}} \mathrm{emb}_h(F, G_i^P) = f(n)^{\underline{k}} \cdot f(n)^{\underline{k}} \cdot \mathrm{emb}(K_{k,k}, (K_{k,k})_i) \cdot \mathrm{emb}(K_{f(n)-k,f(n)-k}, K_{f(n)-k,f(n)-k})$$

where $n^{\underline{k}} \coloneqq n(n-1)\cdots(n-k+1)$ denotes the falling factorial and $(K_{k,k})_i$ the CFI graphs of $K_{k,k}$. By Corollary C.8, $\mathrm{emb}(F, G_0^P) \neq \mathrm{emb}(F, G_1^P)$. Thus, the counting width of $\mathrm{sub}_{K_{f(n),f(n)},n,n}$ on (n,n)-vertex simple graphs is at least k. As k was chosen arbitrarily, the desired implication follows. □

Theorem 7.3 shows that $\mathrm{sub}_{K_{k,k},n,n}$ admits small symmetric circuits if, and only if, k is small or $n-k$ is small. We generalise the backward direction of this argument for all pattern graphs by introducing a new graph parameter which captures non-uniformity and relaxes hereditary treewidth. To that end, we first recall the *vertex cover number* $\mathrm{vc}(F)$ of a graph F which is defined as the minimum size of a set $C \subseteq V(F)$ such that every edge in F is incident to a vertex in C. Vertex cover number, hereditary treewidth, and matching number are functionally equivalent, cf. Section D and [9].

Lemma 7.4. *For every graph F, $\frac{1}{2}\,\mathrm{hdtw}(F) \leq \mathrm{mn}(F) \leq \mathrm{vc}(F) \leq 2\,\mathrm{mn}(F) \leq (\mathrm{hdtw}(F)+2)^2$.*

We relax the graph parameters above as follows:

Definition 7.5. Let $n, m \in \mathbb{N}$. Let F be a simple (n,m)-vertex bipartite graph with bipartition $A \uplus B$. Define the simple (n,m)-vertex bipartite graph $\overline{F}$ via $V(\overline{F}) \coloneqq V(F)$ and $E(\overline{F}) \coloneqq (A \times B) \setminus E(F)$. The *logical vertex cover number of F* is $\mathrm{lvc}_{n,m}(F) \coloneqq \min\{\mathrm{vc}(F), \mathrm{vc}(\overline{F})\}$. For $n' \geq n$ and $m' \geq m$, define $\mathrm{lvc}_{n',m'}(F) \coloneqq \mathrm{lvc}_{n',m'}(F')$ where F' is obtained from F by adding $n'-n$ isolated vertices on the left side and $m'-m$ isolated vertices on the right side. If $n' < n$ or $m' < m$, let $\mathrm{lvc}_{n',m'}(F) \coloneqq 0$.

For an (n,m)-vertex graph F, we have $\mathrm{lvc}_{n,m}(F) = k$ if, and only if, there exists a set $C \subseteq V(F)$ of size at most k such that $F - C$ is an independent set or a biclique. Equipped with this definition, we first show that a bound on the logical cover number of a pattern yields polynomial size symmetric circuits for their subgraph polynomials.

Theorem 7.6. *For every family $(F_{n,m})_{n,m\in\mathbb{N}}$ of simple bipartite graphs, if $\mathrm{lvc}_{n,m}(F_{n,m})$ is bounded, then the $\mathrm{sub}_{F_{n,m},n,m}$ admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of size polynomial in $n+m$.*

Proof. Fix $F \coloneqq F_{n,m}$ for some $n, m \in \mathbb{N}$ and write $A \uplus B$ for the fixed bipartition of F. We assume first that F has no isolated vertices and that removing the logical vertex cover results in an independent set rather than a biclique (that is, the logical vertex cover F is indeed a vertex cover). Let $K = K_A \uplus K_B$ be a vertex cover of F, such that $|K| \leq k$ and $K_A \subseteq A$, $K_B \subseteq B$. We perform the following circuit construction for every $\iota\colon K \hookrightarrow [n] \uplus [m]$. Given ι, let $p_{K,\iota} \coloneqq \prod_{uv\in E(F[K])} x_{\iota(u)\iota(v)}$. This polynomial is represented by a constant-size circuit which is symmetric under $\mathbf{Stab}^\bullet(\iota(K))$, the pointwise stabiliser of the set $\iota(K)$ in $\mathbf{Sym}_n \times \mathbf{Sym}_m$. Now for every type $S \subseteq K_A$ or $S \subseteq K_B$, introduce a fresh variable t_S. If $S \subseteq K_A$, we call S an A-type, otherwise a B-type. Let

$$q_\iota \coloneqq p_{K,\iota} \cdot \prod_{j \in [n]\uplus[m] \setminus \iota(K)} \left(\sum_{S\subseteq K \text{ an A-type}} t_S \cdot \prod_{i\in\iota(S)} x_{ij} + \sum_{S\subseteq K \text{ a B-type}} t_S \cdot \prod_{i\in\iota(S)} x_{ji} \right)$$

The obvious circuit representation of this has size $O((n+m)\cdot 2^k \cdot k)$ (the number of different types is at most 2^k), and the circuit is also $\mathbf{Stab}^\bullet(\iota(K))$-symmetric. Now we view q_ι as a polynomial in the variables $\{t_S \mid S \text{ a type}\}$ and with coefficients in $\mathbb{Q}[\mathcal{X}_{n,m}]$. For a type S, let $\#(S)$ denote the number of vertices in $V(F) \setminus K$ whose $E(F)$-neighbourhood in K is precisely S.

The coefficient of the monomial $\prod_{S \text{ a type}} t_S^{\#(S)}$ in q_ι is the part of $\mathrm{sub}_{F,n,m}$ that sums over all injections $V(F) \hookrightarrow [n] \uplus [m]$ which extend ι. More formally, this coefficient is

$$\mathrm{sub}_{F,n,m;\iota} = \sum_{\substack{\iota'\colon V(F)\hookrightarrow[n]\uplus[m] \\ \iota'|_K = \iota}} \prod_{vw\in E(F)} x_{\iota'(v)\iota'(w)}.$$

Given a symmetric circuit for q_ι, we can compute $\mathrm{sub}_{F,n,m;\iota}$ by interpolation. The polynomial q_ι has at most 2^k variables. In each of these variables, its maximal degree is $n+m-k$. Thus, by Corollary B.2, $\mathrm{sub}_{F,n,m;\iota}$ can be computed with a $\mathbf{Stab}^\bullet(\iota(K))$-symmetric circuit of size $O((n+m)^{2^k+1} \cdot 2^k \cdot k)$.

Let C_ι denote the circuit that realises this. The polynomial $\mathrm{sub}_{F,n,m}$ is then computed by the circuit C that simply sums up all C_ι, for all injections $\iota : K \hookrightarrow [n] \uplus [m]$. The size of C is $O((n+m)^{2^k+k+1} \cdot 2^k \cdot k)$. It remains to argue that C is symmetric. For this, we firstly observe that for any two $\iota_1, \iota_2 \colon K \hookrightarrow [n] \uplus [m]$, the polynomials q_{ι_1} and q_{ι_2} are symmetric to each other via any $(\pi, \sigma) \in \mathbf{Sym}_n \times \mathbf{Sym}_m$ that maps ι_1 to ι_2. Since we use the same points $a_1, ..., a_{n+m-k} \in \mathbb{Q}$ for the interpolation in each subcircuit C_ι, and the coefficients in the linear combination given by Lemma B.1 are the same, C_{ι_1} and C_{ι_2} are also symmetric to each other. This finishes the case where removing the logical vertex cover results in an independent set.

In the other case, the graph after removing the logical vertex cover is a biclique. Then we perform the same construction as above, with the difference that now, $p_{K,\iota} = \prod_{uv \in E(F[K])} x_{\iota(u)\iota(v)} \cdot \prod_{i \in [n] \setminus \iota(K_A), j \in [m] \setminus \iota(K_B)} x_{ij}$. Finally, if isolated vertices are present in F, then we just multiply our circuit with the appropriate constant factor to obtain the subgraph count polynomial. □

We conjecture that the converse of Theorem 7.6 holds. That is, the parameter $\mathrm{lvc}_{n,m}$ is the right parameter for measuring the symmetric circuit complexity of subgraph polynomials.

Conjecture 7.7. *For every family $(F_{n,n})_{n,m \in \mathbb{N}}$ of simple bipartite graphs, the following are equivalent:*

1. $\mathrm{lvc}_{n,m}(F_{n,m})$ *is bounded,*
2. *the counting width of* $\mathrm{sub}_{F_{n,m},n,m}$ *on (n,m)-vertex edge-weighted graphs is bounded,*
3. *the* $\mathrm{sub}_{F_{n,m},n,m}$ *admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of orbit size polynomial in $n+m$,*
4. *the* $\mathrm{sub}_{F_{n,m},n,m}$ *admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of size polynomial in $n+m$.*

The implications $1 \Rightarrow 4 \Rightarrow 3 \Rightarrow 2$ follow from Definition 5.1 and Theorems 5.2 and 7.6. We prove Conjecture 7.7 for families of patterns of sublinear size.

Theorem 7.8. *Let $(F_{n,m})_{n,m \in \mathbb{N}}$ be a family of simple bipartite graphs such that, for all $\nu, \mu \in \mathbb{N}$, there exist $n', m' \in \mathbb{N}$, such that $|F_{\nu n, \mu m}| \leq \min\{n, m\}$ for all $n > n'$ and $m > m'$. The following are equivalent:*

1. $\mathrm{lvc}_{n,m}(F_{n,m})$ *is bounded,*
2. $\mathrm{vc}(F_{n,m})$ *is bounded,*
3. *the counting width of* $\mathrm{sub}_{F_{n,m},n,m}$ *on (n,m)-vertex simple graphs is bounded,*
4. *the counting width of* $\mathrm{sub}_{F_{n,m},n,m}$ *on (n,m)-vertex edge-weighted graphs is bounded,*
5. *the* $\mathrm{sub}_{F_{n,m},n,m}$ *admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of orbit size polynomial in $n+m$,*
6. *the* $\mathrm{sub}_{F_{n,m},n,m}$ *admit $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric circuits of size polynomial in $n+m$.*

Proof. The implications $1 \Rightarrow 6 \Rightarrow 5 \Rightarrow 4 \Rightarrow 3$ follow from Theorems 5.2 and 7.6 and Definition 5.1. By Definition 7.5, the vertex cover number of F is generally an upper bound for $\mathrm{lvc}_{n,m}(F)$, so 2 implies 1. The remaining implication $3 \Rightarrow 2$ follows from Theorems D.1 and 6.9 and Fact D.4. □

In the remainder of this section, we give evidence for the remaining implication.

Conjecture 7.7 predicts that, for every simple (n,m)-vertex bipartite graph F, $\mathrm{sub}_{F,n,m}$ has bounded counting width if, and only if, $\mathrm{sub}_{\overline{F},n,m}$ has bounded counting width. We prove this consequence (independently of the conjecture).

Theorem 7.9. *Let $n, m \in \mathbb{N}$. For every (n,m)-vertex simple bipartite graph F, the polynomials* $\mathrm{sub}_{F,n,m}$ *and* $\mathrm{sub}_{\overline{F},n,m}$ *have the same counting width on (n,m)-vertex edge-weighted graphs.*

Proof. The proof is by constructing, given (n,m)-vertex bipartite edge-weighted graphs G and H such that $G \equiv_{\mathcal{C}^k} H$ and $\mathrm{sub}_{F,n,m}(G) \neq \mathrm{sub}_{F,n,m}(H)$, two (n,m)-vertex bipartite edge-weighted graphs G' and H' such that $G' \equiv_{\mathcal{C}^k} H'$ and $\mathrm{sub}_{\overline{F},n,m}(G') \neq \mathrm{sub}_{\overline{F},n,m}(H')$.

To that end, we turn the edge-weights from $\mathbb{Q}$ in G into formal variables. Let $y_1, \ldots, y_r$ denote formal variables, one for every value appearing as an edge weight in G and H. Let G'' and H'' denote the (n,m)-vertex bipartite graphs whose edges are annotated by these formal variables according to the corresponding values in G and H. Write $C_1 \uplus \cdots \uplus C_r = E(G)$ for the partition of the edges according to this correspondence. Write $V(F) = A \uplus B$ and $V(G) = [n] \uplus [m]$ for the bipartitions. Consider the following polynomials in $\mathbb{Q}[y_1, \ldots, y_r]$:

$$\mathrm{sub}_{F,n,m}(G'') = \frac{1}{|\mathbf{Aut}(F)|} \sum_{\substack{h \colon A \uplus B \to [n] \uplus [m] \\ \text{bijective}}} \prod_{i=1}^{r} y_i^{|h(E(F)) \cap C_i|},$$

$$\mathrm{sub}_{\overline{F},n,m}(G'') = \frac{1}{|\mathbf{Aut}(F)|} \sum_{\substack{h \colon A \uplus B \to [n] \uplus [m] \\ \text{bijective}}} \prod_{i=1}^{r} y_i^{|h(E(\overline{F})) \cap C_i|} = \frac{1}{|\mathbf{Aut}(F)|} \sum_{\substack{h \colon A \uplus B \to [n] \uplus [m] \\ \text{bijective}}} \prod_{i=1}^{r} y_i^{|C_i| - |h(E(F)) \cap C_i|}.$$

The final equality holds since $h(E(\overline{F})) \cap C_i = ([n] \times [m] \setminus h(E(F))) \cap C_i = C_i \setminus (h(E(F)) \cap C_i)$ as F and G have the same number of vertices.

It follows that $\mathrm{sub}_{F,n,m}(G''), \mathrm{sub}_{\overline{F},n,m}(G'') \in \mathbb{Q}[y_1, \dots, y_r]$ are the same polynomials up to a permutation of coefficients. More precisely, the coefficient of $y_1^{n_1} \cdots y_r^{n_r}$ in $\mathrm{sub}_{F,n,m}(G'')$ for $n_i \in \{0, \dots, |C_i|\}$, $i \in [r]$, equals the number of bijective maps $h \colon A \uplus B \to [n] \uplus [m]$ such that $|h(E(F)) \cap C_i| = n_i$ for $i \in [r]$. This number equals the coefficient of $y_1^{|C_1|-n_1} \cdots y_r^{|C_r|-n_r}$ in $\mathrm{sub}_{\overline{F},n,m}(G'')$. Hence, as formal polynomials in $\mathbb{Q}[y_1, \dots, y_r]$,

$$\mathrm{sub}_{F,n,m}(G'') = \mathrm{sub}_{F,n,m}(H'') \iff \mathrm{sub}_{\overline{F},n,m}(G'') = \mathrm{sub}_{\overline{F},n,m}(H'').$$

Finally, the edge-weighted graphs G and H such that $\mathrm{sub}_{F,n,m}(G) \neq \mathrm{sub}_{F,n,m}(H)$ give assignments to the variables $y_1, \dots, y_r$ such that the polynomials $\mathrm{sub}_{F,n,m}(G'')$ and $\mathrm{sub}_{F,n,m}(H'')$ evaluate to different values. Hence, $\mathrm{sub}_{\overline{F},n,m}(G'') \neq \mathrm{sub}_{\overline{F},n,m}(H'')$ and there exist assignments of rationals to $y_1, \dots, y_r$ such that these polynomials evaluate to different values. These assignments correspond to (n, m)-vertex edge-weighted graphs G' and H' such that $\mathrm{sub}_{\overline{F},n,m}(G') \neq \mathrm{sub}_{\overline{F},n,m}(H')$. The graphs G' and H' can be obtained from G and H by replacing the edge weights according to some function. Hence, G' and H' are $\mathcal{C}^k$-equivalent if G and H are. □

Theorem 7.9 allows us to separate counting width on simple graphs and counting width on edge-weighted graphs. This is in stark contrast to Theorem 6.1, which shows that, for single homomorphism polynomials, the simple and edge-weighted counting width coincide.

Example 7.10. Let $n \in \mathbb{N}$ and write $F \coloneqq \overline{nK_{1,1}}$ for the bipartite complement of the matching on (n, n)-vertices. The counting width of the multilinear polynomial $\mathrm{sub}_{F,n,n}$ on (n, n)-vertex simple graphs is 2 while its counting width on (n, n)-vertex edge-weighted graphs is $\Theta(n)$.

Proof. For the first claim, let G be an arbitrary simple (n, n)-vertex bipartite graph with bipartition $V(G) = X \uplus Y$. If G contains a subgraph isomorphic to F, then all its vertices have degree n or $n-1$. This property can be defined in $\mathcal{C}^2$. Write $X' \subseteq X$ and $Y' \subseteq Y$ for the sets of vertices of degree n on each side of the bipartition. By the handshaking lemma, $|X'| = |Y'| =: \ell$. It holds that $\mathrm{sub}(F, G) = \ell!$. Clearly, the number ℓ is definable in $\mathcal{C}^2$. Hence, the counting width of $\mathrm{sub}_{F,n,n}$ is 2.

The second claim follows from Theorem 7.9 and [21, Theorem 7.2] which states that the counting width of the permanent, which equals $\mathrm{sub}_{\overline{F},n,n}$, on simple graphs is $\Theta(n)$. □

8 Symmetric complexity of the immanants

The permanent and determinant are the extreme intractable and tractable case of *immanant* polynomials. Immanants are families of $\mathbf{Sym}_n$-symmetric polynomials $(p_n)_{n \in \mathbb{N}}$ that are typically defined via so-called *irreducible characters* of the symmetric group. An irreducible character $f \colon \mathbf{Sym}_n \to \mathbb{C}$ is a class function, i.e. it only depends on the multiset of cycle lengths of the input permutation. Given such an f, the corresponding immanant is defined as

$$\mathrm{imm}_f = \sum_{\pi \in \mathbf{Sym}_n} f(\pi) \prod_{i \in [n]} x_{i,\pi(i)}$$

We view these as polynomials over the field $\mathbb{C}$. If f is constantly 1, then imm_f is the permanent. The determinant is obtained by letting $f = \mathrm{sgn}$.

Note that while the permanent is $\mathbf{Sym}_n \times \mathbf{Sym}_n$-symmetric, this is not true for every immanant (indeed, not even of the determinant). But for every f, imm_f is $\mathbf{Sym}_n$-symmetric because for any $\sigma \in \mathbf{Sym}_n$, the monomial $\prod_{i \in [n]} x_{\sigma(i),\sigma(\pi(i))}$ encodes a permutation from the same conjugacy class as π, so it has the same value under f. Therefore, we study the complexity of $\mathbf{Sym}_n$-symmetric algebraic circuits for imm_f. The $\mathbf{Sym}_n \times \mathbf{Sym}_m$-symmetric polynomials that we have mostly been working with naturally expressed properties of weighted undirected bipartite graphs; now, $\mathbf{Sym}_n$-symmetric polynomials express properties of weighted *directed* (not necessarily bipartite) graphs (see also Remark 4.1).

In [7], Curticapean shows a complexity dichotomy for the immanants: In the tractable case, imm_f is in VP (i.e. admits polynomial size algebraic circuits) and is computable in polynomial time. In the intractable case, the family of polynomials is not in VP, unless $\mathsf{VFPT} = \mathsf{VW}[1]$, and also, it is not computable in polynomial time, unless $\mathsf{FPT} = \#\mathsf{W}[1]$. The complexity is controlled by a certain parameter of f.

We show that the same parameter of f produces the analogous dichotomy with regards to the complexity of $\mathbf{Sym}_n$-symmetric algebraic circuits for the immanants. The benefit of considering symmetric circuits is that

the lower bound is not conditional on any complexity-theoretic assumptions. We now introduce the relevant parameter of f. As explained in [7], the irreducible characters f correspond naturally to partitions of $[n]$. If λ is a partition of $[n]$, we denote by $\chi^\lambda \colon \mathbf{Sym}_n \to \mathbb{C}$ its corresponding irreducible character, and we also write imm_λ for $\mathrm{imm}_{\chi^\lambda}$. Partitions of $[n]$ are denoted as tuples $(k_1, \ldots, k_s)$, where each entry k_i denotes a part of size k_i. For example, the permanent is $\mathrm{imm}_{(n)}$, and the determinant is $\mathrm{imm}_{(1,\ldots,1)}$. For other partitions, the rule for computing their corresponding irreducible characters is more involved and not needed here (for details, see [7]). If λ is a partition of $[n]$, let $b(\lambda) = n - s$, where s denotes the number of parts.

Formally, [7] considers a family Λ of partitions. It is said that Λ *supports growth* $g \colon \mathbb{N} \to \mathbb{N}$ if for every $n \in \mathbb{N}$ there is a partition $\lambda^{(n)}$ in Λ with $b(\lambda^{(n)}) \geq g(n)$ and size $\Theta(n)$. According to the dichotomy from [7], $(\mathrm{imm}_\lambda)_{\lambda\in\Lambda}$ is tractable if there is a constant that bounds $b(\lambda)$ for all $\lambda \in \Lambda$. Otherwise, if Λ supports growth $\omega(1)$, then the immanant family is (conditionally) intractable. If Λ even supports growth $\omega(n^k)$, then $(\mathrm{imm}_\lambda)_{\lambda\in\Lambda}$ is VNP-complete.

Our analogous result for symmetric circuits is the following.

Theorem 3.7. *Let Λ be a family of partitions.*

1. *If there exists a $k \in \mathbb{N}$ such that $b(\lambda) \leq k$ for all $\lambda \in \Lambda$, then the $(\mathrm{imm}_\lambda)_{\lambda\in\Lambda}$ admit $\mathbf{Sym}_n$-symmetric circuits of polynomial size in n.*
2. *If Λ supports growth $g \in \omega(1)$, then the counting width of $(\mathrm{imm}_\lambda)_{\lambda\in\Lambda}$ on directed $\mathbb{Q}$-edge-weighted n-vertex graphs is unbounded, where n is the integer that is partitioned by the respective λ. Thus, all $\mathbf{Sym}_n$-symmetric circuits for the immanant family have super-polynomial size in n.*
3. *If Λ supports growth $g \in \omega(n^k)$ for a constant k, then the counting width of $(\mathrm{imm}_\lambda)_{\lambda\in\Lambda}$ on directed $\mathbb{Q}$-edge-weighted n-vertex graphs is $\Omega(n)$, so all $\mathbf{Sym}_n$-symmetric circuits have size $2^{\Omega(n)}$.*

This theorem subsumes and generalises the results from [21].

8.1 The tractable case

We show the first part of Theorem 3.7. In [32], Hartmann gives an algorithm for imm_λ that runs in polynomial time if $b(\lambda)$ is bounded. This algorithm is essentially a formula that expresses the immanant in terms of the determinant. For the determinant (over fields of characteristic 0), we know that efficient symmetric circuits exist:

Lemma 8.1 ([21])**.** *The family $(\det_n)_{n\in\mathbb{N}}$ admits $\mathbf{Sym}_n$-symmetric algebraic circuits of polynomial size.*

We now present the relevant details of Hartmann's formula and show that using the above result, we can express it with a symmetric circuit. Let $m \leq n$ and let $I = (i_1, \ldots, i_m)$ be a tuple of natural numbers. Define a class function $f_I \colon \mathbf{Sym}_n \to \mathbb{C}$ as follows.

$$f_I(\pi) \coloneqq \mathrm{sgn}(\pi) \cdot \prod_{\ell\in[m]} \alpha_\ell(\pi)^{i_\ell},$$

where $\alpha_\ell(\pi)$ denotes the multiplicity of the cycle of length ℓ in the cycle decomposition of π.

Our first subgoal is a circuit for imm_{f_I}. Following [32], we define certain matrices and use their determinants to compute imm_{f_I}. A cycle in the graph corresponding to the matrix $(x_{ij})_{1\leq i,j\leq n}$ is a sequence of variables such that each x_{ij} in this sequence is followed by a variable x_{jk}, and the right index of the last variable is equal to the left index of the first variable of the sequence. Given a fixed tuple $I = (i_1, \ldots, i_m)$ of cycle multiplicities, we consider tuples of cycles in the graph $(x_{ij})_{1\leq i,j\leq n}$: Let $(\sigma_1^{(1)}, \sigma_1^{(2)}, \ldots, \sigma_1^{(i_1)}, \sigma_2^{(1)}, \ldots, \sigma_2^{(i_2)}, \ldots, \sigma_m^{(i_m)})$ denote a tuple of (directed) cycles in the graph $(x_{ij})_{1\leq i,j\leq n}$, where the subscript ℓ of each σ denotes the length of the cycle, and the superscript indexes the cycles of length ℓ from 1 up to i_ℓ.

For a given tuple of cycles, we define a matrix $A(\sigma_1^{(1)}, \ldots, \sigma_m^{(i_m)})$ in variables $(x_{ij})_{1\leq i,j\leq n}$ and $(t_\ell^{(k)})$, where ℓ and k range over the corresponding sub- and superscripts of σ. For $i, j \in [n]$, let X_{ij} denote the set of index pairs (k, ℓ) such that the cycle $\sigma_\ell^{(k)}$ contains the variable x_{ij}. The matrix $A(\sigma_1^{(1)}, \ldots, \sigma_m^{(i_m)})$ is defined by letting

$$a_{ij} \coloneqq x_{ij} \prod_{(k,\ell)\in X_{ij}} t_\ell^{(k)}.$$

We extend the action of $\mathbf{Sym}_n$ from the variables x_{ij} to the variables $t_\ell^{(k)}$ by just letting $\pi(t_\ell^{(k)}) = t_\ell^{(k)}$. It can be checked that every $\pi \in \mathbf{Sym}_n$ maps $A \coloneqq A(\sigma_1^{(1)}, \ldots, \sigma_m^{(i_m)})$ to $A' \coloneqq A(\pi(\sigma_1^{(1)}, \ldots, \sigma_m^{(i_m)}))$ such that

$\pi(a_{ij}) = a'_{\pi(i),\pi(j)}$. The following is a $\mathbf{Sym}_n$-invariant polynomial:

$$S \coloneqq \sum_{(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)})\in\mathcal{C}} \det A(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)}),$$

where $\mathcal{C}$ is the set of all tuples of cycles $(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)})$ (with the respective lengths as given by the subscripts) such that the union of their edges is a partial cycle cover of $(x_{ij})_{1\le i,j\le n}$ (i.e. a disjoint union of cycles). Note that $\mathcal{C}$ is $\mathbf{Sym}_n$-invariant. By Lemma 8.1, the polynomial S is computable by a $\mathbf{Sym}_n$-symmetric circuit C_S of size $|\mathcal{C}|\cdot\mathrm{poly}(n)$: Each $\det A(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)})$ in the sum can be represented by a poly-size $\mathbf{Sym}_n$-symmetric circuit $C(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)})$ that is just like the circuit for $\det_n$ where the inputs are modified to be the products a_{ij} instead of single variables x_{ij}. If $\pi\in\mathbf{Sym}_n$ does not fix the tuple of cycles $(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)})$, then the circuit automorphism of $C(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)})$ that π extends to turns $C(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)})$ into $C(\pi(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)}))$. Thus, because $\mathcal{C}$ is $\mathbf{Sym}_n$-invariant, C_S is $\mathbf{Sym}_n$-symmetric.

Claim 8.1a. If we view S as a polynomial in the $t_\ell^{(k)}$-variables, then the coefficient of the monomial $\prod(t_\ell^{(k)})^\ell$ in S (where the product ranges over the entire tuple of variables $(t_1^{(1)},\dots,t_1^{(i_1)},\dots,t_m^{(i_m)})$) is exactly imm_{f_I}.

Proof of Claim. Let M be a monomial in $\det A(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)})$. Then its submonomial consisting of x_{ij}-variables is a monomial in $\det_n$, so it describes a permutation $\pi\in\mathbf{Sym}_n$. Its submonomial in $t_\ell^{(k)}$-variables is equal to $\prod(t_\ell^{(k)})^\ell$ if and only if every cycle in $(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)})$ is a cycle in the cycle decomposition of π (note that $(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)})$ may contain the same cycle multiple times, and then this is also true). Thus, the coefficient of $\prod(t_\ell^{(k)})^\ell$ in S is of the form

$$\sum_{\pi\in P}\mathrm{sgn}(\pi)\cdot\beta_\pi\cdot\prod_{i\in[n]}x_{i\pi(i)},$$

where $P\subseteq\mathbf{Sym}_n$ is the set of all permutations whose cycle decomposition contains at least one cycle of length ℓ, for each $\ell\in[m]$. The coefficient β_π is the number of tuples $(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)})\in\mathcal{C}$ such that each cycle $\sigma_\ell^{(k)}$ appears in the cycle decomposition of π. Since duplicate cycles can appear in tuples in $\mathcal{C}$, we have that $\mathrm{sgn}(\pi)\cdot\beta_\pi = \mathrm{sgn}(\pi)\cdot\prod_{\ell\in[m]}\alpha_\ell(\pi)^{i_\ell} = f_I(\pi)$ because for every cycle length ℓ, we have i_ℓ many slots in each tuple in $\mathcal{C}$, each of which can be filled with one of the $\alpha_\ell(\pi)$ many cycles of π. This proves the claim. ⌟

This means that it remains to compute the coefficient of the monomial $\prod(t_\ell^{(k)})^\ell$ in S, which can be done by the interpolation method.

Claim 8.1b. Let d be the maximum degree of a $t_\ell^{(k)}$-variable in S, and let t denote the number of $t_\ell^{(k)}$-variables. Then $d^t\le m^{\sum_{\ell\in[m]} i_\ell}$.

Proof of Claim. We know $t=\sum_{\ell\in[m]} i_\ell$. It remains to show that $d\le m$. Let M be a monomial in S. Then it is a monomial that appears in some $\det A(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)})$. This means that the exponent of a variable $t_\ell^{(k)}$ in M can be at most $\ell\le m$ because $t_\ell^{(k)}$ corresponds to the cycle $\sigma_\ell^{(k)}$ of length ℓ and can only appear in a monomial in $\det A(\sigma_1^{(1)},\dots,\sigma_m^{(i_m)})$ together with a variable on the cycle $\sigma_\ell^{(k)}$. ⌟

Claim 8.1c. $|\mathcal{C}|\le\prod_{\ell\in[m]}\left(\frac{n!}{(n-\ell)!}\right)^{i_\ell}$.

Proof of Claim. The quantity $\frac{n!}{(n-\ell)!}$ is just the number of different ordered length-ℓ cycles in the n-vertex graph $(x_{ij})_{1\le i,j\le n}$. ⌟

Using Claim 8.1a and Corollary B.2, we can obtain a symmetric circuit C_I for imm_{f_I} from the circuit C_S. With Claims 8.1b and 8.1c it follows that

$$\|C_I\|\le m^{\sum_{\ell\in[m]} i_\ell}\cdot\|C_S\|\le m^{\sum_{\ell\in[m]} i_\ell}\cdot\prod_{\ell\in[m]}\left(\frac{n!}{(n-\ell)!}\right)^{i_\ell}\cdot\mathrm{poly}(n).$$

This finishes the first subgoal of the proof. Now we use C_I to obtain a circuit for imm_λ. The proof of Theorem 1 in [32] continues as follows. Let r denote the size of the largest part of λ, and let $m \coloneqq n-s$, where s is the number of parts of λ. Let $\mathcal{I}$ be the set of all tuples $(k_1,\dots,k_m)$ such that $m-r+1 \leq \sum_{\ell\in[m]} \ell k_\ell \leq m$.

Let η be the function that takes as input a tuple $(k_1,\dots,k_m) \in \mathcal{I}$ and is defined as

$$\eta(k_1,\dots,k_m) \coloneqq \Big\{(c,i_1,\dots,i_m) \mid c\in\mathbb{C}, i_1,\dots,i_m\in\mathbb{N}, \\ \text{the monomial } c\cdot\prod_{\ell\in[m]}\alpha_\ell^{i_\ell} \text{ appears in } \binom{\alpha_1}{k_1}\dots\binom{\alpha_m}{k_m}\Big\}.$$

In the above definition, $\binom{\alpha_1}{k_1}\dots\binom{\alpha_m}{k_m}$ is viewed as a polynomial in formal variables $\alpha_1,\dots,\alpha_m$. The following identity is from [32]:

$$\mathrm{imm}_\lambda = \sum_{(k_1,\dots,k_m)\in\mathcal{I}} \chi(k_1,\dots,k_m)\cdot\Big(\sum_{(c,i_1,\dots,i_m)\in\eta(k_1,\dots,k_m)} c\cdot\mathrm{imm}_{f_{(i_1,\dots,i_m)}}\Big), \tag{14}$$

where $\chi(k_1,\dots,k_m)$ is some $\mathbb{C}$-valued function that we do not need to elaborate on further. In a circuit, we can simply hardwire its respective value. We have

$$|\mathcal{I}| \leq \sum_{m-r+1\leq i\leq m} p(i),$$

where $p(i)$ denotes the number of partitions of $[i]$. As argued in [32], the polynomial $\binom{\alpha_1}{k_1}\cdot\dots\cdot\binom{\alpha_m}{k_m}$ has at most $\prod_{\ell\in[m]}(k_\ell+1) \leq \binom{2m}{m}$ many monomials and the degree of each α_ℓ is at most k_ℓ. Therefore, $|\eta(k_1,\dots,k_m)| \leq \binom{2m}{m}$, and for every tuple $(c,i_1,\dots,i_m)\in\eta(k_1,\dots,k_m)$, it holds $i_\ell \leq k_\ell$ for every $\ell\in[m]$. Using this, we can upper-bound one of the products in the size bound of C_I as follows:

$$m^{\sum_{\ell\in[m]} i_\ell} \leq m^{\sum_{\ell\in[m]} k_\ell} \leq m^m.$$

In total, we can use Equation (14) and plug in the symmetric circuits C_I for the respective $\mathrm{imm}_{f_{(i_1,\dots,i_m)}}$. Thus we obtain a symmetric circuit C_λ for imm_λ with

$$\|C_\lambda\| \in O\Big(\Big(\sum_{m-r+1\leq i\leq m} p(i)\Big)\cdot\binom{2m}{m}\cdot m^m\cdot\prod_{\ell\in[m]}\Big(\frac{n!}{(n-\ell)!}\Big)^{i_\ell}\cdot\mathrm{poly}(n)\Big).$$

From [32], we get the following estimation:

$$\Big(\sum_{m-r+1\leq i\leq m} p(i)\Big)\cdot\binom{2m}{m}\cdot m^m\cdot\prod_{\ell\in[m]}\Big(\frac{n!}{(n-\ell)!}\Big)^{i_\ell} \leq (m+1)e^{3m^{1/2}}e^{2m}\cdot m^m\cdot n^m \\ \leq n^{6(n-s)+4}\cdot(n-s)^{(n-s)}.$$

The last step holds for $n>3$ and uses the fact that $m+1\leq n$ and $m=n-s$. Since we are in the first case of Theorem 3.7, we assume that $m=n-s$ is bounded by a constant k. In particular, the factor $(n-s)^{(n-s)}$ and the exponent of n become constant. So $\|C_\lambda\|$ is in $O(n^{6k+4})$, multiplied with the symmetric complexity of the determinant, which is also polynomial in n.

8.2 The hard case

Let us first focus on the second part of Theorem 3.7. Fix a family Λ of partitions such that $b(\Lambda)$ is unbounded, i.e. it supports growth $g\in\omega(1)$. We exploit the detailed analysis in [8] to show that the counting width of $(\mathrm{imm}_\lambda)_{\lambda\in\Lambda}$ on edge-weighted directed graphs is unbounded. The hardness proof in [7] reduces the problem of counting k-matchings, for growing k, to the immanant. Consider the following family of graph parameters, for a given function $h\colon\mathbb{N}\to\mathbb{N}$. Let $n\in\mathbb{N}$ and let G be an undirected graph with n vertices. Then $\#\mathrm{Match}_{n,h}(G)$ denotes the number of matchings in G consisting of exactly $h(n)$ many edges.

Lemma 8.2. *For any super-constant function $h(n)$ with $h(n)\leq n/2$, the graph parameter $\#\mathrm{Match}_{n,h}$ has unbounded counting width on undirected (n,n)-vertex bipartite graphs with fixed bipartition. Moreover, the number of* perfect *matchings is a graph parameter with counting width $\Omega(n)$ in undirected (n,n)-vertex bipartite graphs with fixed bipartition.*

Proof. Let $k \in \mathbb{N}$ be arbitrary. For a large enough $n \in \mathbb{N}$, we can do the following: Let H'_0, H'_1 be large enough CFI graphs from [21, Theorem 7.2], possibly each augmented with a disjoint copy of a size-t matching, for an appropriate $t \in \mathbb{N}$, so that the size of a perfect matching in each of them is precisely $h(n)$; the additional t-matching may be needed because $h(n) \in \mathbb{N}$ is not necessarily a number that can be the size of a perfect matching in any CFI graph. The graphs H'_0 and H'_1 are $\mathcal{C}^k$-equivalent because the CFI graphs are by [21, Theorem 7.2], and taking the disjoint union with a t-matching in each graph does not change this. The graphs are bipartite (and we can assume the bipartition to be marked with unary relation symbols without breaking $\mathcal{C}^k$-equivalence, cf. Corollary C.9). Let H_0, H_1 be defined by padding H'_0, H'_1, respectively, with isolated vertices so that they both have exactly n vertices. Then still, $H_0 \equiv_{\mathcal{C}^k} H_1$. Moreover, the number of size-$h(n)$ matchings in H_0 and H_1, respectively, is equal to the number of perfect matchings in H'_0 and H'_1, respectively, and these numbers are different by [21] (having added an additional t-matching to the graphs does not change the number of perfect matchings). Therefore, $\#\mathrm{Match}_{n,h}(H_0) \neq \#\mathrm{Match}_{n,h}(H_1)$. Since k was arbitrary, this shows that the counting width of $\#\mathrm{Match}_{n,h}$ is unbounded. For the second part of the lemma, we do not even have to add isolated vertices to H'_0 and H'_1. They are n-vertex bipartite graphs with different numbers of perfect matchings. It also follows from [21] that $H'_0 \equiv_{\mathcal{C}^{o(n)}} H'_1$. □

Now one has to go through the proof of [8, Theorem 2] in some detail: Let $h(n) = \sqrt{g(n)}/24$. Fix an arbitrary $k \in \mathbb{N}$. By Lemma 8.2, there exists an $n \in \mathbb{N}$ and $\mathcal{C}^k$-equivalent undirected bipartite (n,n)-vertex graphs H_0, H_1 such that $\#\mathrm{Match}_{n,h}(H_0) \neq \#\mathrm{Match}_{n,h}(H_1)$. Due to the growth condition on Λ, there is a $\lambda \in \Lambda$ with $b(\lambda) \geq 24h(n) = \sqrt{g(n)}$ and such that λ satisfies certain further properties that we do not need to introduce here. The properties of λ fall into two cases, and in each of them, it is shown in [8] that $\#\mathrm{Match}_{n,h}$ is reducible to imm_λ. In the following, the term *digon* refers to a directed 2-cycle.

First case In the first case, the second part of Lemma 22 from [7] is used. It states that for any given undirected bipartite (n,n)-vertex graph H, there exists a directed edge-weighted graph $G(H)$ with edge weights $\{0, 1, x\}$ (x being a formal variable), such that for the λ at hand we have:

$$\#\mathrm{Match}_{n,h}(H) = c(\lambda, h(n)) \cdot [x^{2h(n)}] \,\mathrm{imm}_\lambda(G(H)) \tag{15}$$

Here, $[x^{2h(n)}] \,\mathrm{imm}_\lambda(G(H))$ denotes the coefficient of the monomial $x^{2h(n)}$ in $\mathrm{imm}_\lambda(G(H))$, and $c(\lambda, h(n)) \in \mathbb{Q}$ is a constant depending only on λ and $h(n)$ but not on the graph H (see [8, Lemma 26]).

The precise construction of $G(H)$ can be found in [8, Lemma 20]: Let $L \uplus R = V(H)$ be the bipartition of H. To obtain $G(H)$, all edges in H are directed from L to R, and all edges $R \times L$ are added to G. A set T of fresh vertices is added (whose size depends only on λ), together with all edges $R \times T$ and $T \times L$. For every $v \in V(H)$, a so-called switch vertex s_v is added, with a self-loop of weight x, and a digon between v and s_v. Finally, a number (that depends only on λ) of padding digons and padding vertices with self-loops is added. For $q \in \mathbb{Q}$, denote by G_q the graph constructed as just described, but with edge weight q instead of x.

Lemma 8.3. *For every $q \in \mathbb{Q}$ and bipartite graphs H_0, H_1 with fixed bipartitions, if $H_0 \equiv_{\mathcal{C}^k} H_1$, then $G_q(H_0) \equiv_{\mathcal{C}^k} G_q(H_1)$.*

Proof. It can be checked that after every construction step of G_q, the Duplicator still has a winning strategy in the bijective k-pebble game played on the two graphs. The first step is directing the edges from L to R, and adding all edges $R \times L$. This new edge relation is definable by a quantifier-free FO-formula because H_0 and H_1 have their bipartitions marked with unary relations. It is clear that $\mathcal{C}^k$-equivalence is preserved through such quantifier-free FO-definitions because any formula speaking about the new graphs can be translated into a formula speaking about the original graphs by replacing the symbol for the edge relation with its defining formula. This does not increase the number of variables, so no k-variable formula can distinguish the new pair of graphs.

The second step is to add a set T of fresh isolated vertices. We can assume them to be coloured with a new unary relation symbol T. It is clear that if the two graphs up to this point are $\mathcal{C}^k$-equivalent, then adding the same number of T-coloured isolated vertices to each graph does not change this fact because Duplicator still has a winning strategy. In the next step, the edges $R \times T$ and $T \times L$ are inserted. These edges are again definable with a quantifier-free formula, using the colours, so again, $\mathcal{C}^k$-equivalence is preserved. The next step is to add a switch vertex s_v for each $v \in L \cup R$, to connect it with v via a digon, and to add a self-loop with weight q (represented as a new binary relation) to every s_v. Here, one can easily define a winning strategy for Duplicator in the bijective k-pebble game on the new pair of graphs, using the Duplicator strategy on the pair before inserting the switch vertices: Any position in the game on the modified structures corresponds to

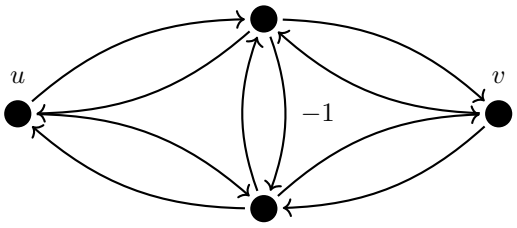


Figure 2: The gadget that replaces an edge uv.

a position in the game on the original structures: If a switch vertex s_v is pebbled, this corresponds to v being pebbled in the original graph. Then if f is Duplicator's chosen bijection in the game on the original graphs, this can be lifted to a bijection on the modified graphs by just mapping s_v to $s_{f(v)}$. It is easy to see that Duplicator wins with this strategy. Finally, the same number of isolated digons and vertices with self-loops is added to both graphs, which clearly does not change the fact that Duplicator has a winning strategy. □

Putting Equation (15) together with the equation $\#\mathrm{Match}_{n,h}(H_0) \neq \#\mathrm{Match}_{n,h}(H_1)$, we have:

$$[x^{2h(n)}]\,\mathrm{imm}_\lambda(G(H_0)) \neq [x^{2h(n)}]\,\mathrm{imm}_\lambda(G(H_1)).$$

It follows that there must exist a $q \in \mathbb{Q}$ such that $\mathrm{imm}_\lambda(G_q(H_0)) \neq \mathrm{imm}_\lambda(G_q(H_1))$, for if we had $\mathrm{imm}_\lambda(G_q(H_0)) = \mathrm{imm}_\lambda(G_q(H_1))$ for every $q \in \mathbb{Q}$, then $\mathrm{imm}_\lambda(G(H_0))$ and $\mathrm{imm}_\lambda(G(H_1))$ would be the same polynomial in $\mathbb{Q}[x]$, contradicting the statement above. By Lemma 8.3, we have $G_q(H_0) \equiv_{\mathcal{C}^k} G_q(H_1)$, which proves that imm_λ is not $\mathcal{C}^k$-invariant on $\mathbb{Q}$-edge-weighted graphs.

Second case In the second case, the properties of λ are different, and the second part of [7, Lemma 28] has to be applied. This describes another graph construction, let us call it $G'(H)$, such that

$$\#\mathrm{Match}_{n,h}(H) = c'(n, |E(H)|, h(n), \lambda) \cdot \mathrm{imm}_\lambda(G'(H)) \tag{16}$$

Here, the constant c' depends on $n = |V(H)|, |E(H)|, h(n)$ and λ (see [8, Lemma 28]), but not on H itself. The graph $G'(H)$ is directed and has edge weights $\{0, 1, -1\}$ if H is unweighted. The construction of $G'(H)$ is the following [8, Lemma 28]: Replace every edge uv in H with an edge gadget consisting of u, v, two fresh vertices, and five digons, as depicted in Figure 2 (see also page 6 of [8]).

This edge gadget is symmetric between u and v. Next, for each vertex $v \in V(H)$, add a fresh 'switch vertex' s_v connected to v with a digon. Add another $2h(n)$ many fresh vertices, called 'receptor vertices', and add an edge between each pair of receptor and switch vertex. Replace each of these new edges with the aforementioned edge gadget. Finally, a number of isolated digons and a number of isolated vertices is added. These numbers depend on $\lambda, |V(H)|, |E(H)|$, and $h(n)$.

Lemma 8.4. *Let $k \geq 4$. If H_0, H_1 are undirected simple bipartite graphs with fixed bipartition, and $H_0 \equiv_{\mathcal{C}^k} H_1$, then $G'(H_0) \equiv_{\mathcal{C}^{k/2}} G'(H_1)$.*

Proof. Again, we verify that every construction step of G' preserves $\mathcal{C}^k$-equivalence. Most of it works just like in Lemma 8.3. The only additional construction step here is the replacement of every edge with the gadget depicted above. Since these edge gadgets contain "internal" vertices, they give Spoiler the possibility to fix an edge using just one pebble. This is why we need to halve the number of pebbles in this construction step to ensure that Duplicator can still win. The argument is the following. We now start with H_0 and H_1 and consider only the construction step where each edge is replaced with the gadget. Consider a position in the bijective k-pebble game on H_0 and H_1. The Duplicator winning strategy gives us a bijection $f\colon V(H_0) \to V(H_1)$ that respects the $\mathcal{C}^k$-types of the vertices, with the $\leq k-1$ pebbles on the board as parameters. This is the bijection that Duplicator plays. If at most $k-2$ pebbles are currently on the board, then in addition to f, there also exists a bijection $f_2\colon V(H_0)^2 \to V(H_1)^2$ on the pairs of vertices which respects the parametrised $\mathcal{C}^k$-types of the pairs. If such a bijection did not exist, then the tuples of pebbled vertices would have different $\mathcal{C}^k$-types and Duplicator would not be able to win from this position. (Note that f and f_2 in general do not agree on the vertices, unless H_0 and H_1 are isomorphic.) With this in mind, we can define a Duplicator winning strategy in the $k/2$-pebble game on the modified graphs: A position on the modified graphs translates back into a pebble position on the original graphs as follows: Pebbles on original vertices are kept, and each pebble on an internal vertex of the gadget of an edge uv is replaced by two pebbles, one

on u and one on v, in the original graph. Before Duplicator gets to specify a bijection in the game on the modified structures, there are at most $k/2-1$ many pebbles on each graph, so the corresponding pebble position in the original graphs has at most $k-2$ many pebbles on each graph. Because Duplicator has a winning strategy in the k-pebble game on the original graphs, there exist a $\mathcal{C}^k$-type preserving bijection f on vertices, and a $\mathcal{C}^k$-type preserving bijection f_2 on the vertex pairs, as argued before. Because the existence of an edge is part of the $\mathcal{C}^k$-type of a pair, f_2 preserves edges. We use f and f_2 to define Duplicator's bijection in the game on the modified graphs: The internal vertices of the edge gadget of an edge e are mapped to the corresponding internal vertices of the gadget of the edge $f_2(e)$. The non-gadget vertices are mapped according to f. Because f and f_2 are type-preserving, after Spoiler's move in the game on the modified graphs, when the position is translated back to the other game, the pebbled tuples again have the same $\mathcal{C}^k$-type, so Duplicator can indeed maintain this invariant and win the game.

The other construction steps can be dealt with as in Lemma 8.3, noting that the introduction of edges between all pairs of receptor and switch vertices is FO-definable if we assume the set of receptor and switch vertices to be coloured with a distinguished colour, respectively. In the final construction step, we just have to note that the number of isolated vertices and digons that are added depend on $\lambda, |V(H)|, |E(H)|$, and $h(n)$, and we have $|V(H_0)| = |V(H_1)|$ and $|E(H_0)| = |E(H_1)|$ because H_0 and H_1 are $\mathcal{C}^k$-equivalent, and $k \geq 2$. □

The constant $c'(n, |E(H)|, h(n), \lambda) \in \mathbb{Q}$ in Equation (16) is in particular equal for $\mathcal{C}^k$-equivalent graphs H_0, H_1, as they have the same number of edges and vertices. Thus, as before, Equation (16) implies that $\mathrm{imm}_\lambda(G'(H_0)) \neq \mathrm{imm}_\lambda(G'(H_1))$. The two graphs are $\mathcal{C}^{k/2}$-equivalent by Lemma 8.4, so imm_λ is not $\mathcal{C}^{k/2}$-invariant on edge-weighted directed graphs. Since k was arbitrary, we find such an example of graphs for every k, so the counting width of imm_λ is unbounded.

The two cases together show the second part of Theorem 3.7 because they prove that the family imm_Λ is not $\mathcal{C}^k$-invariant for any fixed $k \in \mathbb{N}$.

The first part of Theorem 5.2 then implies that imm_Λ does not admit polynomial size symmetric circuits.

The third part of Theorem 3.7 is shown analogously. The proof in [8] is then again divided into two cases, in each of which we get Equations (15) and (16), respectively, but where $\#\mathrm{Match}_{n,h}(H)$ is now replaced with the number of *perfect* matchings. Lemma 8.2 then gives us graphs that differ with respect to the number of perfect matchings and are $\mathcal{C}^{o(n)}$-equivalent. It is easy to check that Lemma 8.3 and Lemma 8.4 are also true for the graph constructions from [8] that are used here (they use exactly the same gadgets as before). So in total, if $g \in \Omega(n^k)$, then it follows that for any sublinear function $f(n)$, and every large enough n, we can find graphs $G(H_0) \equiv_{\mathcal{C}^{f(n)}} G(H_1)$ with $\mathrm{imm}_\lambda(G(H_0)) \neq \mathrm{imm}_\lambda(G(H_1))$. Hence, the counting width of the immanant is $\Omega(n)$.

The $2^{\Omega(n)}$ size lower bound on symmetric circuits follows from the second part of Theorem 5.2.

9 Conclusion

In recent years, a rich theory of symmetric computation has been emerging which has established a tight and surprising relationship between logical definability, circuit complexity and also computation in other models such as linear programming (see [11, 13] for pointers). A central plank of this is the use of variations of the Cai-Fürer-Immerman construction to establish unconditional lower bounds in the context of models of computation with natural symmetries. The work in [20, 21] is a significant example of this, showing an unconditional exponential separation between the complexity of the determinant and the permanent polynomials in the context of symmetric algebraic circuits.

The present work gives a sweeping generalisation of these results in two different directions. First, considering matrix polynomials symmetric under arbitrary permutations of the rows and columns (i.e. the $\mathbf{Sym}_n \times \mathbf{Sym}_m$ action), we give a complete characterisation of the tractable cases in terms of homomorphism polynomials of bounded treewidth, linking this study to other complexity classifications such as [9]. For polynomials on *square* matrices symmetric under the simultaneous application of a permutation to the rows and columns (i.e. the $\mathbf{Sym}_n$ action), we generalise the separation of the determinant and permanent by giving a complete classification of the symmetric complexity of all immanants.

The work raises a number of directions for future research. It suggests that there is a close connection between the counting width of polynomials, which we define, and their symmetric algebraic complexity. We have not established this in full generality and thus the first important direction is to prove or disprove Conjecture 7.7. A step in this direction would be to understand the complexity of linear combinations of

homomorphism polynomials where neither Theorem 6.9 nor Theorem 6.1 applies. That is to say, there is more than one homomorphism polynomial involved and the size of the pattern graphs is not sublinearly bounded.

Our most general results are for matrix polynomials symmetric under $\mathbf{Sym}_n \times \mathbf{Sym}_m$, where we have a complete characterisation of tractable families. The polynomials on square matrices symmetric under the $\mathbf{Sym}_n$ action include many that are not $\mathbf{Sym}_n \times \mathbf{Sym}_n$-symmetric (for instance, the determinant and many other immanants). While we give a complete classification of the immanant polynomials, there are many other such polynomial families. It seems plausible that a complete classification in terms of homomorphism polynomials from directed graphs might be achievable in this case, and we leave it for future work.

It would be interesting to relate this to other methods for establishing lower bounds in algebraic complexity and to natural algorithmic methods in that field. Indeed, one of the striking features of the study of symmetric computation models is that a surprising range of algorithms that are used (for instance in combinatorial optimisation) are indeed symmetric and thus the unconditional lower bounds show the limitations of standard techniques.

Finally, this work may provide the means to prove lower bounds for symmetric circuits for interesting polynomial families beyond the determinant and the permanent. One potential application would be in proof complexity. There has been considerable recent interest in the *Ideal Proof System* [29], in which refutations of unsatisfiable Boolean formulas or systems of polynomial equations take the form of algebraic circuits. One could conceivably use the methods developed here to show that certain families of formulas with natural symmetries do not admit succinct symmetric refutations.

References

[1] 'On Symmetric Circuits and Fixed-Point Logics'. In: *Theory of Computing Systems* 60.3 (2017), pp. 521–551. DOI: 10.1007/s00224-016-9692-2.

[2] 'Monotone Bounded-Depth Complexity of Homomorphism Polynomials'. In: *50th International Symposium on Mathematical Foundations of Computer Science (MFCS 2025)*. Ed. by Paweł Gawrychowski, Filip Mazowiecki and Michał Skrzypczak. Vol. 345. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 19:1–19:18. DOI: 10.4230/LIPIcs.MFCS.2025.19.

[3] 'Choiceless polynomial time'. In: *Annals of Pure and Applied Logic* 100.1-3 (1999), pp. 141–187. DOI: 10.1016/S0168-0072(99)00005-6.

[4] 'A partial k-arboretum of graphs with bounded treewidth'. In: *Theoretical Computer Science* 209.1 (1998), pp. 1–45. DOI: 10.1016/S0304-3975(97)00228-4.

[5] 'Fixed-Parameter Tractable Distances to Sparse Graph Classes'. In: *Algorithmica* 79.1 (2017), pp. 139–158. DOI: 10.1007/s00453-016-0235-7.

[6] 'An optimal lower bound on the number of variables for graph identification'. In: *Combinatorica* 12.4 (1992), pp. 389–410. DOI: 10.1007/BF01305232.

[7] 'A full complexity dichotomy for immanant families'. In: *Proceedings of the 53rd Annual ACM SIGACT Symposium on Theory of Computing*. STOC 2021. New York, NY, USA: Association for Computing Machinery, 2021, pp. 1770–1783. DOI: 10.1145/3406325.3451124.

[8] *A full complexity dichotomy for immanant families*. 2021. arXiv: 2102.04340 [cs.CC]. URL: https://arxiv.org/abs/2102.04340.

[9] 'Homomorphisms Are a Good Basis for Counting Small Subgraphs'. In: *Proceedings of the 49th Annual ACM SIGACT Symposium on Theory of Computing*. STOC 2017. New York, NY, USA: Association for Computing Machinery, 2017, pp. 210–223. DOI: 10.1145/3055399.3055502.

[10] 'On the VNP-Hardness of Some Monomial Symmetric Polynomials'. In: *42nd IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2022)*. Ed. by Anuj Dawar and Venkatesan Guruswami. Vol. 250. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2022, 16:1–16:14. DOI: 10.4230/LIPIcs.FSTTCS.2022.16.

[11] 'Limits of Symmetric Computation'. In: *51st International Colloquium on Automata, Languages, and Programming (ICALP 2024)*. Ed. by Karl Bringmann, Martin Grohe, Gabriele Puppis and Ola Svensson. Vol. 297. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024, 1:1–1:8. DOI: 10.4230/LIPIcs.ICALP.2024.1.

[12] 'Notions of Width: Variables, Pebbles and Supports'. In: *EATCS Bulletin*. The Logic in Computer Science Column 147 (2025). URL: https://bulletin.eatcs.org/index.php/beatcs/article/view/865.

[13] 'Symmetric Computation'. In: *28th EACSL Annual Conference on Computer Science Logic (CSL 2020)*. Ed. by Maribel Fernández and Anca Muscholl. Vol. 152. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2020, 2:1–2:12. DOI: 10.4230/LIPIcs.CSL.2020.2.

[14] 'The Nature and Power of Fixed-point Logic with Counting'. In: *ACM SIGLOG News* (2015), pp. 8–21. DOI: 10.1145/2728816.2728820.

[15] 'Symmetric Proofs in the Ideal Proof System'. In: *50th International Symposium on Mathematical Foundations of Computer Science (MFCS 2025)*. Ed. by Paweł Gawrychowski, Filip Mazowiecki and Michał Skrzypczak. Vol. 345. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 40:1–40:18. DOI: 10.4230/LIPIcs.MFCS.2025.40.

[16] 'Lovász-Type Theorems and Game Comonads'. In: *36th Annual ACM/IEEE Symposium on Logic in Computer Science, LICS 2021, Rome, Italy, June 29 - July 2, 2021*. IEEE, 2021, pp. 1–13. DOI: 10.1109/LICS52264.2021.9470609.

[17] 'Symmetric Algebraic Circuits and Homomorphism Polynomials'. In: *17th Innovations in Theoretical Computer Science Conference (ITCS 2026)*. Ed. by Shubhangi Saraf. Vol. 362. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2026, 46:1–46:15. DOI: 10.4230/LIPIcs.ITCS.2026.46.

[18] 'Definability of semidefinite programming and Lasserre lower bounds for CSPs'. In: *2017 32nd Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)*. 2017 32nd Annual ACM/IEEE Symposium on Logic in Computer Science (LICS). Reykjavik, Iceland: IEEE, 2017, pp. 1–12. DOI: 10.1109/LICS.2017.8005108.

[19] 'Lower Bounds for Symmetric Circuits for the Determinant'. In: *13th Innovations in Theoretical Computer Science Conference, ITCS 2022, January 31 - February 3, 2022, Berkeley, CA, USA*. Ed. by Mark Braverman. Vol. 215. LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2022, 52:1–52:22. DOI: 10.4230/LIPICS.ITCS.2022.52.

[20] 'Symmetric Arithmetic Circuits'. In: *47th International Colloquium on Automata, Languages, and Programming (ICALP 2020)*. Ed. by Artur Czumaj, Anuj Dawar and Emanuela Merelli. Vol. 168. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2020, 36:1–36:18. DOI: 10.4230/LIPIcs.ICALP.2020.36.

[21] 'Symmetric Arithmetic Circuits'. In: *Theory of Computing* 21.14 (2025), pp. 1–32. DOI: 10.4086/toc.2025.v021a014.

[22] 'Symmetric Circuits for Rank Logic'. In: *ACM Trans. Comput. Logic* 23.1 (2021). DOI: 10.1145/3476227.

[23] 'Lovász Meets Weisfeiler and Leman'. In: *45th International Colloquium on Automata, Languages, and Programming (ICALP 2018)*. Ed. by Ioannis Chatzigiannakis, Christos Kaklamanis, Dániel Marx and Donald Sannella. Vol. 107. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2018, 40:1–40:14. DOI: 10.4230/LIPIcs.ICALP.2018.40.

[24] *Computational Invariant Theory*. Encyclopaedia of Mathematical Sciences. Berlin, Heidelberg: Springer Berlin Heidelberg, 2015. DOI: 10.1007/978-3-662-48422-7.

[25] 'Homomorphism Polynomials complete for VP'. In: *Chicago Journal of Theoretical Computer Science* 22.1 (2016), pp. 1–25. DOI: 10.4086/cjtcs.2016.003.

[26] 'On recognizing graphs by numbers of homomorphisms'. In: *Journal of Graph Theory* 64.4 (2010), pp. 330–342. DOI: 10.1002/jgt.20461.

[27] 'Going Deep and Going Wide: Counting Logic and Homomorphism Indistinguishability over Graphs of Bounded Treedepth and Treewidth'. In: *32nd EACSL Annual Conference on Computer Science Logic (CSL 2024)*. Ed. by Aniello Murano and Alexandra Silva. Vol. 288. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024, 27:1–27:17. DOI: 10.4230/LIPIcs.CSL.2024.27.

[28] 'Low-depth algebraic circuit lower bounds over any field'. In: *39th Computational Complexity Conference (CCC 2024)*. Schloss Dagstuhl–Leibniz-Zentrum für Informatik. 2024, pp. 31–1. DOI: 10.4230/LIPIcs.CCC.2024.31.

[29] 'Circuit Complexity, Proof Complexity, and Polynomial Identity Testing: The Ideal Proof System'. In: *J. ACM* 65 (2018), 37:1–37:59. DOI: 10.1145/3230742.

[30] 'Counting Bounded Tree Depth Homomorphisms'. In: *Proceedings of the 35th Annual ACM/IEEE Symposium on Logic in Computer Science*. LICS '20. New York, NY, USA: Association for Computing Machinery, 2020, pp. 507–520. DOI: 10.1145/3373718.3394739.

[31] 'Homomorphism Tensors and Linear Equations'. en. In: *Advances in Combinatorics* (2025). DOI: 10.19086/aic.2025.4.

[32] 'On the complexity of immanants'. In: *Linear and Multilinear Algebra* 18.2 (1985), pp. 127–140. DOI: 10.1080/03081088508817680.

[33] 'Symmetric Formulas for Products of Permutations'. In: *14th Innovations in Theoretical Computer Science Conference, ITCS 2023, January 10-13, 2023, MIT, Cambridge, Massachusetts, USA*. Ed. by Yael Tauman Kalai. Vol. 251. LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2023, 68:1–68:23. DOI: 10.4230/LIPICS.ITCS.2023.68.

[34] 'Logical Hierarchies in PTIME'. In: *Information and Computation* 129.1 (1996), pp. 1–19. DOI: 10.1006/inco.1996.0070.

[35] 'Some Exact Complexity Results for Straight-Line Computations over Semirings'. In: *J. ACM* 29.3 (1982), pp. 874–897. DOI: 10.1145/322326.322341.

[36] 'Violating Constant Degree Hypothesis Requires Breaking Symmetry'. In: *42nd International Symposium on Theoretical Aspects of Computer Science, STACS 2025, March 4-7, 2025, Jena, Germany*. Ed. by Olaf Beyersdorff, Michal Pilipczuk, Elaine Pimentel and Kim Thang Nguyen. Vol. 327. LIPIcs. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2025, 58:1–58:21. DOI: 10.4230/LIPICS.STACS.2025.58.

[37] 'A Selection of Lower Bounds for Arithmetic Circuits'. In: *Perspectives in Computational Complexity*. Ed. by M. Agrawal and V. Arvind. Birkhäuser Basel, 2014.

[38] 'The Weisfeiler-Leman algorithm: an exploration of its power'. In: *ACM SIGLOG News* 7.3 (2020), pp. 5–27. DOI: 10.1145/3436980.3436982.

[39] 'Monotone Arithmetic Complexity of Graph Homomorphism Polynomials'. In: *Algorithmica* 85.9 (2023), pp. 2554–2579. DOI: 10.1007/S00453-023-01108-0.

[40] 'On the Power of the Weisfeiler-Leman Test for Graph Motif Parameters'. In: *The Twelfth International Conference on Learning Representations*. 2024. URL: https://openreview.net/forum?id=FddFxi08J3.

[41] *Large networks and graph limits*. American Mathematical Society colloquium publications volume 60. Providence, Rhode Island: American Mathematical Society, 2012. ISBN: 978-0-8218-9085-1.

[42] 'Operations with structures'. In: *Acta Mathematica Academiae Scientiarum Hungarica* 18.3 (1967), pp. 321–328. DOI: 10.1007/BF02280291.

[43] 'Quantum isomorphism is equivalent to equality of homomorphism counts from planar graphs'. In: *2020 IEEE 61st Annual Symposium on Foundations of Computer Science (FOCS)*. 2020, pp. 661–672. DOI: 10.1109/FOCS46700.2020.00067.

[44] 'The Pebble-Relation Comonad in Finite Model Theory'. In: *Logical Methods in Computer Science* Volume 20, Issue 2 (2024). DOI: 10.46298/lmcs-20(2:9)2024.

[45] 'Homomorphism-Distinguishing Closedness for Graphs of Bounded Tree-Width'. In: *41st International Symposium on Theoretical Aspects of Computer Science (STACS 2024)*. Ed. by Olaf Beyersdorff, Mamadou Moustapha Kanté, Orna Kupferman and Daniel Lokshtanov. Vol. 289. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024, 53:1–53:12. DOI: 10.4230/LIPIcs.STACS.2024.53.

[46] 'Isomorphism types of objects in categories determined by numbers of morphisms'. In: *Acta scientiarum mathematicarum* 35 (1973), pp. 155–160. URL: https://acta.bibl.u-szeged.hu/14440/.

[47] 'Multi-linear formulas for permanent and determinant are of super-polynomial size'. In: *Proceedings of the 36th Annual ACM Symposium on Theory of Computing, Chicago, IL, USA, June 13-16, 2004*. Ed. by László Babai. ACM, 2004, pp. 633–641. DOI: 10.1145/1007352.1007353.

[48] 'Lower Bounds and Separations for Constant Depth Multilinear Circuits'. In: *Proceedings of the 23rd Annual IEEE Conference on Computational Complexity, CCC 2008, 23-26 June 2008, College Park, Maryland, USA*. IEEE Computer Society, 2008, pp. 128–139. DOI: 10.1109/CCC.2008.8.

[49] *Oddomorphisms and homomorphism indistinguishability over graphs of bounded degree*. 2022. arXiv: 2206.10321v1 [math.CO].

[50] 'Graph Minors. XX. Wagner's conjecture'. In: *Special Issue Dedicated to Professor W.T. Tutte* 92.2 (2004), pp. 325–357. DOI: 10.1016/j.jctb.2004.08.001.

[51] 'Equi-Rank Homomorphism Preservation Theorem on Finite Structures'. In: *33rd EACSL Annual Conference on Computer Science Logic (CSL 2025)*. Ed. by Jörg Endrullis and Sylvain Schmitz. Vol. 326. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, 6:1–6:17. DOI: 10.4230/LIPIcs.CSL.2025.6.

[52] 'Subspace-Invariant AC^0 Formulas'. In: *Log. Methods Comput. Sci.* 15.3 (2019). DOI: 10.23638/LMCS-15(3:3)2019.

[53] *The Symmetric Group: Representations, Combinatorial Algorithms, and Symmetric Functions*. 2nd. Springer, 2001.

[54] 'An Algorithmic Meta Theorem for Homomorphism Indistinguishability'. In: *49th International Symposium on Mathematical Foundations of Computer Science (MFCS 2024)*. Ed. by Rastislav Královič and Antonín Kučera. Vol. 306. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024, 82:1–82:19. DOI: 10.4230/LIPIcs.MFCS.2024.82.

[55] 'Homomorphism Indistinguishability'. Dissertation. Aachen: RWTH Aachen University, 2024. DOI: 10.18154/RWTH-2024-11629.

[56] 'Logical equivalences, homomorphism indistinguishability, and forbidden minors'. In: *Information and Computation* 301 (2024), p. 105224. DOI: https://doi.org/10.1016/j.ic.2024.105224.

[57] *Enumerative combinatorics*. Cambridge studies in advanced mathematics 49, 62. Cambridge ; New York: Cambridge University Press, 1997. ISBN: 978-0-521-55309-4.

A Möbius Inversion over Polynomial Rings

Let L be a finite poset and R be a ring. Define the *Möbius function* $\mu\colon L \times L \to R$ of L recursively via

$$\begin{aligned} \mu(s,s) &= 1 && \text{for all } s \in L, \\ \mu(s,u) &= -\sum_{s \le t < u} \mu(s,t) && \text{for all } s < u \text{ in } L. \end{aligned}$$

Lemma A.1 ([57, Proposition 3.7.2])**.** *Let L be a finite poset, R a ring, and M an R-module. Let $f, g\colon L \to M$. Then*

$$g(s) = \sum_{t \ge s} f(t) \text{ for all } s \in L \iff f(s) = \sum_{t \ge s} g(t)\mu(s,t) \text{ for all } s \in L.$$

Proof. Assume the first identity holds. Then

$$\begin{aligned} \sum_{t \ge s} g(t)\mu(s,t) &= \sum_{t \ge s} \mu(s,t) \sum_{u \ge t} f(u) \\ &= \sum_{u \ge s} f(u) \sum_{u \ge t \ge s} \mu(s,t) \\ &= \sum_{u \ge s} f(u)\delta_{u=s} \\ &= f(s). \end{aligned}$$

The converse follows analogously. □

Lemma A.1 is applied to the lattice $\Pi(A)$ of partitions of some finite set A. The Möbius function of the partition lattice is given by the Frucht–Rota–Schützenberger formula [41, (A.2)]. For a partition $\pi \in \Pi(A)$,

$$\mu_\pi \coloneqq (-1)^{|A|-|A/\pi|} \prod_{P \in A/\pi} (|P|-1)!. \tag{17}$$

Lemma A.2. *Let A and I be finite sets. For every map $h\colon A \to I$, let $p_h \in M$. Then, the following hold:*

$$\begin{aligned} \sum_{h\colon A \to I} p_h &= \sum_{\pi \in \Pi(A)} \sum_{h\colon A/\pi \hookrightarrow I} p_{h \circ \pi}, \\ \sum_{h\colon A \hookrightarrow I} p_h &= \sum_{\pi \in \Pi(A)} \mu_\pi \sum_{h\colon A/\pi \to I} p_{h \circ \pi}. \end{aligned}$$

Proof. We first verify the first identity. To that end, note that the set of maps $h\colon A \to I$ can be partitioned according to the partition $\pi \in \Pi(A)$ the map h induces. In other words,

$$\begin{aligned} \{(\pi, h) \mid \pi \in \Pi(A), h\colon A/\pi \hookrightarrow I\} &\to I^A \\ (\pi, h) &\mapsto h \circ \pi \end{aligned}$$

is a bijection. Towards the second identity, we verify the assumptions of Lemma A.1. For a partition $\pi \in \Pi(A)$, let

$$f(\pi) \coloneqq \sum_{h\colon A/\pi \hookrightarrow I} p_{h \circ \pi}, \quad \text{and} \quad g(\pi) \coloneqq \sum_{h\colon A/\pi \to I} p_{h \circ \pi}.$$

Now, the first identity reads as $g(\bot) = \sum_{\pi \in \Pi(A)} f(\pi)$ while the second identity reads as $f(\bot) = \sum_{\pi \in \Pi(A)} \mu_\pi g(\pi)$. Towards the assumptions of Lemma A.1, let $\pi \in \Pi(A)$ be arbitrary. Then

$$\begin{aligned} g(\pi) &= \sum_{h\colon A/\pi \to I} p_{h \circ \pi} \\ &= \sum_{\sigma \in \Pi(A/\pi)} \sum_{h\colon (A/\pi)/\sigma \hookrightarrow I} p_{(h \circ \sigma) \circ \pi} \\ &= \sum_{\substack{\sigma \in \Pi(A) \\ \sigma \ge \pi}} \sum_{h\colon A/\sigma \hookrightarrow I} p_{h \circ \sigma} \\ &= \sum_{\substack{\sigma \in \Pi(A) \\ \sigma \ge \pi}} f(\sigma), \end{aligned}$$

as desired. Lemma A.1 yields the second identity. □

Consider the following corollary. Let F be a bipartite graph with bipartition $A \uplus B = V(F)$. For $n, m \in \mathbb{N}$, consider the *embedding polynomial*

$$\mathrm{emb}_{F,n,m} \coloneqq \sum_{h\colon A\uplus B \hookrightarrow [n]\uplus[m]} \prod_{vw\in E(F)} x_{h(v)h(w)}.$$

Clearly, $\mathrm{emb}_{F,n,m} = |\mathbf{Aut}(F)| \cdot \mathrm{sub}_{F,n,m}$.

Corollary A.3. *Let F be a bipartite multigraph with bipartition $A \uplus B = V(F)$. Let $n, m \in \mathbb{N}$. Then*

$$\mathrm{sub}_{F,n,m} = \frac{1}{|\mathbf{Aut}(F)|} \sum_{\pi\in\Pi(A)} \mu_\pi \sum_{\sigma\in\Pi(B)} \mu_\sigma \hom_{F/(\pi,\sigma),n,m}.$$

Proof. Applying Lemma A.2, we treat the two parts of the bipartition separately.

For maps $h\colon A \to [n]$ and $h'\colon B \to [m]$, define the polynomial

$$p_{h,h'} \coloneqq \prod_{vw\in E(F)} x_{h(v)h'(w)}.$$

Applying Lemma A.2 twice,

$$\begin{aligned}
\mathrm{emb}_{F,n,m} &= \sum_{h\colon A\hookrightarrow[n]} \sum_{h'\colon B\hookrightarrow[m]} p_{h,h'} \\
&= \sum_{\pi\in\Pi(A)} \mu_\pi \sum_{h\colon A/\pi\to[n]} \sum_{h'\colon B\hookrightarrow[m]} p_{h\circ\pi,h'} \\
&= \sum_{\pi\in\Pi(A)} \mu_\pi \sum_{h\colon A/\pi\to[n]} \sum_{\sigma\in\Pi(B)} \mu_\sigma \sum_{h'\colon B/\sigma\to[m]} p_{h\circ\pi,h'\circ\sigma} \\
&= \sum_{\pi\in\Pi(A)} \sum_{\sigma\in\Pi(B)} \mu_\pi\mu_\sigma \hom_{F/(\pi,\sigma),n,m}.
\end{aligned}$$

□

Example A.4. Consider the $F = P_3$, i.e. the path on three vertices where A is a singleton and B is of size two. Then

$$2\,\mathrm{sub}_{F,n,m} = \sum_{i\in[n]} \sum_{\substack{k,j\in[m]\\ k\neq j}} x_{ij}x_{ik}.$$

There is only one partition in $\Pi(A)$, namely $\bot = \top$ with $\mu_\bot = 1$. In $\Pi(B)$, there are two partitions, namely $\bot$ and $\top$ with $\mu_\bot = 1$ and $\mu_\top = -1$. It follows that

$$2\,\mathrm{sub}_{F,n,m} = \hom_{F/(\bot,\bot),n,m} - \hom_{F/(\bot,\top),n,m} = \hom_{P_3,n,m} - \hom_{K_2^2,n,m} = \sum_{i\in[n]} \sum_{k,j\in[m]} x_{ij}x_{ik} - \sum_{i\in[n]} \sum_{k\in[m]} x_{ik}^2$$

where K_2^2 denotes the loopless multigraph with two vertices and two edges. In particular, for $m = 1$, the polynomial $\mathrm{sub}_{F,n,m}$ is the zero polynomial.

B Interpolation

Lemma B.1 (Multivariate Polynomial Interpolation)**.** *Let $\mathbb{K}$ be a field and A be a $\mathbb{K}$-algebra. Let $p \in A[x_1, \ldots, x_t]$ be a multivariate polynomial such that the degree in each of the variables is less than n. Let $a_1, \ldots, a_n \in \mathbb{K}$ be distinct. Then the coefficients of p are $\mathbb{K}$-linear combinations of the $p(a_{j_1}, \ldots, a_{j_t})$ for $j_1, \ldots, j_t \in [n]$. The coefficients of the respective linear combination only depend on $a_1, \ldots, a_n$ and on the coefficient of p to be expressed by the linear combination.*

Note that the polynomial is evaluated at n^t points in $\mathbb{K}^t$. We think of applying the lemma to $\mathbb{K} = \mathbb{Q}$ and $A = \mathbb{Q}[y_1, \ldots, y_t]$.

Proof. Write $p = \sum_{0\le i_1,\ldots,i_t<n} \alpha_{i_1\ldots i_t} x_1^{i_1} \cdots x_t^{i_t}$ for $\alpha_{i_1,\ldots,i_t} \in A$. Then for all $j_1, \ldots, j_t \in [n]$

$$p(a_{j_1}, \ldots, a_{j_t}) = \sum_{0\le i_1,\ldots,i_t<n} a_{j_1}^{i_1} \cdots a_{j_t}^{i_t} \alpha_{i_1\ldots i_t}$$

Define the $\{1,\dots,n\} \times \{0,\dots,n-1\}$-matrix V such that $V_{ji} \coloneqq a_j^i$. By assumption, V is invertible with entries in $\mathbb{K}$. Hence, the Kronecker product $W \coloneqq V \otimes \cdots \otimes V$ of t copies of V is invertible. The above equation now reads as

$$p(a_{j_1},\dots,a_{j_t}) = \sum_{0\le i_1,\dots,i_t<n} W_{j_1\dots j_t,i_1\dots i_t}\alpha_{i_1\dots i_t}$$

Hence, for all $0 \le i_1,\dots,i_t < n$,

$$\alpha_{i_1\dots i_t} = \sum_{j_1,\dots,j_t\in[n]} W^{-1}_{i_1\dots i_t,j_1\dots j_t} p(a_{j_1},\dots,a_{j_t}).$$

Thus, the coefficients $\alpha_{i_1\dots i_t}$ are linear combinations of the $p(a_{j_1},\dots,a_{j_t})$ with coefficients in $\mathbb{K}$. □

Corollary B.2. *Let $\mathbb{F}$ be a field. Let $p \in \mathbb{F}[x_1,\dots,x_t,y_1,\dots y_r]$. Let n be an upper bound on the degree of any of the variables x_i in p. Let $q \in \mathbb{F}[y_1,\dots y_r]$ be the coefficient of a monomial in the x_i-variables in p, when p is viewed as a polynomial in $(\mathbb{F}[y_1,\dots y_r])[x_1,\dots,x_t]$. If p admits a circuit representation of size s, then q admits a circuit representation of size at most $n^t \cdot s$. If the circuit for p is Γ-symmetric for a group Γ that acts on the y_i-variables and fixes the x_i-variables pointwise, then so is the the circuit for q.*

Proof. Lemma B.1 tells us that we can express the desired q as a linear combination of n^t many different evaluations of $p(x_1,\dots,x_t)$. Therefore, we just need n^t many copies of the circuit for p, where different constants are hardwired to the inputs in each copy. If the circuit for p is symmetric under a group Γ that acts on the y-variables and fixes the x-variables pointwise, then substituting the x-variables with constants does not change these symmetries, so the resulting circuit are a sum of Γ-symmetric circuits. □

C CFI Graphs

In this section, we recall the CFI graphs studied in [49] and some of their properties. In the literature, this construction is known as CFI graphs without internal vertices. Throughout, we highlight properties of bipartite CFI graphs.

For a connected simple graph G and $v \in V(G)$, write $E(v) \subseteq E(G)$ for the set of edges incident to v.

Definition C.1. For a connected simple graph G and a vector $U \in \mathbb{Z}_2^{V(G)}$, define the CFI graph G_U via

$$V(G_U) \coloneqq \left\{ (v,T) \;\middle|\; v \in V(G), T \in \mathbb{Z}_2^{E(v)}, \sum_{e\in E(v)} T(e) = U(v) \right\},$$
$$E(G_U) \coloneqq \{(v,T)(u,S) \mid vu \in E(G), T(uv) + S(uv) = 0\}.$$

The set of vertices of the form $(v,T) \in V(G_U)$ associated with $v \in V(G)$ is also called the *CFI gadget* for v. We write $\gamma(G) \coloneqq \sum_{v\in V(G)} 2^{\deg_G(v)-1}$ for the size of G_U. Even though we have defined one graph G_U for every vector $U \in \mathbb{Z}_2^{V(G)}$, there are in fact only two such graphs up to isomorphism.

Lemma C.2 ([49, Corollary 3.7])**.** *For a connected simple graph G and a vector $U \in \mathbb{Z}_2^{V(G)}$, the following are equivalent:*

1. $\sum_{v\in V(G)} U(v) = 0$,
2. $G_0 \cong G_U$,
3. $\hom(G,G_0) = \hom(G,G_U)$.

By virtue of Lemma C.2, we define G_1 as any of the graph G_U for $\sum_{v\in V(G)} U(v) = 1$. The graphs G_0 and G_1 are the *even* and the *odd CFI graphs* of G.

In [49], a characterisation of the graphs F was given that are such that $\hom(F,G_0) = \hom(F,G_1)$ for some connected graph G. We recall this characterisation for the purpose of proving Theorem 6.3.

Definition C.3 ([49, Definition 3.9])**.** Let F and G be simple graphs and $\varphi\colon F \to G$ be a homomorphism. A vertex $a \in V(F)$ is *φ-even / φ-odd* if $|N_F(a) \cap \varphi^{-1}(v)|$ is even / odd for all $v \in N_G(\varphi(a))$. The map φ is an *oddomorphism* if

1. every vertex $a \in V(F)$ is φ-odd or φ-even and
2. every fibre $\varphi^{-1}(v) \subseteq V(F)$ for $v \in V(G)$ contains an odd number of φ-odd vertices.

The map φ is a *weak oddomorphism* if there exists a subgraph $F' \subseteq F$ such that $\varphi|_{F'} \colon F' \to G$ is an oddomorphism.

For example, the identity map $G \to G$ is an oddomorphism.

Theorem C.4 ([49, Theorem 3.13])**.** *Let G be a connected simple graph. For every simple graph F,*

$$\hom(F, G_0) \geq \hom(F, G_1)$$

with strict inequality if, and only if, there exists a weak oddomorphism $F \to G$.

The details of Definition C.3 are irrelevant for this article. In the following observation, we state some straightforward properties of weak oddomorphisms.

Observation C.5. *Let F and G be simple graphs. Every weak oddomorphism $\varphi \colon F \to G$ is surjective on vertices and edges.*

Proof. For every $v \in V(G)$, $\varphi^{-1}(v)$ contains an odd number of φ-odd vertices. Hence, $\varphi^{-1}(v)$ is non-empty and φ is surjective on vertices. Let $uv \in E(G)$ be an edge and $a \in \varphi^{-1}(v)$ be some φ-odd vertex. Then a has an odd number of neighbours in $\varphi^{-1}(u)$. In particular, there is a neighbour $b \in \varphi^{-1}(u)$ of a and $\varphi(ab) = uv$. Hence, φ is surjective on edges. □

Theorem C.6 ([45, Lemma 12, Corollary 13])**.**
1. *Let G be a connected simple graph and $k \in \mathbb{N}$. If $\operatorname{tw}(G) \geq k$, then G_0 and G_1 are $\mathcal{C}^k$-equivalent.*
2. *For all simple graphs F and G for which there exists a weak oddomorphism $F \to G$, we have $\operatorname{tw}(F) \geq \operatorname{tw}(G)$.*

C.1 Bipartite CFI Graphs

In this section, the results stated above are adjusted for bipartite graphs with fixed bipartition. Recall Definition C.1. The map $\rho \colon G_U \to G$ given by $(v, T) \mapsto v$ is a homomorphism. Hence, if G is bipartite, then so is G_U. If the fixed bipartition of G is $A \uplus B$, then we fix the bipartition $\rho^{-1}(A) \uplus \rho^{-1}(B)$ of G_U. We write $\gamma_A(G) \coloneqq \sum_{v \in A} 2^{\deg_G(v)-1}$ and $\gamma_B(G) \coloneqq \sum_{v \in B} 2^{\deg_G(v)-1}$ for the size of left and the right side of the bipartition of G_U.

Corollary C.7. *Let G be a connected simple bipartite graph. For every simple bipartite graph F,*

$$\hom(F, G_0) \geq \hom(F, G_1)$$

with strict inequality if, and only if, there exists a bipartition-respecting weak oddomorphism $F \to G$.

Note that the identity map $G \to G$ is a bipartition-respecting weak oddomorphism. Hence, $\hom(G, G_0) > \hom(G, G_1)$.

Proof. For a homomorphism $\psi \colon F \to G$, write $\hom_\psi(F, G_i)$ for the number of homomorphisms $h \colon F \to G_i$ such that $\rho h = \psi$. Note that

$$\hom(F, G_i) = \sum_{\substack{\psi \colon F \to G \\ \psi \text{ respects bipartitions}}} \hom_\psi(F, G_i).$$

By [49, Theorem 3.6], $\hom_\psi(F, G_0) \geq \hom_\psi(F, G_1)$ for every homomorphism $\psi \colon F \to G$. This implies that $\hom(F, G_0) \geq \hom(F, G_1)$ where F, G_0, and G_i are regarded as graphs with fixed bipartition.

It holds that $\hom(F, G_0) > \hom(F, G_1)$ if, and only if, there exists a bipartition-respecting homomorphism $\psi \colon F \to G$ such that $\hom_\psi(F, G_0) > \hom_\psi(F, G_1)$. By the proof of [49, Theorem 3.13], $\hom_\psi(F, G_0) > \hom_\psi(F, G_1)$ if, and only if, ψ is a weak oddomorphism. □

Corollary C.8. *For every connected simple bipartite graph G, $\operatorname{emb}(G, G_0) > \operatorname{emb}(G, G_1)$.*

Proof. Write $A \uplus B$ for the fixed bipartition of G. By Corollary A.3, for $i \in \{0, 1\}$,

$$\operatorname{emb}(G, G_i) = \sum_{\pi \in \Pi(A)} \sum_{\sigma \in \Pi(B)} \mu_\pi \mu_\sigma \hom(G/(\pi, \sigma), G_i).$$

Here, $G/(\pi,\sigma)$ can be assumed to be simple by discarding possible multiedges. The number of vertices in $G/(\pi,\sigma)$ is $|A/\pi|$ on the left side and $|B/\sigma|$ on the right side. By Corollary C.7 and Observation C.5,

$$\begin{aligned}\operatorname{emb}(G,G_0)-\operatorname{emb}(G,G_1) &= \mu_\perp\sigma_\perp\left(\hom(G/(\perp,\perp),G_0)-\hom(G/(\perp,\perp),G_1)\right)\\ &= \hom(G,G_0)-\hom(G,G_1)\end{aligned}$$

where $\perp$ denotes the discrete partitions of A and B. □

Corollary C.9. *Let G be a connected simple bipartite graph and $k\in\mathbb{N}$. If $\operatorname{tw}(G)\geq k$, then G_0 and G_1 are $\mathcal{C}^k$-equivalent as bipartite graphs with fixed bipartition.*

Proof. Note that the Duplicator strategy constructed in [45, Lemmas 11 and 12] respects the gadgets of the CFI graphs. Thus, this strategy allows Duplicator to win even when the bipartitions are fixed. □

D Hereditary Treewidth, Vertex Cover Number, and Matching Number

In this section, we show that the three graph parameters hereditary treewidth, vertex cover number, and matching number are functionally equivalent. That is, we prove Lemma 7.4.

A *matching* of F is a set $M\subseteq E(F)$ such that for all distinct $e_1,e_2\in M$ it holds that $e_1\cap e_2=\emptyset$. Define the *matching number* $\operatorname{mn}(F)$ of F as the size of the largest matching in F.

Theorem D.1. *For every graph F, $\frac{1}{2}\operatorname{hdtw}(F)\leq\operatorname{mn}(F)\leq\binom{\operatorname{hdtw}(F)+2}{2}$.*

We first consider the upper bound:

Lemma D.2. *For every graph F, $\operatorname{mn}(F)\leq\binom{\operatorname{hdtw}(F)+2}{2}$.*

Proof. Let F be a graph. By [9, Fact 3.4], if F' is a graph with at most $\operatorname{mn}(F)$ edges and no isolated vertices then $\operatorname{tw}(F')\leq\operatorname{hdtw}(F)$. Let $k\in\mathbb{N}$ be such that $\binom{k+2}{2}\geq\operatorname{mn}(F)\geq\binom{k+1}{2}$. Then F' can be taken to be the $(k+1)$-clique, which has treewidth k. Hence, $\operatorname{hdtw}(F)\geq k$. Thus, $\binom{\operatorname{hdtw}(F)+2}{2}\geq\binom{k+2}{2}\geq\operatorname{mn}(F)$. □

By [9, p. 220], one gets an asymptotic linear upper bound on $\operatorname{mn}(F)$ in terms of $\operatorname{hdtw}(F)$.

Lemma D.3. *For every graph F, $\operatorname{hdtw}(F)\leq 2\operatorname{mn}(F)$.*

Proof. We first show that $\operatorname{tw}(F)\leq 2\operatorname{mn}(F)$ by constructing a tree decomposition of width $\operatorname{mn}(F)$ from a maximum matching.

Let M be a maximum matching in F, i.e. a matching of cardinality $\operatorname{mn}(F)$. Write $V\coloneqq\bigcup_{e\in M}e$ for the set of vertices incident to matched edges. Every edge $e\in V(F)$ is incident to at least one vertex in V. Let $W\coloneqq V(F)\setminus V$ denote the set of vertices not incident to any matched edge.

Define a tree decomposition β over the star graph with centre x and tips indexed by $w\in W$ as follows. Let $\beta(x)\coloneqq V$ and $\beta(w)\coloneqq\{w\}\cup V$. Since no two vertices in W are adjacent, this is a valid tree decomposition of width $2\operatorname{mn}(F)$.

Now consider a surjective homomorphism $h\colon F\to F'$. Let $V'\coloneqq h(V)\subseteq V(F')$ and write $W'\coloneqq h(W)\setminus V\subseteq V(F')$. Define a tree decomposition β' for F' over the star graph with centre x' and tips indexed by $w'\in W'$. Let $\beta'(x')\coloneqq V'$ and $\beta(w')\coloneqq\{w'\}\cup V'$. Every bag is of size at most $2\operatorname{mn}(F)+1$ and the bags cover the entire graph F' as h is an edge- and vertex-surjective homomorphism. It is a valid decomposition because all $w_1'\neq w_2'$ in W' are non-adjacent since their preimages under h are non-adjacent. □

A *vertex cover* is a set $C\subseteq V(F)$ such that every edge $e\in E(F)$ is incident to C, i.e. $e\cap C\neq\emptyset$. Write $\operatorname{vc}(F)$ for the size of the smallest vertex cover in F.

Fact D.4. *For every graph F, $\operatorname{mn}(F)\leq\operatorname{vc}(F)\leq 2\operatorname{mn}(F)$.*

Theorem D.1 and Fact D.4 yield Lemma 7.4.

E The Non-Uniform Homomorphism Distinguishing Closure

In this appendix, we prove some further properties of the non-uniform homomorphism distinguishing closure and conduct a comparison to the homomorphism distinguishing closure introduced by *Oddomorphisms and homomorphism indistinguishability over graphs of bounded degree* [49].

Firstly note that it does not matter if the definition of $\mathrm{cl}_n(\mathcal{F})$ is replaced with the following:

$$\mathrm{cl}_{\leq n}(\mathcal{F}) \coloneqq \{F \mid \forall G, H \text{ on } \leq n \text{ vertices. } G \equiv_{\mathcal{F}} H \implies \hom(F,G) = \hom(F,H)\}.$$

Lemma E.1. *For $\mathcal{F}$ as in Proviso 6.2 and all $n \in \mathbb{N}$, $\mathrm{cl}_n(\mathcal{F}) = \mathrm{cl}_{\leq n}(\mathcal{F})$.*

Proof. Clearly, $\mathrm{cl}_{\leq n}(\mathcal{F}) \subseteq \mathrm{cl}_n(\mathcal{F})$. For the converse direction, let G and H be graphs on at most n vertices. Suppose that $G \equiv_{\mathcal{F}} H$. By Proviso 6.2, $K_1 \in \mathcal{F}$ and hence G and H both have $m \leq n$ vertices. By [56, Theorem 7], $G + (m-n)K_1 \equiv_{\mathcal{F}} H + (m-n)K_1$. Let $F \in \mathrm{cl}_n(\mathcal{F})$ and write $F = F' + \ell K_1$ where F' does not contain isolated vertices and $\ell \geq 0$. Then

$$\begin{aligned}
\hom(F,G) &= \hom(F' + \ell K_1, G) \\
&= \hom(F', G) \cdot m^{\ell} \\
&= \hom(F', G + (m-n)K_1) \cdot m^{\ell} \\
&= \hom(F, G + (m-n)K_1) \cdot \frac{m^{\ell}}{n^{\ell}}
\end{aligned}$$

Hence, $\hom(F,G) = \hom(F,H)$, i.e. $F \in \mathrm{cl}_{\leq n}(\mathcal{F})$. □

Corollary E.2. *For $\mathcal{F}$ as in Proviso 6.2 and all $n \in \mathbb{N}$, $\mathrm{cl}_{n+1}(\mathcal{F}) \subseteq \mathrm{cl}_n(\mathcal{F})$.*

Proof. By Lemma E.1, $\mathrm{cl}_{n+1}(\mathcal{F}) = \mathrm{cl}_{\leq n+1}(\mathcal{F}) \subseteq \mathrm{cl}_{\leq n}(\mathcal{F}) = \mathrm{cl}_n(\mathcal{F})$ where the middle inclusion is immediate from the definition. □

Lemma E.3. *For $\mathcal{F}$ as in Proviso 6.2, $\mathrm{cl}(\mathcal{F}) = \bigcap_{n\in\mathbb{N}} \mathrm{cl}_n(\mathcal{F})$.*

Proof. Clearly, $\mathrm{cl}(\mathcal{F}) \subseteq \bigcap_{n\in\mathbb{N}} \mathrm{cl}_n(\mathcal{F})$. For the converse inclusion, let $F \in \bigcap_{n\in\mathbb{N}} \mathrm{cl}_n(\mathcal{F})$ and G and H such that $G \equiv_{\mathcal{F}} H$. Since $K_1 \in \mathcal{F}$, the graphs G and H have the same number of vertices n, say. Since $F \in \mathrm{cl}_n(\mathcal{F})$, G and H have the same number of homomorphisms from F. Hence, $F \in \mathrm{cl}(\mathcal{F})$. □